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The stochastic parametrix method for the Backward diffusion SPDE

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Alla mia famiglia

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Introduction

Stochastic partial differential equations (SPDEs) arise in many applications in probability theory and represent a generalization of classical partial differential equations, where the evolution of the system is influenced by random noise. Within this context, Backward Stochastic Partial Differential Equations (BSPDEs) are a specific type of stochastic differential equations that are solved backward in time. These equations naturally arise in various areas of applied mathematics, particularly in stochastic control theory, mathematical finance and mathematical physics. The idea behind BSPDEs is to describe the current value of a stochastic system based on a final condition imposed at a future time.

BSPDEs find applications in multiple contexts. In mathematical finance, they are used for the valuation of complex financial derivatives, such as options with payoffs dependent on the entire path of the underlying asset. In stochastic control, BSPDEs naturally appear in the analysis of Hamilton-Jacobi-Bellman (HJB) equations, enabling the modelling of dynamic optimization problems under contractions. In mathematical physics, these equations are used to model diffusive phenomena influenced by random noise, such as fluid dynamics in turbulent environments and the transport of particles in random media.

The purpose of my thesis is to prove existence, regularity and estimates from above of the stochastic fundamental solution of a class of SPDEs by using a classical tool from PDEs' theory, the parametrix method. In this work, we consider the backward diffusion SPDE, that is

$$-du_s(x) = L_s u_s(x) dt + G_s^k u_s(x) \star dW_s^k, \quad x \in \mathbb{R}^d, \quad (1)$$

where L_s is the second-order operator

$$L_s u_s(x) = \frac{1}{2} a_s^{ij}(x) \partial_{ij} u_s(x) + b_s^j(x) \partial_j u_s(x), \quad a_s^{ij}(x) = (\sigma \sigma^*)^{ij}(x), \quad (2)$$

and G_s^k is the first-order operator

$$G_s^k u_s(x) = \sigma_s^{ik}(x) \partial_i u_s(x), \quad k = 1, \dots, d_1. \quad (3)$$

where $b = b_s(x):]0, T[\times \mathbb{R}^d \longrightarrow \mathbb{R}^d$, $\sigma = \sigma_s(x):]0, T[\times \mathbb{R}^d \longrightarrow \mathcal{L}(\mathbb{R}^{d_1}; \mathbb{R}^d) \equiv \mathbb{R}^{d \times d_1}$ and W is a d_1 -dimensional Brownian motion. Throughout the thesis, the summation convention over repeated indices is enforced regardless of whether they stand at the same level or at different ones. The coefficient b_s, σ_s are intended to be non-random.

Several authors have studied the Cauchy problem for backward SPDEs. Under coercivity conditions analogous to uniform ellipticity for PDEs, there exists a complete theory in Sobolev spaces (see [38], [4]) and in the space of Bessel potentials (see [14] and [15]). Classical solutions in Hölder classes were first considered in [26] and [40].

In the last decades, the use of analytical or PDE techniques in the study of SPDEs has

become widespread. For instance, the results in [2], [28], [46] are based on classical methods of deterministic PDEs, such as Duhamel principle and a priori Schauder estimates. A further remarkable example is given by the recent series of papers by Krylov [16], [18], [17] where the Hörmander's theorem for SPDEs is proved; see also the recent results in [37].

In this thesis we extend a classical tool, the well-known *parametrix method* for the construction of the fundamental solution of PDEs with Hölder continuous coefficients. We consider an analogy of the Duhamel principle to construct the fundamental solution of (1). These notions are shown to coincide with *Donsker's delta functions*, generalized Wiener functionals, which have been studied in the frame of Malliavin calculus. In this thesis we follow the techniques exposed in [45] that, to the best of our knowledge, is the unique work on the use of the parametrix method on this class of SPDEs.

The main idea of the parametrix method is to construct the fundamental solution $p(s, x; t, y)$ of

$$\left(\frac{\partial}{\partial s} + L_s \right) = 0, \quad x \in \mathbb{R}^d,$$

by using successive approximations based on the explicit fundamental solution of the parabolic equation with constant coefficients

$$\left(\frac{\partial}{\partial s} + \bar{L}^{t,y} \right) = 0, \quad x \in \mathbb{R}^d, \text{ a.e. } s \in [0, t), \quad (4)$$

where

$$\bar{L}^{t,y} = \frac{1}{2} (\sigma \sigma^*)_t^{ij} (y) \partial_{ij}, \quad (5)$$

is the operator acting in the x -variable obtained by freezing the coefficients of L_s in (3.2) at a fixed point $(t, y) \in (0, T] \times \mathbb{R}^d$. The parametrix $\bar{p}$ is obtained from the fundamental solution to (3.5) by evaluating it at the same point (t, y) . Let

$$H(\tau, \xi; t, y) := (L_\tau - \bar{L}^{t,y}) \bar{p}(\tau, \xi; t, y),$$

and we set the time-space convolution operator

$$(p \otimes H)(s, x; t, y) := \int_s^t \int_{\mathbb{R}^d} p(s, x; \tau, \xi) H(\tau, \xi; t, y) d\xi d\tau. \quad (6)$$

Thus the Duhamel representation suggests to find p as

$$p(s, x; t, y) = \bar{p}(s, x; t, y) + \sum_{r=1}^{\infty} (\bar{p} \otimes H^{\otimes r})(s, x; t, y). \quad (7)$$

In the stochastic setting the difficulty comes from the fact that, defining the stochastic convolution, the integrands in general are not well-defined as in the framework of classical Itô-based stochastic calculus. In [13] the authors represent the solution for SPDE in divergence form through the Duhamel formula which involves an anticipative stochastic integral, the so-called Skorohod integral. Hence, we can interpret the stochastic convolutions as a series of iterated Skorohod integrals. Following [45], we define the stochastic convolution of the Skorokhod integral through the Wiener chaos expansion.

Moreover, fix $x \in \mathbb{R}^d$ and consider the diffusion defined by the SDE

$$\begin{cases} dX_t^{s,x} = b_t(X_t^{s,x})dt + \sigma_t^i(X_t^{s,x})dW_t^i, \\ X_s^{s,x} = x. \end{cases} \quad (8)$$

Provided the coefficients b, σ are sufficiently smooth in x , (8) admits a unique solution $X = X_t^{s,x}$ that is a stochastic flow of diffeomorphisms and it is possible to define

$$\delta_y(X_t^{s,x}), \quad (9)$$

where $\delta_y(x)$ is the Dirac delta function with the pole at $y \in \mathbb{R}^d$. We will give a rigorous mathematical meaning of this formal expression in Chapter 1. $\delta_y(X_t^{s,x})$ was first defined by H. H. Kuo in [23], he called it *Donsker's delta function* and we shall see it can be defined as a generalized Wiener functional in the frame of Malliavin calculus. By Krylov's theorem, we can easily verify that the unique solution to (1) with terminal value $u_t(x) = \varphi(x)$ is given by $\varphi(X_t^{s,x})$, hence the fundamental solution is given by (9) in the sense of distributions. Furthermore, we will show that, in the case of C^2 -coefficients, the stochastic fundamental solution to (1) coincides with the Donsker's delta function $\delta_y(X_t^{s,x})$.

The thesis is organized as follows: in Chapter 1 we give some preliminaries and the basic definitions; Chapter 2 deals with the backward diffusion SPDE and its solution; in Chapter 3 we define the parametrix and construct Γ , the candidate stochastic fundamental solution. Finally, in Chapter 4 we show that Γ is actually the fundamental solution to (1) and coincides with the Donsker's delta function.

Notations

$x = (x_1, \dots, x_d)$	point in $\mathbb{R}^d$
$L^2([0, T]^n)$	space of square integrable Borel real functions on $[0, T]^n$
$\tilde{L}^2([0, T]^n)$	space of symmetric square integrable Borel real functions on $[0, T]^n$
$\mathbb{F}_{\tau, t}$	σ – field generated by the Brownian motion $\{W_v - W_\tau : \tau \leq v \leq t\}$
∂_j	j^{th} partial derivative
∂^α	$\partial_1^{\alpha_1} \dots \partial_n^{\alpha_n}$ with multi-index $\alpha \in \mathbb{N}^n$
∇	$(\partial_1, \dots, \partial_d)$
∇^2	Hessian matrix with respect to the spatial variable
$C_0(\mathbb{R}^d)$	space of continuous functions f on $\mathbb{R}^d$ which vanish to infinity
$\mathcal{S}(\mathbb{R}^d)$	Schwartz space of rapidly decreasing smooth $C^\infty(\mathbb{R}^d)$ real functions
$\mathcal{S}'(\mathbb{R}^d)$	space of tempered distributions on $\mathbb{R}^d$
$C_p^\infty(\mathbb{R}^d)$	space of all infinitely continuously differentiable functions $f: \mathbb{R}^d \rightarrow \mathbb{R}$ such that f and all of its partial derivatives have at most polynomial growth
$C_{0,T}(\mathbb{R}^d)$	space of all real-valued Borel measurable functions $f = f_s(x)$ on $[0, T] \times \mathbb{R}^d$ that are continuous in x
$C_{0,T}^k(\mathbb{R}^d)$	space of all real-valued Borel measurable functions $f = f_s(x)$ on $[0, T] \times \mathbb{R}^d$ that are k -times differentiable w.r.t. x
$\text{b}C_{0,T}^k(\mathbb{R}^d)$	space of functions $f \in C_{0,T}^k(\mathbb{R}^d)$ such that $\sum_{ \beta \leq k} \sup_{s \in [0, T]} \sup_{x \in \mathbb{R}^d} \partial^\beta f_s(x) < \infty$
$C_{0,T}^\beta(\mathbb{R}^d)$	space of functions $f = f_s(x)$ which are β – Hölder continuous in x uniformly in s
C	positive constant, only dependent on the quantities in the assumptions, in the context of the proofs its value may update from line to line

Chapter 1

Preliminaries

1.1 The Wiener-Itô chaos expansion

For ease of exposition, we first consider the one-dimensional case; in the following, we will generalize the results of this section to the multidimensional case. Let $(W_t)_{t \in [0, T]}$ be a one-dimensional Brownian motion on the complete Wiener space $(\Omega, \mathcal{F}, \mathbb{P})$, where $\Omega = C_0([0, T]; \mathbb{R})$ is the Banach space of continuous functions $\omega(\cdot)$ such that $\omega(0) = 0$, $\mathbb{P}$ is the Wiener measure and $W_t(\omega) = \omega(t)$. Let $(\mathcal{F}_t^W)_{t \in [0, T]}$ be the filtration generated by $\{W_s, 0 \leq s \leq t\}$, augmented by all the $\mathbb{P}$ -zero measure events.

Definition 1.1.1. A real function $g: [0, T]^n \rightarrow \mathbb{R}$ is called *symmetric* if

$$g(t_{\sigma_1}, \dots, t_{\sigma_n}) = g(t_1, \dots, t_n), \quad (1.1)$$

for all permutations $\sigma = (\sigma_1, \dots, \sigma_n)$ of $(1, 2, \dots, n)$.

Let $L^2([0, T]^n)$ be the space of square integrable Borel real functions on $[0, T]^n$ such that

$$\|g\|_{L^2([0, T]^n)}^2 := \int_{[0, T]^n} g^2(t_1, \dots, t_n) dt_1 \dots dt_n < \infty. \quad (1.2)$$

Let $\tilde{L}^2([0, T]^n) \subset L^2([0, T]^n)$ be the space of *symmetric* square integrable Borel real functions on $[0, T]^n$. Let us consider the n -dimensional simplex

$$S_n[0, T] := \{(t_1, \dots, t_n) \in [0, T]^n : 0 \leq t_1 \leq t_2 \leq \dots \leq t_n \leq T\}$$

Note that this set $S_n[0, T]$ occupies the fraction $\frac{1}{n!}$ of the whole n -dimensional box $[0, T]^n$. Therefore, if $g \in \tilde{L}^2([0, T]^n)$ then $g|_{S_n[0, T]} \in L^2(S_n[0, T])$ and

$$\|g\|_{L^2([0, T]^n)}^2 = n! \int_{S_n[0, T]} g^2(t_1, \dots, t_n) dt_1 \dots dt_n = n! \|g\|_{L^2(S_n[0, T])}^2, \quad (1.3)$$

where $\|\cdot\|_{L^2(S_n[0, T])}$ and $\|\cdot\|_{L^2([0, T]^n)}$ denote the norm induced by $L^2([0, T]^n)$ on $L^2(S_n[0, T])$ and $L^2([0, T]^n)$ respectively, where $L^2(S_n[0, T])$ is the space of square integrable functions on

$S_n[0, T]$. If f is a real function on $[0, T]^n$, the *symmetrization* $\tilde{f}$ is defined by

$$\tilde{f}(t_1, \dots, t_n) = \frac{1}{n!} \sum_{\sigma} f(t_{\sigma_1}, \dots, t_{\sigma_n}), \quad (1.4)$$

where the sum is taken over all permutations $\sigma = (\sigma_1, \dots, \sigma_n)$ of $(1, \dots, n)$. Note that $\tilde{f} = f$ if and only if f is symmetric.

Definition 1.1.2. Let $f \in L^2(S_n[0, T])$. We can define the n -th multiple stochastic integral as

$$\begin{aligned} J_n(f) &:= \int_{S_n[0, T]} f(t_1, \dots, t_n) dW_{t_1} \dots dW_{t_n} \\ &:= \int_0^T \int_0^{t_n} \dots \int_0^{t_3} \int_0^{t_2} f(t_1, \dots, t_n) dW_{t_1} dW_{t_2} \dots dW_{t_{n-1}} dW_{t_n}. \end{aligned} \quad (1.5)$$

Remark 1.1.3. Note that at each iteration the corresponding Itô integral with respect to dW_{t_i} is well-defined, being the integrand $\int_0^{t_i} \dots \int_0^{t_2} f(t_1, \dots, t_n) dW_{t_1} \dots dW_{t_{i-1}}$, $t_i \in [0, t_{i+1}]$, a stochastic process that is $(\mathcal{F}_t^W)$ -adapted and square integrable with respect to $d\mathbb{P} \times dt_i$. In fact, $(\int_0^{t_2} f(t_1, \dots, t_n) dW_{t_1})_{t_2}$ is a $(\mathcal{F}_t^W)$ -adapted process with continuous trajectories and by Itô isometry and, Fubini theorem it holds

$$\begin{aligned} \mathbb{E} \left[\int_0^T \left(\int_0^{t_2} f(t_1, \dots, t_n) dW_{t_1} \right)^2 dt_2 \right] &= \int_0^T \mathbb{E} \left[\left(\int_0^{t_2} f(t_1, \dots, t_n) dW_{t_1} \right)^2 \right] dt_2 \\ &= \int_0^T \int_0^{t_2} f^2(t_1, \dots, t_n) dt_1 dt_2, \end{aligned}$$

which is finite for almost every $(t_3, \dots, t_n)$, hence $(\int_0^{t_3} \int_0^{t_2} f(t_1, \dots, t_n) dW_{t_1} dW_{t_2})_{t_3}$ is a continuous martingale in $L^2(\Omega)$. By iterating this procedure, after $n-1$ steps we obtain that (1.5) is a continuous martingale and $J_n(f) \in L^2(\Omega)$.

In particular, by applying the Itô isometry iteratively we have the following result.

Proposition 1.1.4. Let $m, n \in \mathbb{N}$, $g \in L^2(S_m[0, T])$, $h \in L^2(S_n[0, T])$. The following relations hold true:

$$\mathbb{E}[J_m(g)J_n(h)] = \begin{cases} 0 & \text{if } n \neq m, \\ (g, h)_{L^2(S_n[0, T])} & \text{if } n = m, \end{cases} \quad (1.6)$$

where $(g, h)_{L^2(S_n[0, T])}$ is the inner product of $L^2(S_n[0, T])$. In particular, we have

$$\|J_n(h)\|_{L^2(\Omega)} = \|h\|_{L^2(S_n[0, T])} \quad (1.7)$$

Proof. See [30] proposition 1.4. □

Let $g \in \tilde{L}^2([0, T]^n)$, since

$$\int_{[0, T]^n} g(t_1, \dots, t_n) dt_1 \dots dt_n = n! \int_{S_n[0, T]} g(t_1, \dots, t_n) dt_1 \dots dt_n,$$

this equality motivates the following definition.

Definition 1.1.5. Let $g \in \tilde{L}^2([0, T]^n)$. We define

$$I_n(g) := \int_{[0, T]^n} g(t_1, \dots, t_n) dW_{t_1} \dots dW_{t_n} := n! J_n(g). \quad (1.8)$$

We also call $I_n(g)$ the n -th multiple stochastic integral. When there is no ambiguity, we will simply call $I(g)$ to mean the 1st multiple stochastic integral $I_1(g)$ and in particular it coincides with the Itô integral $\int_0^T g(t) dW_t$.

Remark 1.1.6. Note that the map $g \mapsto I_n(g)$ is linear and from (1.7) and (1.8) we have

$$\|I_n(g)\|_{L^2(\Omega)}^2 = \mathbb{E}[I_n^2(g)] = \mathbb{E}[(n!)^2 J_n^2(g)] = (n!)^2 \|g\|_{L^2(S_n[0, T])}^2 = n! \|g\|_{L^2([0, T]^n)}^2, \quad (1.9)$$

for all $g \in \tilde{L}^2([0, T]^n)$; hence, the image $I_n(\tilde{L}^2([0, T]^n))$ is closed. Moreover, if $g \in \tilde{L}^2([0, T]^m)$ and $h \in \tilde{L}^2([0, T]^n)$, we have

$$\mathbb{E}[I_m(g)I_n(h)] = \begin{cases} 0 & \text{if } n \neq m, \\ n!(g, h)_{L^2([0, T]^n)} & \text{if } n = m, \end{cases} \quad (1.10)$$

with $(g, h)_{L^2([0, T]^n)} = n!(g, h)_{L^2(S_n[0, T])}$.

There is a useful formula due to Itô for the computation of the multiple stochastic integrals. This formula relies on the relationship between Hermite polynomials and the Gaussian distribution density. Let $h_n(x)$, $x \in \mathbb{R}$, denote the n -th Hermite polynomial, which is defined by

$$h_n(x) = (-1)^n e^{\frac{x^2}{2}} \frac{d^n}{dx^n} \left(e^{-\frac{x^2}{2}} \right), \quad n \in \mathbb{N}_{\geq 1}, \quad (1.11)$$

and $h_0(x) = 1$. Thus, the first Hermite polynomials are

$$\begin{aligned} h_0(x) &= 1 \\ h_1(x) &= x \\ h_2(x) &= x^2 - 1 \\ h_3(x) &= x^3 - 3x \\ h_4(x) &= x^4 - 6x^2 + 3 \\ h_5(x) &= x^5 - 10x^3 + 15x \\ &\dots \end{aligned}$$

The generating function of $h_n(x)$ is

$$\exp\left(tx - \frac{t^2}{2}\right) = \sum_{n=0}^{\infty} \frac{t^n}{n!} h_n(x). \quad (1.12)$$

Definition 1.1.7. For each $n \geq 1$ we denote by $\mathcal{H}_n$ the closed linear subspace of $L^2(\Omega, \mathcal{F}, \mathbb{P})$ generated by the random variables $\{h_n(I(f)) : f \in L^2([0, T]), \|f\|_{L^2([0, T])} = 1\}$. The space $\mathcal{H}_n$ is called the Wiener chaos of order n .

Theorem 1.1.8. The space $L^2(\Omega, \mathcal{F}, \mathbb{P})$ can be decomposed into the infinite orthogonal sum of the subspaces $\mathcal{H}_n$:

$$L^2(\Omega, \mathcal{F}, \mathbb{P}) = \bigoplus_{n=0}^{\infty} \mathcal{H}_n$$

Proof. See [29] theorem 1.1.1. □

Definition 1.1.9. We denote by $\mathcal{J}_n: L^2(\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow \mathcal{H}_n$ the orthogonal projection on the n -th Wiener chaos $\mathcal{H}_n$.¹

Definition 1.1.10. Let $f \in \tilde{L}^2([0, T]^n)$, $g \in \tilde{L}^2([0, T]^m)$. We define the tensor product of f and g by

$$(f \boxtimes g)(t_1, \dots, t_{n+m}) := f(t_1, \dots, t_n)g(t_{n+1}, \dots, t_{n+m}),$$

notice that $(f \otimes g) \in L^2([0, T]^{n+m})$, and define the k -th tensor power of f by

$$f^{\boxtimes k}(t_1, \dots, t_{kn}) := f(t_1, \dots, t_n)f(t_{n+1}, \dots, t_{2n}) \dots f(t_{(k-1)n+1}, \dots, t_{kn}),$$

so $f^{\boxtimes k} \in L^2([0, T]^{kn})$. For any $1 \leq r \leq \min\{n, m\}$, we define the contraction of r indices of f and g by

$$(f \boxtimes_r g)(t_1, \dots, t_{n+m-2r}) := \int_{[0, T]^r} f(t_1, \dots, t_{n-r}, s_1, \dots, s_r) g(t_{n+1}, \dots, t_{n+m-r}, s_1, \dots, s_r) ds_1 \dots ds_r. \quad (1.13)$$

Note that $(f \boxtimes_r g) \in L^2([0, T]^{n+m-2r})$. By convention we put $(f \boxtimes_0 g)(t_1, \dots, t_{n+m}) := (f \boxtimes g)(t_1, \dots, t_{n+m})$.

Proposition 1.1.11. Let $f \in \tilde{L}^2([0, T]^n)$, $g \in \tilde{L}^2([0, T]^m)$. Then

$$I_n(f)I_m(g) = \sum_{r=0}^{\min\{n, m\}} r! \binom{n}{r} \binom{m}{r} I_{n+m-2r}(f \boxtimes_r g). \quad (1.14)$$

Proof. See [29], proposition 1.1.3. □

¹Be careful not to confuse the symbols $\mathcal{J}_n$ and J_n . However, we use the same letter since (as we shall see in Theorem 1.1.14) they are strictly connected.

The next result gives the relationship between Hermite polynomials and multiple stochastic integrals.

Theorem 1.1.12. *Let $g \in L^2([0, T])$ and h_n be the n -th Hermite polynomial. Then*

$$I_n(g^{\boxtimes n}) = \|g\|_{L^2}^n h_n\left(\frac{I(g)}{\|g\|_{L^2}}\right), \quad (1.15)$$

where $\|g\|_{L^2} = \|g\|_{L^2([0, T])}$, in particular $I_n(g^{\boxtimes n}) \in \mathcal{H}_n$ by Theorem 1.1.8.

Proof. See [29] proposition 1.1.4. □

Since linear combination of functions of the form $\sum_{i=1}^k c_i g_i^{\boxtimes n}$ are dense in $\tilde{L}^2([0, T]^n)$, we easily deduce the following corollary.

Corollary 1.1.13. *The n -th multiple stochastic integral I_n maps $\tilde{L}^2([0, T]^n)$ onto the Wiener chaos $\mathcal{H}_n$ defined in 1.1.7, and in particular it holds $I_n(\tilde{L}^2([0, T]^n)) = \mathcal{H}_n$.²*

As a consequence of Theorem 1.1.8 and Corollary 1.1.13, we deduce the following version of the Wiener chaos expansion.

Theorem 1.1.14 (The Wiener-Itô chaos expansion). *Let $F \in L^2(\Omega, \mathcal{F}_T^W, \mathbb{P})$. Then there exists a unique sequence $\{f_n\}_{n=0}^\infty$ of kernel functions $f_n \in \tilde{L}^2([0, T]^n)$ such that*

$$F = \sum_{n=0}^{\infty} I_n(f_n), \quad (1.16)$$

where the convergence is in $L^2(\Omega)$. Moreover, we have the isometry

$$\|F\|_{L^2(\Omega)}^2 = \sum_{n=0}^{\infty} n! \|f_n\|_{L^2([0, T]^n)}^2. \quad (1.17)$$

Here $f_0 = \mathbb{E}[F]$ and I_0 is the identity map on the constants.

Remark 1.1.15. *In particular, with the notation of theorem 1.1.14 it holds*

$$\mathcal{J}_n F = I_n(f_n)$$

for any $F \in L^2(\Omega, \mathcal{F}, \mathbb{P})$.

Example 1.1.16. *Let us write the Wiener-Itô chaos expansion of the $L^2(\Omega)$ random variable $\exp\left(I(g) - \frac{1}{2}\|g\|_{L^2}^2\right)$, for $g \in L^2([0, T])$. Set $u = t\sqrt{\lambda}$, from the generating function of Hermite*

²Due to the orthogonality between multiple stochastic integrals of different order, we have $I_n(\tilde{L}^2([0, T]^n))$ is orthogonal to $\mathcal{H}_m$, for $n \neq m$. So the inclusion is an equality.

polynomials (1.12) we get

$$\begin{aligned}
 \exp\left(tx - \frac{t^2\lambda}{2}\right) &= \exp\left(u \frac{x}{\sqrt{t}} - \frac{u^2}{2}\right) \\
 &= \sum_{n=0}^{\infty} \frac{u^n}{n!} h_n\left(\frac{x}{\sqrt{\lambda}}\right) \\
 &= \sum_{n=0}^{\infty} \frac{t^n \lambda^{n/2}}{n!} h_n\left(\frac{x}{\sqrt{\lambda}}\right).
 \end{aligned} \tag{1.18}$$

If we choose $x = I(g)$, $\lambda = \|g\|_{L^2}^2$ and $t = 1$ in (1.18), we get

$$\begin{aligned}
 \exp\left(I(g) - \frac{1}{2}\|g\|_{L^2}^2\right) &= \sum_{n=0}^{\infty} \frac{\|g\|_{L^2}^n}{n!} h_n\left(\frac{I(g)}{\|g\|_{L^2}}\right) \\
 &= \sum_{n=0}^{\infty} \frac{1}{n!} I_n(g^{\otimes n}),
 \end{aligned}$$

by using formula (1.15). Hence $f_n = \frac{1}{n!} g^{\otimes n}$ are the functions involved in the Wiener-Itô chaos expansion of the r.v.

1.2 The multidimensional case

Now we are in the position to generalize the results of the previous section to the multidimensional case and to an arbitrary sub-interval of $[0, T]$. For fixed $0 \leq s < t \leq T$, in the following we denote by $L^2(\Omega, \mathcal{F}_{s,t}^W)$, $\mathcal{H}_n[s, t]$ the closed subspace consisting of all $F \in L^2(\Omega)$, respectively $F \in \mathcal{H}_n$, which are measurable with respect to the σ -field generated by $\{W_\tau - W_s : s \leq \tau \leq t\}$. We extend notations and results introduced before on $L^2(\Omega, \mathcal{F}_{s,t}^W)$ for a sub-interval $[s, t] \subseteq [0, T]$. Let

$$S_n[s, t] = \{(t_1, \dots, t_n) \in [s, t]^n : s \leq t_1 \leq t_2 \leq \dots \leq t_n \leq t\}$$

be n -dimensional simplex on $[s, t]$. Let $L^2([s, t])$, $\|\cdot\|_{L^2([s, t])}$ be the restriction on $[s, t]$ of $L^2([0, T])$ functions, respectively the norm induced by $L^2([0, T])$ on $L^2([s, t])$. In this case the multiple stochastic integral $I_n(f)$ for $f \in \tilde{L}^2([s, t]^n)$ can be expressed as

$$\begin{aligned}
 I_n(f) &= \int_{[s, t]^n} f(t_1, \dots, t_n) dW_{t_1} \dots dW_{t_n} \\
 &= n! \int_{S_n[s, t]} f(t_1, \dots, t_n) dW_{t_1} \dots dW_{t_n} \\
 &= n! \int_s^t \int_s^{t_n} \dots \int_s^{t_3} \int_s^{t_2} f(t_1, \dots, t_n) dW_{t_1} dW_{t_2} \dots dW_{t_{n-1}} dW_{t_n}.
 \end{aligned}$$

Moreover, we can generalize previous computations to the case when we take into account $(W_t)_{t \in [0, T]}$, that is a d_1 -dimensional Brownian motion on the Wiener space $(\Omega, \mathcal{F}, (\mathcal{F}_t^W), \mathbb{P})$, where now $\Omega = C_0([0, T]; \mathbb{R}^{d_1})$ and $\mathbb{P}$ is the d_1 -dimensional Wiener measure. In this case the multiple stochastic integral $I_n(f)$ is defined for $f \in \tilde{L}^2([0, T]^n; \mathbb{R}^{d_1 n})$, where $\tilde{L}^2([0, T]^n; \mathbb{R}^{d_1 n})$ is the space of square integrable functions $f = f^{(i_1, \dots, i_n)}(t_1, \dots, t_n)$ which are symmetric in the variables $(t_j, i_j) \in [0, T] \times \{1, 2, \dots, d_1\}$. Then $I_n(f)$ is given by a sum of multiple stochastic

integrals as:

$$I_n(f) := \sum_{(i_1, \dots, i_n)} \int_{[0, T]^n} f^{(i_1, \dots, i_n)}(t_1, \dots, t_n) dW_{t_1}^{i_1} \dots dW_{t_n}^{i_n} \quad (1.19)$$

$$:= n! \sum_{(i_1, \dots, i_n)} \int_{S_n[0, T]} f^{(i_1, \dots, i_n)}(t_1, \dots, t_n) dW_{t_1}^{i_1} \dots dW_{t_n}^{i_n}, \quad (1.20)$$

where the indices $(i_1, \dots, i_n)$ range over all n -tuples of $1, 2, \dots, d_1$, i.e.

$$(i_1, \dots, i_n) \in \{1, 2, \dots, d_1\}^n.$$

Then, in the general case for a random variable in $L^2(\Omega)^{s, t}$, Theorem 1.1.14 can be restated as follows.

Corollary 1.2.1. *Let $0 \leq s < t \leq T$. Let $F \in L^2(\Omega)^{s, t}$. Then there exists a unique sequence $\{f_n\}_{n=0}^\infty$ of functions $f_n \in \tilde{L}^2([s, t]^n; \mathbb{R}^{d_1^n})$ such that*

$$F = \sum_{n=0}^{\infty} I_n(f_n), \quad (1.21)$$

where the convergence is in $L^2(\Omega)$,

$$I_n(f_n) = \sum_{(i_1, \dots, i_n)} \int_{[s, t]^n} f^{(i_1, \dots, i_n)}(t_1, \dots, t_n) dW_{t_1}^{i_1} \dots dW_{t_n}^{i_n}, \quad (1.22)$$

and

$$\|I_n(f_n)\|_{L^2(\Omega)}^2 = n! \sum_{(i_1, \dots, i_n)} \int_{[s, t]^n} |f^{(i_1, \dots, i_n)}(t_1, \dots, t_n)|^2 dt_1 \dots dt_n \quad (1.23)$$

$$= n! \sum_{(i_1, \dots, i_n)} \|f_n^{(i_1, \dots, i_n)}\|_{L^2([s, t]^n)}^2. \quad (1.24)$$

1.3 The Malliavin derivative

The *Malliavin calculus*, also known as *stochastic calculus of variation*, was first introduced by Paul Malliavin in [27] as an infinite-dimensional integration by parts technique. The purpose of this calculus was to prove the results about the smoothness of densities of solutions of stochastic differential equations driven by a Brownian motion. For a complete discussion about Malliavin calculus and its application to stochastic differential equations, we refer to [29].

Definition 1.3.1. *Let $C_p^\infty(\mathbb{R}^{d_1})$ be the set of all infinitely continuously differentiable functions $f: \mathbb{R}^{d_1} \rightarrow \mathbb{R}$ such that f and all of its partial derivatives have at most polynomial growth. We say that F is a smooth Brownian functional if it has the form*

$$F = f(I(h_1), \dots, I(h_d)), \quad (1.25)$$

where $h_1, \dots, h_d \in L^2([0, T])$, $I(h_i) = \int_0^T h_i(s) W_s$, $(W_s)_{s \in [0, T]}$ is a d_1 -dimensional Brownian motion and the function $f(x) = f(x_1^1, \dots, x_1^{d_1}; \dots; x_d^1, \dots, x_d^{d_1})$ belongs to $C_p^\infty(\mathbb{R}^{dd_1})$. The class of smooth Brownian functional is denoted by $\mathcal{S}$.

Definition 1.3.2. The Malliavin derivative of a smooth random variable F of the form (1.25) is the d_1 -dimensional stochastic process given by

$$(D_t F)^j := \sum_{i=1}^d \frac{\partial f}{\partial x_i^j} (I(h_1), \dots, I(h_d)) h_i(t), \quad (1.26)$$

for $t \in [0, T]$ and $j = 1, 2, \dots, d_1$; see definition 1.2.1 in [29].

The Malliavin derivative $D_t F$ can be regarded as a random variable taking values in the Hilbert space $L^2([0, T]; \mathbb{R}^{d_1})$, thus it is an element of $L^2([0, T] \times \Omega; \mathbb{R}^{d_1}) \simeq L^2(\Omega; L^2([0, T]; \mathbb{R}^{d_1}))$, here we shall identify them. To grasp the intuitive meaning of the definition, we propose the following reasoning. For simplicity, assume $d_1 = 1$ and let $F = f(I(h_1), \dots, I(h_d))$ be a smooth Brownian functional; for $h \in L^2([0, T])$ and $\varepsilon > 0$ we consider the following perturbation of the Brownian path:

$$\{W_t\}_{t \in [0, T]} \mapsto \left\{ W_t + \varepsilon \int_0^t h(s) ds \right\}_{t \in [0, T]}.$$

Under this perturbation,

$$I(h_i) = \int_0^T h_i(t) dW_t \mapsto \int_0^T h_i(t) (dW_t + \varepsilon h(t) dt) = I(h_i) + \varepsilon \langle h_i, h \rangle_{L^2},$$

for each $i = 1, \dots, d$ and we have

$$F = f(I(h_1), \dots, I(h_d)) \mapsto F_\varepsilon := f(I(h_1) + \varepsilon \langle h_1, h \rangle_{L^2}, \dots, I(h_d) + \varepsilon \langle h_d, h \rangle_{L^2}).$$

If we now differentiate F_ε , with respect to ε at zero, we get

$$\begin{aligned} \left. \frac{dF_\varepsilon}{d\varepsilon} \right|_{\varepsilon=0} &= \left. \frac{d}{d\varepsilon} f(I(h_1) + \varepsilon \langle h_1, h \rangle_{L^2}, \dots, I(h_d) + \varepsilon \langle h_d, h \rangle_{L^2}) \right|_{\varepsilon=0} \\ &= \sum_{i=1}^d \frac{\partial f}{\partial x_i} (I(h_1), \dots, I(h_d)) \langle h_i, h \rangle_{L^2} \\ &= \left\langle \sum_{i=1}^d \frac{\partial f}{\partial x_i} (I(h_1), \dots, I(h_d)) h_i, h \right\rangle_{L^2} \\ &= \langle DF, h \rangle_{L^2}. \end{aligned}$$

Then, for any $h \in L^2([0, T])$, the scalar product $\langle DF, h \rangle_{L^2}$ coincides with the directional derivative of F in the direction of the element $\int_0^\cdot h(s) ds$, that is an element of the Cameron-Martin space. From the definition (1.26), it is easy to deduce the following properties.

Proposition 1.3.3. The following hold:

1. $D_t I(h) = h(t)$ for all $h \in L^2([0, T])$, $D_t W_s = \mathbf{1}_{[0, s]}(t)$ and $D_t a = 0$ for all $a \in \mathbb{R}^{d_1}$;
2. if $a, b \in \mathbb{R}$ and $F, G \in \mathcal{S}$, then $D_t(aF + bG) = aD_t F + bD_t G$;

3. if $F, G \in \mathcal{S}$, then $D_t(FG) = D_tF \cdot G + F \cdot D_tG$;
4. in general, the stochastic process $(D_tF)_{t \in [0, T]}$ is not adapted to the filtration $(\mathcal{F}_t^W)_{t \in [0, T]}$ generated by $\{W_t, 0 \leq t \leq T\}$. When $F = f(W_{t_1}, \dots, W_{t_d})$, with $0 < t_1 < \dots < t_d < T$, we have that F is $\mathcal{F}_{t_d}$ -measurable and

$$(D_tF)^j = \sum_{i=1}^d \frac{\partial f}{\partial x_i^j}(W_{t_1}, \dots, W_{t_d}) \mathbf{1}_{[0, t_i]}(t),$$

is identically zero for all $t > t_d$, $j = 1, \dots, d_1$.

Proposition 1.3.4. *The operator D_t is closable from $L^p(\Omega)$ to $L^p(\Omega; L^2([0, T]))$ for any $p \geq 1$, that is if $\{F_n\}_{n \in \mathbb{N}}$ is a sequence in $\mathcal{S}$ converging to zero in $L^p(\Omega)$ and the sequence $\{D_tF_n\}_{n \in \mathbb{N}}$ converges to η in $L^p(\Omega; L^2([0, T]))$, then $\eta = 0$, $\mathbb{P} \times dt$ -almost surely.*

Proof. See [29] proposition 1.2.1. □

Definition 1.3.5. *For $p \geq 1$, we denote by $\mathbb{D}^{1,p}$ the closure of the class of smooth Brownian functional $\mathcal{S}$ with respect to the seminorm*

$$\|F\|_{1,p} := \left(\mathbb{E}[|F|^p] + \mathbb{E}\left[\|DF\|_{L^2([0, T]; \mathbb{R}^{d_1})}^p\right] \right)^{\frac{1}{p}}.$$

Thus $\mathbb{D}^{1,p}$ consists of all $F \in L^p(\Omega)$ such that there exists $\{F_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}$ with the properties that

- $F_n \rightarrow F$ in $L^p(\Omega)$;
- $\{D_tF_n\}_{n \in \mathbb{N}}$ is convergent in $L^p(\Omega; L^2([0, T]; \mathbb{R}^{d_1}))$.

In this case, we set

$$D_tF := \lim_{n \rightarrow \infty} D_tF_n, \quad \text{in } L^p(\Omega; L^2([0, T]; \mathbb{R}^{d_1})),$$

and call DF Malliavin derivative of F .

The closability of the Malliavin derivative, Proposition 1.3.4, makes the definition of the space $\mathbb{D}^{1,p}$ well-posed. The Malliavin derivative satisfies on the space $\mathbb{D}^{1,p}$ the properties valid in $\mathcal{S}$. Moreover, for $p = 2$, $\mathbb{D}^{1,2}$ is a Hilbert space with scalar product

$$\langle F, G \rangle_{1,2} := \mathbb{E}[FG] + \mathbb{E}[\langle DF, DG \rangle_{L^2([0, T]; \mathbb{R}^{d_1})}].$$

Since, for fixed $t \in [0, T]$, the random variable D_tF with $F \in \mathcal{S}$ is still a smooth Brownian functional, it can therefore be differentiated in the Malliavin sense. So, in general, the k -th Malliavin derivative of F is the iterated derivative

$$(D_{t_1, \dots, t_k}^k F)^{j_1, \dots, j_k} = \sum_{i_1, \dots, i_k=1}^d \frac{\partial^k f}{\partial x_{i_1}^{j_1} \dots \partial x_{i_k}^{j_k}}(I(h_1), \dots, I(h_d)) h_1(t_1) \dots h_k(t_k), \quad (1.27)$$

where $t_1, \dots, t_k \in [0, T]$ and $j_1, \dots, j_k = 1, \dots, d_1$. Here $D_{t_1, \dots, t_k}^k F$ can be regarded as a $L^2([0, T]; \mathbb{R}^{d_1})^{\boxtimes k}$ -valued random variable. By proceeding in the same way as in definition 1.3.5, we can define $\mathbb{D}^{k,p}$ the space of k -times differentiable random variables where the k -derivatives are in $L^p(\Omega)$.

Definition 1.3.6. For $p \geq 1$ and $k \in \mathbb{N}$, we denote by $\mathbb{D}^{k,p}$ the closure of the class of smooth Brownian functionals $\mathcal{S}$ with respect to the seminorm

$$\|F\|_{k,p} := \left(\mathbb{E}[|F|^p] + \sum_{m=1}^k \mathbb{E} \left[\|D^m F\|_{L^2([0,T]; \mathbb{R}^{d_1})^{\boxtimes m}}^p \right] \right)^{\frac{1}{p}}, \quad (1.28)$$

The following chain of inclusions follows immediately from the definition:

$$\mathbb{D}^{k,p} \subset \mathbb{D}^{k',p'} \subset L^p(\Omega) \quad \text{if } k' \leq k \text{ and } p' \leq p.$$

We also set for notational convenience $\mathbb{D}^{0,p} := L^p(\Omega)$. In the following, we assume $d_1 = 1$ for simplicity of notation.

Example 1.3.7. Some examples of elements which belong to $\mathbb{D}^{1,p}$ for any $p \geq 1$, but not to $\mathcal{S}$ are:

1. $|W_t|$: we have

$$\mathcal{S} \ni F_n := \sqrt{\frac{1}{n} + W_t^2} \longrightarrow |W_t| \quad \text{in } L^p(\Omega).$$

Therefore,

$$D_s F_n := \frac{W_t \mathbf{1}_{[0,t]}(s)}{\sqrt{\frac{1}{n} + W_t^2}} \longrightarrow \text{sgn}(W_t) \mathbf{1}_{[0,t]}(s) \quad \text{in } L^p(\Omega; L^2([0, T])),$$

$$\text{thus, } D_s |W_t| = \text{sgn}(W_t) \mathbf{1}_{[0,t]}(s).$$

2. $\exp(W_t)$: we have

$$\mathcal{S} \ni G_n := \left(1 + \frac{W_t}{n}\right)^n \longrightarrow \exp(W_t) \quad \text{in } L^p(\Omega).$$

Therefore,

$$D_s G_n := \left(1 + \frac{W_t}{n}\right)^{n-1} \mathbf{1}_{[0,t]}(s) \longrightarrow \exp(W_t) \mathbf{1}_{[0,t]}(s) \quad \text{in } L^p(\Omega; L^2([0, T])),$$

$$\text{thus, } D_s \exp(W_t) = \exp(W_t) \mathbf{1}_{[0,t]}(s).$$

3. $\exp\left(I(g) - \frac{1}{2}\|g\|_{L^2}^2\right)$ for $g \in L^2([0, T])$: we have

$$\mathcal{S} \ni H_n := \left(1 + \frac{I(g)}{n}\right)^n \exp\left(-\frac{1}{2}\|g\|_{L^2}^2\right) \longrightarrow \exp\left(I(g) - \frac{1}{2}\|g\|_{L^2}^2\right) \quad \text{in } L^p(\Omega).$$

Therefore,

$$D_s H_n := \left(1 + \frac{I(g)}{n}\right)^{n-1} g(s) \cdot \exp\left(-\frac{1}{2}\|g\|_{L^2}^2\right) \longrightarrow \exp\left(I(g) - \frac{1}{2}\|g\|_{L^2}^2\right) \cdot g(s) \quad \text{in } L^p(\Omega; L^2([0, T])),$$

$$\text{thus, } D_s \exp\left(I(g) - \frac{1}{2}\|g\|_{L^2}^2\right) = \exp\left(I(g) - \frac{1}{2}\|g\|_{L^2}^2\right) \cdot g(s).$$

Since an arbitrary $F \in L^2(\Omega)$ can be expressed by its chaos expansion

$$F = \sum_{n=0}^{\infty} I_n(f_n),$$

given by formula (1.16), where $f_n \in \tilde{L}^2([0, T]^n)$ for all $n \in \mathbb{N}$, then it is natural to ask if we can express the Malliavin derivative of F (if it exists) by means of this.

Theorem 1.3.8. *Let $F \in L^2(\Omega)$ be given by the Wiener-Itô chaos expansion $F = \sum_{n=0}^{\infty} I_n(f_n)$, according to theorem 1.1.14. Then $F \in \mathbb{D}^{1,2}$ if and only if*

$$\sum_{n=1}^{\infty} n n! \|f_n\|_{L^2([0, T]^n)}^2 < \infty, \quad (1.29)$$

and if this is the case we have

$$D_t F = \sum_{n=1}^{\infty} n I_{n-1}(f_n(\cdot, t)). \quad (1.30)$$

where the notation $I_{n-1}(f_n(\cdot, t))$ means that the $(n-1)$ -multiple stochastic integral is taken with respect to the $n-1$ first variables $t_1, \dots, t_{n-1}$ of $f_n(t_1, \dots, t_{n-1}, t)$, that is, t is fixed and kept outside the integration.

Proof. See [30] theorem A.22. □

Note that formula (1.30) is appealing because it has some resemblance to the derivative of a monomial:

$$\frac{d}{dx} x^n = n x^{n-1}.$$

By iterating derivatives of F , by using formula (1.30), we get the general formula for the k -th Malliavin derivative of F .

Corollary 1.3.9. *For $k \geq 1$, let $F \in \mathbb{D}^{k,2}$ be given by the Wiener-Itô chaos expansion $F = \sum_{n=0}^{\infty} I_n(f_n)$, according to theorem 1.1.14. Then*

$$D_{t_1, \dots, t_k}^k F = \sum_{n=k}^{\infty} n(n-1) \cdots (n-k+1) I_{n-k}(f_n(\cdot, t_1, \dots, t_k)), \quad (1.31)$$

for $t_1, \dots, t_k \in [0, T]$.

Finally, we obtain the Stroock formula which gives another way to compute the f_n kernel functions involved in the Wiener-Itô chaos expansion.

Proposition 1.3.10 (Stroock formula). *Let $F \in \mathbb{D}^{\infty,2} = \bigcap_{k=0}^{\infty} \mathbb{D}^{k,2}$ be given by the Wiener-Itô chaos expansion $F = \sum_{n=0}^{\infty} I_n(f_n)$, according to Theorem 1.1.14. Then*

$$f_n(t_1, \dots, t_n) = \mathbb{E} \left[\frac{D_{t_1, \dots, t_n}^n F}{n!} \right]. \quad (1.32)$$

Proof. Let $n \geq 1$. Take the expectation of expression (1.31), by the L^2 -convergence we can pass the series under the expectation, thus we get

$$\mathbb{E} [D_{t_1, \dots, t_n}^n F] = \sum_{m=n}^{\infty} m(m-1) \cdots (m-n+1) \mathbb{E} [I_{m-n}(f_m(\cdot, t_1, \dots, t_n))] = n! f_n(t_1, \dots, t_n),$$

since $\mathbb{E} [I_{m-n}(f_m(\cdot, t_1, \dots, t_n))] = 0$ for any $m > n$ and I_0 is the identity map on constants. $\square$

Note that formula (1.32) is appealing because it has some resemblance to the Taylor formula for smooth functions.

Example 1.3.11. *Now, by using formula (1.32) we can compute the Wiener-Itô chaos expansion of the $L^2(\Omega)$ random variable $\exp(I(g) - \frac{1}{2}\|g\|_{L^2}^2)$ for $g \in L^2([0, T])$. From example 3 in 1.3.7 we have*

$$D_s \exp \left(I(g) - \frac{1}{2} \|g\|_{L^2}^2 \right) = \exp \left(I(g) - \frac{1}{2} \|g\|_{L^2}^2 \right) \cdot g(s),$$

which belongs to $L^2(\Omega)$. Thus its n -th Malliavin derivative is

$$D_{s_1, \dots, s_n}^n \exp \left(I(g) - \frac{1}{2} \|g\|_{L^2}^2 \right) = \exp \left(I(g) - \frac{1}{2} \|g\|_{L^2}^2 \right) \cdot g^{\otimes n}(s_1, \dots, s_n). \quad (1.33)$$

Hence $\exp(I(g) - \frac{1}{2}\|g\|_{L^2}^2) \in \mathbb{D}^{\infty,2}$ and by the Stroock formula (1.32) we get

$$f_n = \mathbb{E} \left[\frac{D^n F}{n!} \right] = \frac{g^{\otimes n}}{n!} \mathbb{E} \left[\exp \left(I(g) - \frac{1}{2} \|g\|_{L^2}^2 \right) \right].$$

In order to compute the expectation, define the process

$$Y_t := \exp \left(\int_0^t g(s) dW_s - \frac{1}{2} \int_0^t g(s)^2 ds \right).$$

The Novikov condition is satisfied, since

$$\exp \left(\frac{1}{2} \int_0^T g(s)^2 ds \right) < \infty,$$

then the process Y_t is a martingale and $\mathbb{E}[Y_T] = \mathbb{E}[Y_0] = 1$. Thus the Wiener-Itô chaos expansion

of $\exp\left(I(g) - \frac{1}{2}\|g\|_{L^2}^2\right) = \sum_{n=0}^{\infty} I_n(f_n)$ in $L^2(\Omega)$ is given by

$$f_n = \frac{g^{\otimes n}}{n!},$$

according to the computation in 1.1.16.

1.4 The Skorohod integral

The aim of the present section is to introduce and state the properties of the adjoint operator of the Malliavin derivative D . We will also refer to it as the Skorohod integral since it coincides with the extension of the Itô integral to integrands that are not necessarily $(\mathcal{F}^W)$ -adapted. This was introduced for the first time by A. Skorohod in 1975. Since $\mathbb{D}^{1,2}$ is an Hilbert space we can provide the following definition.

Definition 1.4.1. Let $\text{Dom}(\delta)$ be the collection of stochastic processes $u \in L^2([0, T] \times \Omega; \mathbb{R}^{d_1}) \simeq L^2(\Omega; L^2([0, T]; \mathbb{R}^{d_1}))$ such that

$$|\mathbb{E}[\langle DF, u \rangle]| \leq C_u \|F\|_{L^2}, \quad \text{for all } F \in \mathbb{D}^{1,2}, \quad (1.34)$$

where C_u is a positive constant depending on u . If $u \in \text{Dom}(\delta)$, we define $\delta(u)$ to be the unique element of $L^2(\Omega)$ characterized by the identity

$$\mathbb{E}[F\delta(u)] = \mathbb{E}[\langle DF, u \rangle], \quad \text{for all } F \in \mathbb{D}^{1,2}. \quad (1.35)$$

The random variable $\delta(u)$ is called divergence of u or Skorohod integral of u and $\text{Dom}(\delta)$ is the domain of the divergence operator δ .

Proposition 1.4.2. The following hold:

1. $L^2([0, T]) \subset \text{Dom}(\delta)$ and $\delta(h) = I(h)$, for all $h \in L^2([0, T])$;
2. $\mathbb{D}^{1,2} \subset \text{Dom}(\delta)$;
3. if $F \in \mathbb{D}^{1,2}$ and $h \in L^2([0, T])$, then $Fh \in \text{Dom}(\delta)$ and

$$\delta(Fh) = F\delta(h) - \langle DF, h \rangle;$$

4. if $u, v \in \text{Dom}(\delta)$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha u + \beta v \in \text{Dom}(\delta)$ and

$$\delta(\alpha u + \beta v) = \alpha\delta(u) + \beta\delta(v);$$

5. if $u \in \text{Dom}(\delta)$, then $\mathbb{E}[\delta(u)] = 0$;
6. δ is a closed operator, i.e. if $\{u^n\}_{n \in \mathbb{N}} \subset \text{Dom}(\delta)$ is such that $u^n \rightarrow u$ in $L^2([0, T] \times \Omega; \mathbb{R}^{d_1})$ and $\delta(u^n) \rightarrow U$ in $L^2(\Omega)$, then $u \in \text{Dom}(\delta)$ and $\delta(u) = U$.

Proof. See [24], proposition 3.4. □

Theorem 1.4.3. Let $u \in L^2([0, T] \times \Omega; \mathbb{R}^{d_1})$ be a $\mathcal{F}_T$ -measurable stochastic process. Let its Wiener-Itô chaos expansion be $u_t = \sum_{n=0}^{\infty} I_n(f_n(\cdot, t))$, as given by theorem 1.1.14. Then $u \in \text{Dom}(\delta)$ if and only if

$$\sum_{n=0}^{\infty} (n+1)! \|\tilde{f}_n\|_{L^2([0, T]^{n+1})}^2 < \infty \quad (1.36)$$

and if this is the case we have

$$\delta(u) = \sum_{n=0}^{\infty} I_{n+1}(\tilde{f}_n). \quad (1.37)$$

Remark 1.4.4. Observe that the $(n+1)$ -dimensional kernel $f_n^{i_1, \dots, i_n, i}(\cdot, t) \in L^2([0, T]^{n+1}; \mathbb{R}^{d_1(n+1)})$ is symmetric in the first n variables and depends on the parameter $t \in [0, T]$, then $\tilde{f}_n$ the symmetrization of f_n in all its variables will be simply given by

$$\tilde{f}_n^{i_1, \dots, i_n, i}(t_1, \dots, t_n, t) = \quad (1.38)$$

$$= \frac{1}{n+1} \left[f_n^{i_1, \dots, i_n, i}(t_1, \dots, t_n, t) + \sum_{k=1}^n f_n^{i_1, \dots, i_{k-1}, i, i_{k+1}, \dots, i_n, i_k}(t_1, \dots, t_{k-1}, t, t_{k+1}, \dots, t_n, t_k) \right]. \quad (1.39)$$

Proof. See [29], proposition 1.3.7. □

Lemma 1.4.5. Let $u \in L^2([0, T] \times \Omega; \mathbb{R}^{d_1})$ be a $\mathcal{F}_T$ -measurable stochastic process. Let its Wiener-Itô chaos expansion be $u_t = \sum_{n=0}^{\infty} I_n(f_n(\cdot, t))$, as given by theorem 1.1.14. Then u is $(\mathcal{F}^W)$ -adapted if and only if, for each $n \in \mathbb{N}$,

$$f_n(t_1, \dots, t_n, t) = 0 \text{ a.s. in } [0, T]^n, \text{ if } t < \max_{1 \leq k \leq n} t_k. \quad (1.40)$$

Proof. See [30], lemma 2.8. □

The following theorem reveals that the divergence operator δ coincides with an extension of the Itô integral to non adapted stochastic processes introduced by Skorohod in 1975. This is why the divergence operator is also called Skorohod integral.

Theorem 1.4.6. Let $u \in L^2([0, T] \times \Omega; \mathbb{R}^{d_1})$ be a $(\mathcal{F}^W)$ -adapted stochastic process. Then $u \in \text{Dom}(\delta)$ and its Skorohod integral coincides with the Itô integral, that is

$$\delta(u) = \int_0^T u_t dW_t. \quad (1.41)$$

Proof. See [30], theorem 2.9. □

Remark 1.4.7. When we want to stress that $\delta(u)$ is a generalization of the Itô integral to non adapted process u , we utilize the symbol

$$\int_0^T u_t \delta W_t.$$

Example 1.4.8. Let us compute the Skorohod integral $\int_0^T W_T \delta W_t$ by means of theorem 1.37. The Wiener-Ito chaos expansion of the integrand $u_t = W_T = \int_0^T 1 dW_t$, $t \in [0, T]$, is given by

$$f_0 = 0, \quad f_1 = 1, \quad f_n = 0, \quad \text{for } n \in \mathbb{N}_{\geq 2}.$$

Hence

$$\int_0^T W_T \delta W_t = \delta(u) = I_2(\tilde{f}_1) = I_2(1) = 2! J_2(1) = 2 \int_0^T \int_0^{t_2} 1 dW_{t_1} dW_{t_2} = W_T^2 - T,$$

where we used the Itô formula for the last equality. Note that, even if the integrand does not depend on t , we have

$$\int_0^T W_T \delta W_t \neq W_T \int_0^T \delta W_t,$$

since

$$\int_0^T \delta W_t = \delta(1) = I_1(\tilde{1}) = J_1(1) = \int_0^T 1 dW_t = W_T.$$

1.5 Sobolev spaces and formal Wiener chaos

Definition 1.5.1. Let $F \in L^2(\Omega)$. We define the Ornstein-Uhlenbeck operator L as follows:

$$LF := \sum_{n=0}^{\infty} -n \mathcal{J}_n F,$$

provided that the series converges in $L^2(\Omega)$. The domain of the operator is the set

$$\text{Dom } L = \{F \in L^2(\Omega, \mathbb{P}), F = \sum_{n=0}^{\infty} I_n(f_n) : \sum_{n=0}^{\infty} n^2 \|\mathcal{J}_n F\|_{L^2(\Omega)}^2 < \infty\}.$$

The next proposition explains the relationship between the operators D , δ and L .

Proposition 1.5.2. For $F \in L^2(\Omega)$, the statement $F \in \text{Dom } L$ is equivalent to $F \in \text{Dom } \delta D$, i.e.

$F \in \mathbb{D}^{1,2}$ and $DF \in \text{Dom } \delta$. In this case

$$LF = -\delta DF.$$

Proof. See [29], proposition 1.4.3. □

Remark 1.5.3. In particular, $\text{Dom } L \subset \mathbb{D}^{1,2}$ and L is an unbounded self-adjoint operator in $L^2(\Omega)$, that is $\mathbb{E}[F \cdot LG] = \mathbb{E}[LF \cdot G]$.

Proof. See [29], proposition 1.4.4. □

Definition 1.5.4. Let $\mathcal{S}$ be the set of smooth Brownian functionals, for any $1 < p < \infty$ and $s \in \mathbb{R}$, we define the seminorm

$$\|F\|_{s,p} := \|(I - L)^{\frac{s}{2}} F\|_{L^p(\Omega)}, \quad F \in \mathcal{S},$$

Note that $(I - L)^{\frac{s}{2}} F = \sum_{n=0}^{\infty} (1+n)^{\frac{s}{2}} \mathcal{J}_n F$. We define the Sobolev space $\mathbb{D}^{s,p}$ as the closure of $\mathcal{S}$ with respect to the seminorm $\|\cdot\|_{s,p}$.

Remark 1.5.5. $\|F\|_{0,p} = \|F\|_{L^p(\Omega)}$ and $\mathbb{D}^{0,p} = L^p(\Omega)$. $\mathbb{D}^{s,p} \subseteq \mathbb{D}^{s',p'}$ if $p' \leq p$ and $s' \leq s$. For $s > 0$ the operator $(I - L)^{-\frac{s}{2}}$ is an isometric isomorphism in the norm $\|\cdot\|_{s,p}$ between $L^p(\Omega)$ and $\mathbb{D}^{s,p}$, and between $\mathbb{D}^{-s,p}$ and $L^p(\Omega)$ for any $p > 1$. As a consequence, the dual of $\mathbb{D}^{s,p}$ is $\mathbb{D}^{-s,q}$ where $\frac{1}{p} + \frac{1}{q} = 1$.

Remark 1.5.6. The family of seminorms $\|\cdot\|_{k,p}$ defined in (1.28) and $\|\cdot\|_{k,p}$ defined in (1.5.4) for $k \in \mathbb{N}$ are equivalent and thus, the spaces $\mathbb{D}^{k,p}$ defined in 1.3.6 and 1.5.4 coincide and we can use the same notation without ambiguity. This result follows from the Meyer's inequalities, see theorem 1.5.1 [29].

The spaces $\mathbb{D}^{s,p}$ are natural and typical ones which correspond to Bessel potential spaces in classical analysis. So, roughly, $F \in \mathbb{D}^{s,p}$ if and only if the derivatives of F up to the s -th order belong to $L^p(\Omega)$ and therefore, p and s are the integrability index and the differentiability index of the Sobolev space, respectively. Similarly, as in Schwartz theory of distribution, we define the following spaces.

Definition 1.5.7. We define the space of test Wiener functionals by

$$\mathbb{D}^{\infty} := \bigcap_{s>0} \bigcap_{1<p<\infty} \mathbb{D}^{s,p},$$

equipped with the projective topology, that is, $F_n \rightarrow F$ as $n \rightarrow \infty$, in $\mathbb{D}^{\infty}$ if and only if $\|F_n - F\|_{s,p} \rightarrow 0$ as $n \rightarrow \infty$, for all $s > 0, 1 < p < \infty$.

Its dual space, the space of generalized Wiener functionals, is given by

$$\mathbb{D}^{-\infty} := \bigcup_{s>0} \bigcup_{1<p<\infty} \mathbb{D}^{-s,p}.$$

equipped with the inductive topology, that is, $F_n \rightarrow F$ in $\mathbb{D}^{-\infty}$ as $n \rightarrow \infty$ if and only if there exist $s > 0$ and $1 < p < \infty$ such that $\|F_n - F\|_{-s,p} \rightarrow 0$ as $n \rightarrow \infty$.

$\mathbb{D}^\infty$ is an algebra and $\mathbb{D}^{-\infty}$ is a $\mathbb{D}^\infty$ -module. Note that if $s > 0$ the elements of $\mathbb{D}^{-s,p}$ may not be ordinary random variables and are interpreted as generalized random variables. The interest of the space $\mathbb{D}^{-\infty}$ is that it contains the composition of Schwartz distributions with smooth and nondegenerate random variables, as we shall show in the following.

Example 1.5.8. Let $(W_t)_{t \in [0,T]}$ be a one-dimensional Brownian motion. For every $0 < \gamma < \frac{1}{2}$ and $p \in \mathbb{N}_{p \geq 2}$ such that $\gamma < \frac{1}{2} - \frac{1}{2p}$, the random variable defined by

$$W_{p,\gamma}^{2p} := \int_{[0,T]^2} \frac{|W_t - W_s|^{2p}}{|t - s|^{1+2p\gamma}} dt ds,$$

belongs to $\mathbb{D}^\infty$, see [1].

Definition 1.5.9. Given $F \in \mathbb{D}^{-\infty}$, we define the generalized expectation of F by

$$\mathbb{E}[F] = \langle F, 1 \rangle, \quad (1.42)$$

where $\langle \cdot, \cdot \rangle$ is the natural coupling between $\mathbb{D}^{-\infty}$ and $\mathbb{D}^\infty$. Note that if $F \in L^2(\Omega)$, then the generalized expectation coincides with the usual expectation, since by the Wiener-Itô chaos expansion of F , $\mathbb{E}[I_n(f_n)] = 0$ for all $n \geq 1$.

Definition 1.5.10. Let L_0 be the space of finite order Wiener chaos, that is

$$L_0 := \{F \in L^2(\Omega) : \exists n_0 \in \mathbb{N} \text{ such that } \mathcal{J}_n F = 0, \quad \forall n \geq n_0\}$$

Let L'_0 be its dual space, the space of formal Wiener chaos, defined by

$$L'_0 := \{T : T : L_0 \rightarrow \mathbb{R} \text{ linear and continuous on each } \mathcal{H}_n, n \in \mathbb{N}\}.$$

equipped with the projective topology, that is, $F^k \rightarrow F$ as $k \rightarrow \infty$ in L'_0 if and only if $\mathcal{J}_n F^k \rightarrow \mathcal{J}_n F$ in $L^2(\Omega)$ as $k \rightarrow \infty$, for every $n \in \mathbb{N}$.

Remark 1.5.11. For a given $T \in L'_0$, T restricted on each $\mathcal{H}_n$ is given by an element $F_n \in \mathcal{H}_n$ in the sense that $T(G) = \langle F_n, G \rangle_{L^2(\Omega)}$ for every $G \in \mathcal{H}_n$. Hence T is determined by a sequence $F = \{F_n\}_{n=0}^\infty$. Conversely, any such sequence determines $T \in L'_0$ and, therefore, L'_0 can be

identified with the space of all such sequences. By this reason, an element in L'_0 is called a formal Wiener chaos. For $F = \{F_n\}_{n=0}^\infty \in L'_0$, we write with abuse of notation $F_n = \mathcal{J}_n F$, $n \in \mathbb{N}$. The constant $\mathcal{J}_0 F$ is called the generalized expectation of the formal Wiener chaos F .

Theorem 1.5.12. $\mathbb{D}^{-\infty} \hookrightarrow L'_0$ and the inclusion is continuous, in the sense that $X \in \mathbb{D}^{-\infty}$ determines a unique $F_X \in L'_0$ such that $F_X(G) = \langle X, G \rangle$ for all $G \in L_0$ and the map $\mathbb{D}^{-\infty} \ni X \rightarrow F_X \in L'_0$ is injective, where $\langle \cdot, \cdot \rangle$ is the natural coupling between $\mathbb{D}^{-\infty}$ and $\mathbb{D}^\infty$. In particular, the notions of generalized expectations, when restricted on $\mathbb{D}^{-\infty}$, coincide with each other.

Proof. See [45] proposition 2.1. □

We have the following inclusions

$$L_0 \subset \mathbb{D}^\infty \subset \mathbb{D}^{s,p} \subset L^p(\Omega) \subset \mathbb{D}^{-s,p} \subset \mathbb{D}^{-\infty} \subset L'_0, \quad \text{for all } s > 0, 1 < p < \infty.$$

1.6 Donsker's delta function

The notion of *Donsker's delta functions* has been introduced first by Kuo [23] and then, in the frame of Malliavin calculus, by Watanabe [43]. Let F be a d -dimensional random vector. The probability density of F at $x \in \mathbb{R}^d$ can be formally defined as the generalized expectation $\mathbb{E}[\delta_x(F)] = \langle \delta_x(F), 1 \rangle$, provided that $\delta_x(F) \in \mathbb{D}^{-\infty}$, where δ_x denotes the Dirac function at x . In fact, formally we have

$$\mathbb{E}[f(F)] = \langle f(F), 1 \rangle = \left\langle \int_{\mathbb{R}^d} f(x) \delta_x(F) dx, 1 \right\rangle = \int_{\mathbb{R}^d} f(x) \langle \delta_x(F), 1 \rangle dx$$

where the integral $\int_{\mathbb{R}^d} f(x) \delta_x(F) dx$ is $\mathbb{D}^{-\infty}$ -valued so it must be considered in the sense of Bochner. Thus, if the function $x \mapsto \langle \delta_x(F), 1 \rangle$ is smooth then it is the density of F . In the following, we shall make this computation rigorous.

Definition 1.6.1. Let $\mathcal{S}(\mathbb{R}^d)$ be the Schwartz space of rapidly decreasing $C^\infty(\mathbb{R}^d)$ real functions. Let $\mathcal{S}'(\mathbb{R}^d)$ be its dual, the space of tempered distributions.

Let $\mathcal{S}_{2k}$ be the closure of $\mathcal{S}(\mathbb{R}^d)$ with respect to the seminorm

$$\|f\|_{2k} := \|(1 + |x|^2 - \Delta)^k f\|_\infty, \quad f \in \mathcal{S}(\mathbb{R}^d),$$

for $k \in \mathbb{Z}$. Since $\|\cdot\|_{2k}$ is a sequence of increasing seminorms, then we have

$$\mathcal{S}_{2k+2} \subset \mathcal{S}_{2k} \subset \dots \mathcal{S}_2 \subset \mathcal{S}_0 \subset \mathcal{S}_{-2} \subset \dots \mathcal{S}_{-2k} \subset \mathcal{S}_{-2k-2}.$$

Moreover,

$$\mathcal{S}(\mathbb{R}^d) = \bigcap_{k \geq 1} \mathcal{S}_{2k}, \quad \mathcal{S}'(\mathbb{R}^d) = \bigcup_{k \geq 1} \mathcal{S}_{-2k}.$$

Let F be a d -dimensional random vector, for $T \in \mathcal{S}'(\mathbb{R}^d)$ we want to define the composite $T \circ F = T(F)$ as an element in $\mathbb{D}^{-\infty}$. If $T = \varphi \in \mathcal{S}(\mathbb{R}^d)$, then $\varphi(F)$ is obviously a generalized Wiener functional such that $\varphi(F) \in \bigcap_{p \geq 1} L^p(\Omega) \subset \mathbb{D}^{-\infty}$.

Example 1.6.2. We show that the Dirac Delta δ_x belongs to $\mathcal{S}_{-2k}$ for $k = \lfloor \frac{d}{2} \rfloor + 1$.

The space $\mathcal{S}_{-2k}$ is the dual of $\mathcal{S}_{2k}$, this means that δ_x belongs to $\mathcal{S}_{-2k}$ if and only if there exists a constant C such that for all $f \in \mathcal{S}(\mathbb{R}^d)$,

$$|f(x)| = |\langle \delta_x, f \rangle| \leq C \|f\|_{2k}.$$

Let $L_k = (1 + |x|^2 - \Delta)^k$. We want to show that there is a tempered distribution G_k (a fundamental solution) such that $L_k G_k = \delta_0$. Then, we can say that $L_k G_k(x - y) = \delta_x(y)$, so that $G_k(x - y)$ is the fundamental solution centered at x . For $k = \lfloor \frac{d}{2} \rfloor + 1$, the fundamental solution G_k of L_k is a continuous function. In fact, let $\mathcal{F}$ denote the Fourier transform. It holds

$$\mathcal{F}(L_k f) = \mathcal{F}((1 + |x|^2 - \Delta)^k f) = (1 - \Delta_\xi + |\xi|^2)^k \mathcal{F}(f),$$

and also $\mathcal{F}(\delta_0) = 1$. Since $L_k G_k = \delta_0$, we have $\mathcal{F}(L_k G_k) = \mathcal{F}(\delta_0)$, thus $(1 - \Delta_\xi + |\xi|^2)^k \mathcal{F}(G_k) = 1$ and therefore, $\mathcal{F}(G_k) = \frac{1}{(1 - \Delta_\xi + |\xi|^2)^k}$. For large $|\xi|$, the term $|\xi|^2$ dominates, so $\mathcal{F}(G_k) \approx \frac{1}{|\xi|^{2k}}$. To ensure that G_k is a continuous function, we need $\mathcal{F}(G_k)$ to be integrable, meaning $\int |\mathcal{F}(G_k)| d\xi < \infty$. In $\mathbb{R}^d$, the integral of $|\xi|^{-2k}$ behaves like $\int_{r>1} r^{-2k} r^{d-1} dr$ and this integral converges if and only if $-2k + d - 1 < -1$, which simplifies to $2k > d$. Therefore, we require $k = \lfloor \frac{d}{2} \rfloor + 1$. Now we have

$$\langle \delta_x, f \rangle = f(x) = \langle \delta_0, f(x + \cdot) \rangle = \langle L_k G_k, f(x + \cdot) \rangle = \langle G_k, L_k f(x + \cdot) \rangle.$$

Therefore,

$$|f(x)| \leq \int_{\mathbb{R}^d} |G_k(y)| |(L_k f)(x + y)| dy \leq \|L_k f\|_\infty \int_{\mathbb{R}^d} |G_k(y)| dy.$$

Since G_k is continuous and decays sufficiently rapidly, $\int_{\mathbb{R}^d} |G_k(y)| dy$ is finite. Then

$$|f(x)| \leq C \|L_k f\|_\infty = C \|f\|_{2k}.$$

Definition 1.6.3. Let $\Phi \in \mathbb{D}^{-\infty}$. If there exist $1 < p < \infty$, $s > 0$ and $k \in \mathbb{N}$ such that $T \in \mathcal{S}_{-2k}$, $\Phi \in \mathbb{D}^{-s,p}$ and the following holds: for any sequence $\{\varphi_n\}_{n \in \mathbb{N}}$ in $\mathcal{S}(\mathbb{R}^d)$ such that

$$\|\varphi_n - T\|_{-2k} \longrightarrow 0, \quad \text{as } n \rightarrow \infty,$$

it holds that

$$\|\varphi_n(F) - \Phi\|_{-s,p} \longrightarrow 0, \quad \text{as } n \rightarrow \infty,$$

then we say that $T(F)$ is defined in $\mathbb{D}^{-\infty}$ and $T(F) = \Phi$.

Definition 1.6.4. Let δ_x be the delta distribution at $x \in \mathbb{R}^d$, by Example 1.6.2 we know $\delta_x \in \mathcal{S}_{-2k}$, for $k = \lfloor \frac{d}{2} \rfloor + 1$. The composite $\delta_x(F)$, when it can be defined as in Definition 1.6.3, is called Donsker's delta function at x .

Definition 1.6.5. A random vector $F = (F_1, \dots, F_d)$ is called non degenerate if

1. $F_i \in \mathbb{D}^\infty$,
2. the Malliavin matrix γ_F of F defined by $\gamma_F := (\langle DF_i, DF_j \rangle)_{1 \leq i, j \leq d}$ is invertible a.s. and

$$(\det \gamma_F)^{-1} \in \bigcap_{1 < p < \infty} L^p(\Omega).$$

where DF_i is the Malliavin derivative of the i -th component of F .

Theorem 1.6.6 (Watanabe). Let F be a non degenerate d -dimensional random vector, $\delta_x \in \mathcal{S}'(\mathbb{R}^d)$ be the delta distribution at $x \in \mathbb{R}^d$. Then for $k = \lfloor \frac{d}{2} \rfloor + 1$ we have

1. $\delta_x(F)$ is defined as in Definition 1.6.4 as an element in $\mathbb{D}^{-2k, p}$ for any $p > 1$,
2. the map $x \mapsto \delta_x(F)$ belongs to $\text{bC}^{2j}(\mathbb{R}^d; \mathbb{D}^{-2k-2j, p})$, for any $j \in \mathbb{N}$,
3. for any $G \in \mathbb{D}^{2k+2j, q}$, with $\frac{1}{p} + \frac{1}{q} = 1$, the map $x \mapsto \langle \delta_x(F), G \rangle$ belongs to $C^{2j}(\mathbb{R}^d)$ and for any $f \in C_0(\mathbb{R}^d)$ we have

$$\langle f(F), G \rangle = \int_{\mathbb{R}^d} f(x) \langle \delta_x(F), G \rangle dx. \quad (1.43)$$

Proof. See [29] corollary 2.1.3, [44] theorem 2.1. □

Remark 1.6.7. In particular, from theorem 1.6.6, by choosing $G \equiv 1 \in \mathbb{D}^\infty$, the left-hand side of (1.43) is the generalized expectation $\mathbb{E}[f(F)]$ and the function $x \mapsto \langle \delta_x(F), 1 \rangle$ is the density of F which is infinitely differentiable and belongs to $\mathcal{S}(\mathbb{R}^d)$.

Remark 1.6.8. We observe that theorem 1.6.9 gives an alternative proof of the theorem of Malliavin (1978), which states that under the assumptions on F of theorem 1.6.9, the law of F on $\mathbb{R}^d$ has smooth density with respect to Lebesgue measure.

These results can be made sharp by fractionalizing the differentiability indices as follows.

Corollary 1.6.9. Let F be a non degenerate d -dimensional random vector, $\delta_x \in \mathcal{S}'(\mathbb{R}^d)$ be the delta distribution at $x \in \mathbb{R}^d$, $1 < p < \infty$. Then for $\alpha > \frac{d}{q}$, where $\frac{1}{p} + \frac{1}{q} = 1$, we have

1. $\delta_x(F)$ is defined as an element in $\mathbb{D}^{-\alpha, p}$,

2. for any $G \in \mathbb{D}^{\alpha,q}$, the map $x \mapsto \langle \delta_x(F), G \rangle$ belongs to $C^\beta(\mathbb{R}^d)$ for any $0 < \beta < \alpha - \frac{d}{q}$.

where $C^\beta(\mathbb{R}^d)$ is the space of $\lfloor \beta \rfloor$ -times continuously differentiable functions on $\mathbb{R}^d$ whose $\lfloor \beta \rfloor$ -th derivatives are uniformly Hölder continuous with exponent $\beta - \lfloor \beta \rfloor$.

Proof. See [44] theorem 2.1. □

1.7 Backward Itô calculus

In this section we collect some basic definitions about *backward Itô calculus* which are concerned in the *backward diffusion SPDE*, see Chapter 2. We refer to [38] for a complete discussion about backward Itô calculus and BSDPE. Let W be a d_1 -dimensional Brownian motion on the Wiener space $(\Omega, \mathcal{F}, \mathcal{F}_t^W, \mathbb{P})$, where $(\mathcal{F}_t^W)$ denotes the standard Brownian filtration augmented with the $\mathbb{P}$ -null sets $\mathcal{N}$. We define

$$\mathcal{F}_T^{W,t} = \sigma(\mathcal{G} \cup \mathcal{N}), \quad \mathcal{G}_t = \sigma(W_r - W_t, t \leq r \leq T), \quad t \in [0, T],$$

the augmented σ -algebra of Brownian increments between t and T . Notice that $(\mathcal{F}_T^{W,t})_{t \in [0, T]}$ is a decreasing family of σ -algebras. Then the process

$$\mathbf{W}_t := W_T - W_{T-t}, \quad t \in [0, T],$$

is a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P}, \mathbf{F})$ where

$$\mathbf{F}_t := \mathcal{F}_T^{W, T-t}, \quad t \in [0, T],$$

is the *backward Brownian filtration*.

Definition 1.7.1. *The backward stochastic Itô integral is defined as*

$$\int_s^t u_r \star dW_r := \int_{T-t}^{T-s} u_{T-r} d\mathbf{W}_r, \quad 0 \leq s \leq t \leq T, \quad (1.44)$$

under the assumptions on u for which the RHS of (1.44) is defined in the usual Itô sense, that is

1. $t \mapsto u_{T-t}$ is $\mathbf{F}$ -progressively measurable, thus u_t is $\mathcal{F}_T^{W,t}$ -measurable for any $t \in [0, T]$;
2. $u \in L^2([0, T])$ a.s.

For practical purposes, if u is continuous, the backward integral is the limit

$$\int_s^t u_r \star dW_r := \lim_{|\pi| \rightarrow 0^+} \sum_{k=1}^n u_{t_k} (W_{t_k} - W_{t_{k-1}}) \quad (1.45)$$

in probability, where $\pi = \{s = t_0 < t_1 < \dots < t_n = t\}$ denotes a partition of $[s, t]$.

Definition 1.7.2. A d_1 -dimensional backward Itô process is a process of the form

$$X_s = X_T + \int_s^T b_t dt + \int_s^T \sigma_t \star dW_t, \quad s \in [0, T],$$

also written in differential form as

$$-dX_s = b_s ds + \sigma_s \star dW_s. \quad (1.46)$$

Theorem 1.7.3 (Backward Itô formula). Let $\varphi = \varphi(t, x) \in C_{0,T}^{1,2}(\mathbb{R}^d)$ and let X be the backward Itô process defined in (1.46). Then

$$-d\varphi(s, X_s) = \left(\partial_s \varphi(s, X_s) + \frac{1}{2} (\sigma \sigma^*)^{ij} \partial_{ij} \varphi(s, X_s) + b_s^i \partial_i \varphi(s, X_s) \right) ds + \sigma_s^{ik} \partial_i \varphi(s, X_s) \star dW_s^k. \quad (1.47)$$

Chapter 2

The backward diffusion SPDE

In this thesis we study the following equation, called *backward diffusion SPDE* or *Krylov equation*, according to [38]. Consider the backward SPDE

$$-du_s(x) = L_s u_s(x) ds + G_s^k u_s(x) \star dW_s^k, \quad x \in \mathbb{R}^d \quad (2.1)$$

where L_s is the second-order operator

$$L_s = \frac{1}{2}(\sigma \sigma^*)^{ij}_s(x) \partial_{ij} + b_s^j(x) \partial_j, \quad (2.2)$$

and G_s is the first-order operator

$$G_s^k = \sigma_s^{ik}(x) \partial_i, \quad k = 1, \dots, d_1, \quad (2.3)$$

and $b = b_s(x): [0, T] \times \mathbb{R}^d \longrightarrow \mathbb{R}^d$, $\sigma = \sigma_s(x): [0, T] \times \mathbb{R}^d \longrightarrow \mathcal{L}(\mathbb{R}^{d_1}; \mathbb{R}^d) \equiv \mathbb{R}^{d \times d_1}$ are supposed to be not random.

Definition 2.0.1. A solution u on $[0, T]$ for (2.1) is a random field $u_s(x)$ taking values in $L^\infty([0, T], C^2(\mathbb{R}^d))$ such that with probability one satisfies

$$u_s(x) = u_t(x) + \int_s^t L_\tau u_\tau(x) d\tau + \int_s^t G_\tau^k u_\tau(x) \star dW_\tau^k, \quad (2.4)$$

for $x \in \mathbb{R}^d$, $t \in (0, T]$ and $s \in [0, t]$.

Problem (2.1) is called the backward diffusion equation or backward equation of the forward equation, as in [20] Krylov proved that it describes the dynamics of the diffusion process $X_t^{s,x}$ in the backward time variable s . We will explain it in the next.

Definition 2.0.2. A fundamental solution for (2.1) on $[0, T] \times \mathbb{R}^d$ is a function $\Gamma = \Gamma(s, x; t, y)$ defined for $0 \leq s < t \leq T$ and $x, y \in \mathbb{R}^d$, such that for any $(t, y) \in (0, T] \times \mathbb{R}^d$ and for any $\varphi \in \text{bC}^2(\mathbb{R}^d)$ we have

1. the function defined by

$$u_s(x) := \int_{\mathbb{R}^d} \Gamma(s, x; t, y) \varphi(y) dy,$$

is a solution to the backward SPDE (2.1) in the sense of definition 2.0.1 on $[0, t)$,

2. $\lim_{s \rightarrow t^-} u_s(x) = \varphi(x)$, $\mathbb{P}$ -a.s.

Now we state the standing assumptions on the coefficients. We stress the fact that, to define the solution to equation (2.1) we need to impose stronger regularity assumptions, that is Assumption 1, while to construct the stochastic parametrix it is sufficient Assumption 1bis.

Assumption 1 (Regularity). *For every $i = 1, \dots, d$ and $j = 1, \dots, d_1$ we have*

$$b^i, \sigma^{ij} \in L^\infty([0, T], \text{bC}^2(\mathbb{R}^d)),$$

Assumption 1bis (Regularity). *For every $i = 1, \dots, d$ and $j = 1, \dots, d_1$ we have*

$$b^i, \sigma^{ij} \in L^\infty([0, T], \text{bC}^\beta(\mathbb{R}^d)),$$

i.e. b^i, σ^{ij} are β -Hölder continuous in the space variable x , uniformly in time.

Assumption 2 (Non-degeneracy). *The matrix $\sigma \sigma^*$ satisfies the uniformly parabolic condition, that is there exists a positive constant λ_0 such*

$$\frac{1}{\lambda_0} |\xi|^2 \leq \langle (\sigma \sigma^*)_t(x) \xi, \xi \rangle \leq \lambda_0 |\xi|^2, \quad t \in [0, T], x, \xi \in \mathbb{R}^d.$$

It is well known (see, for instance, Theorem 4.6.5 in [22]) that, under Assumption 1, 2, equation

$$\begin{cases} dX_t^{s,x} = b_t(X_t^{s,x}) dt + \sigma_t(X_t^{s,x}) dW_t, & t \geq s, \\ X_s^{s,x} = x. \end{cases}$$

admits a strong solution $X = X_t^{s,x}$ that is a stochastic flow of diffeomorphisms.

Theorem 2.0.3 (Backward diffusion SPDE). *Under Assumptions 1,2, denote by $\tau \mapsto X_\tau^{s,x}$ the solution of the SDE*

$$\begin{cases} dX_\tau^{s,x} = b_\tau(X_\tau^{s,x}) d\tau + \sigma_\tau(X_\tau^{s,x}) dW_\tau, & s \leq \tau \leq t \\ X_s^{s,x} = x. \end{cases} \quad (2.5)$$

Then the process $(s, x) \mapsto X_t^{s,x}$ solves the backward SPDE

$$\begin{cases} -dX_t^{s,x} = L_s X_t^{s,x} ds + G_s^k X_t^{s,x} \star dW_s^k, \\ X_t^{t,x} = x, \end{cases} \quad (2.6)$$

Remark 2.0.4. *The regularity assumption 1 of Theorem 2.0.3 on the coefficients is by no means optimal: in [38], Theorem 5.1, authors prove that flow $(s, x) \mapsto X_t^{s,x}$ is a generalized solution of (2.6) if $b, \sigma \in \text{bC}_{0,T}^1(\mathbb{R}^d)$.*

Proof. For illustrative purposes we only consider the autonomous case and $d_1 = 1$. A general proof can be found in [38], Proposition 5.3. Here we follow the "direct" approach proposed in [42] and re-formulated in [33], Theorem 13. Under Assumption 1, the solution $(s, x) \mapsto X_t^{s,x}$ of SDE (2.5) is a stochastic flow of diffeomorphism that supports second derivatives in the classical sense. We use the Taylor expansion for C^2 -functions:

$$f(\delta) - f(0) = \delta f'(0) + \frac{\delta^2}{2} f''(\lambda\delta), \quad \lambda \in [0, 1]. \quad (2.7)$$

We have

$$\begin{aligned} X_t^{s,x} - x &= X_t^{s,x} - X_t^{t,x} \\ &= \sum_{k=1}^n (X_t^{s_{k-1},x} - X_t^{s_k,x}) \end{aligned}$$

(by the flow property)

$$= \sum_{k=1}^n \left(X_t^{s_k, X_{s_k}^{s_{k-1},x}} - X_t^{s_k,x} \right)$$

(by (2.7) with $f(\delta) = X_t^{s_k, x+\delta}$ and $\delta = \Delta_k X := X_{s_k}^{s_{k-1},x} - x$)

$$= \sum_{k=1}^n \left(\Delta_k X \partial_x X_t^{s_k,x} + \frac{(\Delta_k X)^2}{2} \partial_{xx} X_t^{s_k, x+\lambda_k \Delta_k X} \right), \quad (2.8)$$

for some $\lambda_k = \lambda_k(\omega) \in [0, 1]$. Now, we have

$$\Delta_k X = X_{s_k}^{s_{k-1},x} - x = \int_{s_{k-1}}^{s_k} b(X_\tau^{s_{k-1},x}) d\tau + \int_{s_{k-1}}^{s_k} \sigma(X_\tau^{s_{k-1},x}) dW_\tau.$$

Thus, setting

$$\Delta_k s = s_k - s_{k-1}, \quad \Delta_k W = W_{s_k} - W_{s_{k-1}}, \quad \tilde{\Delta}_k X = b(x) \Delta_k s + \sigma(x) \Delta_k W,$$

by standard estimates for solutions of SDEs, we have

$$\begin{aligned}\Delta_k X - \tilde{\Delta}_k X &= \int_{s_{k-1}}^{s_k} (b(X_\tau^{s_{k-1},x}) - b(x)) d\tau + \int_{s_{k-1}}^{s_k} (\sigma(X_\tau^{s_{k-1},x}) - \sigma(x)) dW_\tau = O(\Delta_k s), \\ \partial_{xx} X_t^{s_k, x+\lambda_k \Delta_k X} - \partial_{xx} X_t^{s_k, x} &= O(\Delta_k s),\end{aligned}$$

in the square mean, that is, more precisely,

$$\mathbb{E} \left[|\Delta_k X - \tilde{\Delta}_k X|^2 + |\partial_{xx} X_t^{s_k, x+\lambda_k \Delta_k X} - \partial_{xx} X_t^{s_k, x}|^2 \right] \leq c(1 + |x|^2)(\Delta_k s)^2,$$

with c depending only on T and the Lipschitz constants of b, σ . From (2.8) we get

$$X_t^{s,x} - x = \sum_{k=1}^n \left(\tilde{\Delta}_k X \partial_x X_t^{s_k, x} + \frac{(\tilde{\Delta}_k X)^2}{2} \partial_{xx} X_t^{s_k, x} \right) + O(\Delta_k s).$$

Next, we recall SDE (2.5) and notice that $\partial_x X_t^{s_k, x}, \partial_{xx} X_t^{s_k, x} \in m\mathcal{F}_t^{s_k, W}$. Thus, passing to the limit, we have

$$\begin{aligned}\sum_{k=1}^n \tilde{\Delta}_k X \partial_x X_t^{s_k, x} &\longrightarrow \int_s^t b(x) \partial_x X_t^{\tau, x} d\tau + \int_s^t \sigma(x) \partial_x X_t^{\tau, x} \star dW_\tau, \\ \sum_{k=1}^n \frac{(\tilde{\Delta}_k X)^2}{2} \partial_{xx} X_t^{s_k, x} &\longrightarrow \int_s^t \sigma^2(x) \partial_{xx} X_t^{\tau, x} d\tau,\end{aligned}$$

in the square mean sense and this concludes the proof. $\square$

Corollary 2.0.5 (Invariance of the backward diffusion SPDE). *Under assumptions 1,2, let $\varphi \in \text{bC}^2(\mathbb{R}^d)$ and X as in (2.5). Define $u_s(x) := \varphi(X_t^{s,x})$. Then $u_s(x)$ is a solution of the backward SPDE*

$$\begin{cases} -du_s(x) = L_s u_s(x) ds + G_s^k u_s(x) \star dW_s^k, \\ u_t(x) = \varphi(x). \end{cases} \quad (2.9)$$

Proof. To fix ideas, we first consider the $d_1 = 1$ case. By the backward SPDE (2.6) and the backward Itô formula 1.7.3, we have

$$\begin{aligned}-d\varphi(X_t^{s,x}) &= \left(\frac{\sigma_s^2(x)}{2} \varphi''(X_t^{s,x}) (\partial_x X_t^{s,x})^2 + \frac{\sigma_s^2(x)}{2} \varphi'(X_t^{s,x}) \partial_{xx} X_t^{s,x} + b_s(x) \varphi'(X_t^{s,x}) \partial_x X_t^{s,x} \right) ds + \\ &\quad + \sigma_s(x) \varphi'(X_t^{s,x}) \partial_x X_t^{s,x} \star dW_s\end{aligned}$$

(by the identities $\partial_x \varphi(X_t^{s,x}) = \varphi'(X_t^{s,x}) \partial_x X_t^{s,x}$ and $\partial_{xx} \varphi(X_t^{s,x}) = \varphi''(X_t^{s,x})(\partial_x X_t^{s,x})^2 + \varphi'(X_t^{s,x}) \partial_{xx} X_t^{s,x}$)

$$= \left(\frac{\sigma_s^2(x)}{2} \partial_{xx} \varphi(X_t^{s,x}) + b_s(x) \partial_x \varphi(X_t^{s,x}) \right) ds + \sigma_s(x) \partial_x \varphi(X_t^{s,x}) \star dW_s,$$

and this proves the thesis. In general dimension d_1 , we have

$$\partial_{x_h} \varphi(X_t^{s,x}) = (\nabla \varphi)(X_t^{s,x}) \partial_{x_h} X_t^{s,x}, \quad (2.10)$$

$$\partial_{x_h x_k} \varphi(X_t^{s,x}) = (\partial_{ij} \varphi)(X_t^{s,x}) (\partial_{x_h} X_t^{s,x})_i (\partial_{x_k} X_t^{s,x})_j + (\nabla \varphi)(X_t^{s,x}) (\partial_{x_h x_k} X_t^{s,x}), \quad (2.11)$$

and by (2.6) and 1.7.3

$$\begin{aligned} -d\varphi(X_t^{s,x}) &= \left(\frac{1}{2} ((\nabla X_t^{s,x}) \sigma_s(x) ((\nabla X_t^{s,x}) \sigma_s(x))^*)_{ij} (\partial_{ij} \varphi)(X_t^{s,x}) (\partial_{ij} \varphi)(X_t^{s,x}) \right) ds + \\ &\quad + \left(\frac{1}{2} (\sigma_s(x) \sigma_s^*(x))_{ij} \partial_{ij} X_t^{s,x} + b_s(x) \nabla X_t^{s,x} \right) (\nabla \varphi)(X_t^{s,x}) ds + \\ &\quad + (\nabla \varphi)(X_t^{s,x}) (\nabla X_t^{s,x}) \sigma_s(x) \star dW_s \end{aligned}$$

(by identities (2.10) and (2.11))

$$= \left(\frac{1}{2} (\sigma_s(x) \sigma_s^*(x))_{ij} \partial_{ij} \varphi(X_t^{s,x}) + b_s(x) \nabla \varphi(X_t^{s,x}) \right) ds + \nabla \varphi(X_t^{s,x}) \sigma_s(x) \star dW_s.$$

□

From Corollary 2.0.5 follows that the function

$$u_s(x) = \varphi(X_t^{s,x}),$$

is a solution of the backward diffusion (2.1) with terminal condition φ . The aim of this thesis is to construct the fundamental solution $\Gamma = \Gamma(s, x; t, y)$ for (2.1) by using the parametrix method. Hence, we expect that the fundamental solution in the sense of definition 2.0.2 is given formally by

$$\Gamma(s, x; t, y) = \delta_y(X_t^{s,x}). \quad (2.12)$$

In Theorem 1.6.9 we defined $\delta_y(F)$ for F non-degenerate random vector. Under Assumptions 1, 2, Watanabe proved in [44] that the solution to the SDE (2.5) $X_t^{s,x}$ belongs to $\mathbb{D}^{2,\infty} := \bigcap_{p>1} \mathbb{D}^{2,p}$, therefore (2.12) can be defined, for each $0 \leq s < t \leq T$ and $x, y \in \mathbb{R}^d$, as a generalized Wiener functional. In particular we have:

Theorem 2.0.6 (Watanabe). *Under Assumptions 1,2, for every $0 \leq s < t \leq T$, $x, y \in \mathbb{R}^d$, let $X_t^{s,x}$ be the solution to the SDE (2.5), then the Donsker delta function $\delta_y(X_t^{s,x})$ can be defined in the sense of Definition 1.6.4 and we have*

$$1. \quad \delta_y(X_t^{s,x}) \in \mathbb{D}^{-\alpha,p}, \text{ if } \frac{d(p-1)}{p} < \min \{\alpha, 1\},$$

2. the map $y \mapsto \delta_y(X_t^{s,x})$ is bounded and continuous,
3. if $G \in \mathbb{D}^{\alpha,\infty} = \bigcap_{p>1} \mathbb{D}^{\alpha,p}$ for $\alpha > 0$, then the map $y \mapsto \langle \delta_y(X_t^{s,x}), G \rangle$ is continuous.

Proof. See [44], Theorem 3.1. □

Under assumptions 1,2, the fundamental solution to (2.1) is given by $\delta_y(X_t^{s,x}) \in \mathbb{D}^{-\alpha,p}$, with $\frac{d(p-1)}{p} < \min\{\alpha, 1\}$ and for every $\varphi \in \text{bC}^2(\mathbb{R}^2)$ satisfies

$$\int_{\mathbb{R}^d} \delta_y(X_t^{s,x}) \varphi(y) dy = \varphi(X_t^{s,x}).$$

In this thesis, we shall try to define the fundamental solution by appealing to the parametrix method and without using the solution $X_t^{s,x}$ of SDE (2.5) explicitly.

Chapter 3

The stochastic parametrix method

In this chapter, we extend the parametrix method for the construction of the fundamental solution of PDEs with Hölder continuous coefficients to study the backward diffusion SPDE (2.1). There are two main problems that one faces when trying to apply the parametrix method to the backward diffusion SPDE: the lack of a Duhamel principle and the lack of super-parabolicity condition on the coefficients b, σ .

We follow the work of Watanabe [45] and start by defining the parametrix to construct the fundamental solution and prove some estimates. This is done by adapting the classical parametrix method, which goes back to Levi [25]. It consists on the approximation of the fundamental solution of a differential equation through an iterative process. We remark that the only results available in the literature based on the parametrix method are proved under the assumption of Hölder regularity of the coefficients in space-variable x . Throughout this chapter we work under Hölder regularity of the coefficients, therefore we replace Assumption 1 with Assumption 1bis.

3.1 The Duhamel representation for the deterministic problem

In order to study the backward SPDE (2.1) let us start by considering the deterministic parabolic PDE associated to (2.1) without noise:

$$\left(\frac{\partial}{\partial s} + L_s \right) u_s(x) = 0, \quad (3.1)$$

where

$$L_s u_s(x) = \frac{1}{2} (\sigma \sigma^*)_{ij}^s(x) \partial_{ij} u_s(x) + b_s^j(x) \partial_j u_s(x), \quad (3.2)$$

appears in the backward SPDE (2.1). It is well known that, under Assumptions 2, 1bis, there exists a unique solution to the backward Cauchy problem

$$\begin{cases} \left(\frac{\partial}{\partial s} + L_s \right) u_s(x) = 0, & x \in \mathbb{R}^d, \text{ a.e. } s \in [0, t], \\ u_t(x) = \varphi(x), & x \in \mathbb{R}^d, \end{cases} \quad (3.3)$$

and it is represented as

$$u_s(x) = \int_{\mathbb{R}^d} p(s, x; t, y) \varphi(y) dy,$$

where $p = p(s, x; t, y)$ is the fundamental solution to (3.3). By well-known results, see e.g. [5], the fundamental solution p to (3.3) exists and is smooth for $0 \leq s < t \leq T$. Moreover, p solves

$$\begin{cases} \left(\frac{\partial}{\partial t} - L_t^* \right) p(s, x; t, y) = 0, & x, y \in \mathbb{R}^d, \text{ a.e. } t \in (s, T], \\ p(s, x; t, \cdot) \xrightarrow[t \rightarrow s^+]{} \delta_x(\cdot), & x \in \mathbb{R}^d, \end{cases} \quad (3.4)$$

where L^* stands for the adjoint of L and acts on the y variable. The main idea of the parametrix method is to construct the fundamental solution $p(s, x; t, y)$ by using successive approximations based on the explicit fundamental solution of the parabolic equation with constant coefficients

$$\left(\frac{\partial}{\partial s} + \bar{L}^{t,y} \right) = 0, \quad x \in \mathbb{R}^d, \text{ a.e. } s \in [0, t), \quad (3.5)$$

where

$$\bar{L}^{t,y} = \frac{1}{2} (\sigma \sigma^*)^i_j(y) \partial_{ij}, \quad (3.6)$$

is the operator acting in the x -variable obtained by freezing the coefficients of L_s in (3.2) at a fixed point $(t, y) \in (0, T] \times \mathbb{R}^d$. The parametrix is obtained from the fundamental solution to by evaluating it at the same point (t, y) . It is well-known that equation (3.5) has a Gaussian fundamental solution which is given explicitly as follows.

Definition 3.1.1. Let $A = (A^{ij})_{1 \leq i, j \leq d}$ be a constant, symmetric and positive definite matrix.

We denote by

$$\Gamma^{\text{heat}}(A, x) = \frac{1}{\sqrt{(2\pi)^d \det A}} \exp \left(-\frac{1}{2} \langle A^{-1} x, x \rangle \right), \quad x \in \mathbb{R}^d,$$

the d -dimensional Gaussian kernel with covariance matrix A . When A is the identity matrix, $A = I_d$, for simplicity we denote

$$\Gamma^\lambda(s, x) := \Gamma^{\text{heat}}(\lambda s I_d, x), \quad s \in \mathbb{R}, x \in \mathbb{R}^d, \quad (3.7)$$

that is the fundamental solution to the heat equation $\partial_t u_t(x) = \frac{\lambda}{2} \Delta u_t(x)$.

Clearly Γ^{heat} is a smooth function and satisfies

$$\left(\frac{\partial}{\partial s} - \frac{1}{2} \sum_{i,j=1}^d A^{ij} \partial_{ij} \right) \Gamma^{\text{heat}}(sA, x) = 0, \quad s > 0, x \in \mathbb{R}^d,$$

Now, we denote by

$$\bar{p}^{\tau, \xi}(s, x; t, y) := \Gamma^{\text{heat}}((t-s)(\sigma \sigma^*)_\tau(\xi), y-x), \quad (3.8)$$

Notice that $\bar{p}$ is well defined for $0 \leq s < t \leq T$ in virtue of Assumption 2 and solves

$$\left(\frac{\partial}{\partial s} + \bar{L}^{\tau, \xi} \right) \bar{p}^{\tau, \xi}(s, x; t, y) = 0,$$

for any $x, y \in \mathbb{R}^d$. Finally, we define the parametrix for the fundamental solution to (3.3) as

$$\bar{p}(s, x; t, y) := \bar{p}^{t,y}(s, x; t, y). \quad (3.9)$$

By formal derivation and the equations (3.4), (3.5) we obtain the Duhamel representation:

$$\begin{aligned} p(s, x; t, y) - \bar{p}(s, x; t, y) &= \int_s^t \frac{\partial}{\partial \tau} \int_{\mathbb{R}^d} p(s, x; \tau, \xi) \bar{p}(\tau, \xi; t, y) d\xi d\tau \\ &= \int_s^t \int_{\mathbb{R}^d} \left(L_\tau^* p(s, x; \tau, \xi) \bar{p}(\tau, \xi; t, y) - p(s, x; \tau, \xi) \bar{L}^{t,y} \bar{p}(\tau, \xi; t, y) \right) d\xi d\tau \\ &= \int_s^t \int_{\mathbb{R}^d} \left(p(s, x; \tau, \xi) L_\tau \bar{p}(\tau, \xi; t, y) - p(s, x; \tau, \xi) \bar{L}^{t,y} \bar{p}(\tau, \xi; t, y) \right) d\xi d\tau \\ &= \int_s^t \int_{\mathbb{R}^d} p(s, x; \tau, \xi) \left(L_\tau - \bar{L}^{t,y} \right) \bar{p}(\tau, \xi; t, y) d\xi d\tau. \end{aligned} \quad (3.10)$$

For notational convenience, we write

$$H(\tau, \xi; t, y) := \left(L_\tau - \bar{L}^{t,y} \right) \bar{p}(\tau, \xi; t, y),$$

and we set the time-space convolution operator¹

$$(p \otimes H)(s, x; t, y) := \int_s^t \int_{\mathbb{R}^d} p(s, x; \tau, \xi) H(\tau, \xi; t, y) d\xi d\tau. \quad (3.11)$$

Thus, from the Duhamel representation (3.10), we have

$$p(s, x; t, y) = \bar{p}(s, x; t, y) + (p \otimes H)(s, x; t, y). \quad (3.12)$$

For $N \geq 2$, by iterating $N - 1$ -times the identity (3.12), we obtain

$$p(s, x; t, y) = \bar{p}(s, x; t, y) + \sum_{r=1}^{N-1} (\bar{p} \otimes H^{\otimes r})(s, x; t, y) + (p \otimes H^{\otimes N})(s, x; t, y), \quad (3.13)$$

where $H^{\otimes l}$, $l \geq 1$, stands for the iterated convolutions of H . Hence, when N tends to infinity we formally obtain a representation of p as a series of convolution kernels. Friedman in [5] got a Gaussian upper bound over a compact time interval to the series

$$\sum_{r=1}^{\infty} (\bar{p} \otimes H^{\otimes r})(s, x; t, y),$$

thus, as N tends to infinity, the sum (3.13) converges compact uniformly in $(s, x; t, y)$.

3.2 The Duhamel representation for the stochastic problem

Our purpose is to consider an analogy of the above Duhamel representation formula in the case of a backward SPDE. Let us look for a parametrix for the SPDE (2.1) by proceeding in the

¹Be careful not to confuse symbols $\otimes$ for convolution and $\boxtimes$ for tensor product

same way to what we have done in the deterministic heat equation. Let L_s , G_s , the operators defined as in (2.2), (2.3), in Chapter 2. Let

$$\bar{L}^{\tau,\xi} = \frac{1}{2}(\sigma\sigma^*)_{\tau}^{ij}(\xi)\partial_{ij}, \quad \bar{G}^{k,\tau,\xi} = \sigma_{\tau}^{ik}(\xi)\partial_i, \quad (3.14)$$

be the operators acting in the x -variable obtained by freezing the coefficients of L_s , respectively G_s , at a fixed point $(\tau, \xi) \in (s, T] \times \mathbb{R}^d$. Consider the following SDE

$$\begin{cases} dX_r^{s,x} = \sigma_{\tau}(\xi)dW_r, & s \leq r \leq t \\ X_s^{s,x} = x, \end{cases} \quad (3.15)$$

which is solved by the process $(x + \sigma_{\tau}(\xi)(W_r - W_s))_{r \in [s,t]}$. Then, by Corollary 2.0.5, the backward SPDE

$$\begin{cases} -dv_s(x) = \bar{L}^{\tau,\xi}v_s(x)ds + \bar{G}^{k,\tau,\xi}v_s(x) \star dW_s^k, \\ v_t(x) = \varphi(x), \end{cases} \quad (3.16)$$

has a solution v that admits the representation $v_s^{\tau,\xi}(x) = \varphi(x + \sigma_{\tau}(\xi)(W_t - W_s))$, for $0 \leq s \leq t$, $x \in \mathbb{R}^d$. Hence, the fundamental solution to (3.16) is given by $\Gamma^{\tau,\xi}(s, x; t, y) = \delta_y(x + \sigma_{\tau}(\xi)(W_t - W_s))$ as an element in the space $\mathbb{D}^{-\infty}$. In fact, under Assumptions 1bis and 2, by Theorem 1.6.9 we have

$$\delta_y(x + \sigma_{\tau}(\xi)(W_t - W_s)) \in \mathbb{D}^{-\alpha,2},$$

for $\alpha > \frac{d}{2}$ and the map

$$(s, x; t, y) \in D \mapsto \delta_y(x + \sigma_{\tau}(\xi)(W_t - W_s)),$$

is bounded and continuous, where $D = \{(s, x; t, y) : 0 \leq s < t \leq T, x, y \in \mathbb{R}^d\}$. Thus we define the parametrix for the backward SPDE (2.1) as

$$Z(s, x; t, y) := \Gamma^{t,y}(s, x; t, y) = \delta_y(x + \sigma_t(y)(W_t - W_s)). \quad (3.17)$$

According to the notation introduced in 1.5.11 for elements in L'_0 , we define $Z_n(s, x; t, y) := \mathcal{J}_n Z(s, x; t, y) \in \mathcal{H}_n[s, t]$, thus we identify $Z(s, x; t, y) = \{Z_n(s, x; t, y)\}_{n=0}^{\infty}$.

Definition 3.2.1. *In this setting, we define $H(s, x; t, y) = \{H_n(s, x; t, y)\}_{n=0}^{\infty}$ as an element in L'_0 as*

$$H_n(s, x; t, y) := \left(L_s - \bar{L}^{t,y} \right) Z_n(s, x; t, y), \quad (3.18)$$

as in the deterministic case, and $A^k(s, x; t, y) = \{A_n^k(s, x; t, y)\}_{n=0}^{\infty}$ as

$$A_n^k(s, x; t, y) := \left(G_s^k - \bar{G}^{k,t,y} \right) Z_n(s, x; t, y), \quad (3.19)$$

which are defined precisely in each component of Wiener chaos $\mathcal{H}_n[s, t]$, for every $n \in \mathbb{N}$.

Finally, we expect to obtain the Duhamel representation for the fundamental solution

$\Gamma = \Gamma(s, x; t, y)$ to (2.1) as solution of

$$\Gamma(s, x; t, y) = Z(s, x; t, y) + (\Gamma \otimes H)(s, x; t, y) + \sum_{k=1}^{d_1} (\Gamma \otimes_k A^k)(s, x; t, y), \quad (3.20)$$

We shall define $\otimes$ as a time-space convolution and $\otimes_k$, for $k = 1, \dots, d_1$, as a suitable stochastic time-space convolution that will be rigorously defined in the following section. Intuitively,

$$(\Gamma \otimes H)(s, x; t, y) = \int_s^t \int_{\mathbb{R}^d} \Gamma(s, x; \tau, \xi) H(\tau, \xi; t, y) d\xi d\tau, \quad (3.21)$$

and

$$(\Gamma \otimes_k A^k)(s, x; t, y) = \int_s^t \int_{\mathbb{R}^d} \Gamma(s, x; \tau, \xi) A^k(\tau, \xi; t, y) d\xi dW_\tau^k. \quad (3.22)$$

Note that an anticipative stochastic integral is involved in the definition of (3.22), since $\{H(\tau, \xi; t, y)\}_{\tau \in [s, t]}$ is not $\mathcal{F}_\tau^W$ -adapted. In order to overcome this difficulty, we may make use of the Skorohod operator δ in the frame of Malliavin calculus. However, this kind of anticipative integral can be well-defined, as we stated in theorem 1.37, in the frame of Wiener chaos. To explain the goal of the method, we introduce some notation that will be defined in the following. We write

$$\Lambda(Z)(s, x; t, y) = (Z \otimes H)(s, x; t, y) + \sum_{k=1}^{d_1} (Z \otimes_k A^k)(s, x; t, y),$$

hence, by iterating N -times the identity (3.20), when N tends to infinity we expect to obtain a representation of Γ as a series

$$\Gamma(s, x; t, y) = Z(s, x; t, y) + \sum_{r=1}^{\infty} \Lambda^r(Z)(s, x; t, y),$$

provided that the series converges in L'_0 compact uniformly in $(s, x; t, y) \in D$, where $\Lambda^r(Z) = \Lambda(\Lambda^{r-1}(Z))$. The following theorem allows us to give an explicit expression for $Z_n(s, x; t, y)$.

Theorem 3.2.2. *For multi-index $\alpha = (\alpha_1, \dots, \alpha_{d_1}) \in \mathbb{N}^{d_1}$, we set $|\alpha| = \sum_{k=1}^{d_1} \alpha_k$ and $\alpha! = \prod_{k=1}^{d_1} \alpha_k!$. According to the notation introduced in 1.5.11, we can decompose $Z(s, x; t, y)$ in L'_0 as a formal Wiener chaos as a sequence of elements in L_0 as follows*

$$Z(s, x; t, y) = \{Z_n(s, x; t, y)\}_{n=0}^{\infty} \quad \text{in } L'_0, \quad (3.23)$$

for all $(s, x; t, y) \in D$, where

$$Z_n(s, x; t, y) = \bar{p}(s, x; t, y) \sum_{|\alpha|=n} \frac{1}{\alpha!} \prod_{k=1}^{d_1} h_{\alpha_k} \left(\left[\sigma_t(y)^{-1} \frac{y-x}{\sqrt{t-s}} \right]^k \right) \prod_{k=1}^{d_1} h_{\alpha_k} \left(\frac{W_t^k - W_s^k}{\sqrt{t-s}} \right), \quad (3.24)$$

for each $n \in \mathbb{N}$, where $\bar{p}(s, x; t, y)$ is the parametrix to the backward Cauchy problem defined in (3.9).

Proof. For notational convenience, we give the proof for case $d_1 = d = 1$. The proof is articulated in two steps. In the first one we prove the algebraic identity of the series and we proceed formally. In the second one, we prove the boundedness of the L^2 -norm of the n -th term, hence by orthogonality the sequence defines an element in L'_0 . We write the Dirac delta distribution in the Fourier representation as

$$\delta_y(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \exp(i\zeta(x-y)) d\zeta, \quad x, y \in \mathbb{R}. \quad (3.25)$$

Moreover, recalling that the generating function of Hermite polynomials is

$$\exp\left(tv - \frac{t^2}{2}\right) = \sum_{n=0}^{\infty} \frac{t^n}{n!} h_n(v),$$

for $t = i\zeta$, we have

$$\exp(i\zeta v) = \exp\left(-\frac{\zeta^2}{2}\right) \sum_{n=0}^{\infty} \frac{i^n \zeta^n}{n!} h_n(v). \quad (3.26)$$

Hence we have

$$\exp(i\zeta(v-z)) = \sum_{n=0}^{\infty} \frac{i^n \zeta^n}{n!} h_n(v) \exp\left(-\frac{\zeta^2}{2} - i\zeta z\right). \quad (3.27)$$

In order to compute the following integral

$$I_n = \int_{\mathbb{R}} \exp\left(-\frac{\zeta^2}{2} - i\zeta z\right) \zeta^n d\zeta = \exp\left(-\frac{z^2}{2}\right) \int_{\mathbb{R}} \exp\left(-\frac{1}{2}(\zeta + iz)^2\right) \zeta^n d\zeta, \quad (3.28)$$

we define the following generating function for ζ^n :

$$I(s) := \int_{\mathbb{R}} \exp\left(-\frac{1}{2}(\zeta + iz)^2\right) \exp(s\zeta) d\zeta, \quad (3.29)$$

since it can be written as a series of powers of s . By the change of variable $\zeta = -iz + \xi$, we have

$$\begin{aligned} I(s) &= \int_{\mathbb{R}} \exp\left(-\frac{1}{2}\xi^2\right) \exp(s(-iz + \xi)) d\xi \\ &= \exp(-isz) \exp\left(\frac{s^2}{2}\right) \int_{\mathbb{R}} \exp\left(-\frac{1}{2}(\xi - s)^2\right) d\xi \\ &= \exp(-isz) \exp\left(\frac{s^2}{2}\right) \sqrt{2\pi} \\ &= \exp\left((-iz)s + \frac{s^2}{2}\right) \sqrt{2\pi}. \end{aligned}$$

It holds

$$I_n = \exp\left(-\frac{z^2}{2}\right) \frac{d^n}{ds^n} I(s) \Big|_{s=0},$$

from the generating function for Hermite polynomials and by the definition, we obtain

$$\begin{aligned}
I_n &= \sqrt{2\pi} \exp\left(-\frac{z^2}{2}\right) \frac{d^n}{ds^n} \exp\left((-iz)s + \frac{s^2}{2}\right) \Big|_{s=0} \\
&= \sqrt{2\pi} \exp\left(-\frac{z^2}{2}\right) \cdot h_n(-iz) \\
&= \sqrt{2\pi} \exp\left(-\frac{z^2}{2}\right) (-1)^n \exp\left(\frac{(-iz)^2}{2}\right) \frac{d^n}{d(-iz)^n} \exp\left(-\frac{(-iz)^2}{2}\right) \\
&= \sqrt{2\pi} \exp\left(-\frac{z^2}{2}\right) (-i)^n (-1)^n \exp\left(\frac{z^2}{2}\right) \frac{d^n}{dz^n} \exp\left(-\frac{z^2}{2}\right) \\
&= \sqrt{2\pi} \exp\left(-\frac{z^2}{2}\right) (-i)^n h_n(z),
\end{aligned}$$

by the complex derivation property. We integrate expression (3.27) and obtain

$$\begin{aligned}
\int_{\mathbb{R}} \exp(i\zeta(v-z)) d\zeta &= \sum_{n=0}^{\infty} \frac{i^n}{n!} h_n(v) I_n \\
&= \sum_{n=0}^{\infty} \frac{i^n}{n!} h_n(v) \sqrt{2\pi} \exp\left(-\frac{z^2}{2}\right) (-i)^n h_n(z) \\
&= \sqrt{2\pi} \exp\left(-\frac{z^2}{2}\right) \sum_{n=0}^{\infty} \frac{1}{n!} h_n(z) h_n(v).
\end{aligned}$$

Let $v = \frac{W_t - W_s}{\sqrt{t-s}}$ and $z = \frac{y-x}{\sigma_t(y)\sqrt{t-s}}$, thus

$$\int_{\mathbb{R}} \exp\left(i\zeta \frac{x + \sigma_t(y)(W_t - W_s) - y}{\sigma_t(y)\sqrt{t-s}}\right) d\zeta = \sqrt{2\pi} \exp\left(-\frac{(y-x)^2}{2\sigma_t^2(y)(t-s)}\right) \sum_{n=0}^{\infty} \frac{1}{n!} h_n\left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}}\right) h_n\left(\frac{W_t - W_s}{\sqrt{t-s}}\right).$$

Finally, replacing ζ by $\frac{\zeta}{\sigma_t(y)\sqrt{t-s}}$ in the integral and dividing by 2π , we get

$$\begin{aligned}
&\frac{1}{2\pi} \int_{\mathbb{R}} \exp(i\zeta(x + \sigma_t(y)(W_t - W_s) - y)) d\zeta = \\
&= \frac{1}{\sqrt{2\pi\sigma_t^2(y)(t-s)}} \exp\left(-\frac{(y-x)^2}{2\sigma_t^2(y)(t-s)}\right) \sum_{n=0}^{\infty} \frac{1}{n!} h_n\left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}}\right) h_n\left(\frac{W_t - W_s}{\sqrt{t-s}}\right)
\end{aligned}$$

The series on the right-hand side does not converges in $L^2(\Omega)$, so we can identify $Z(s, x; t, y) = \delta_y(x + \sigma_t(y)(W_t - W_s))$ in L'_0 with the sequence $\{Z_n(s, x; t, y)\}_{n=0}^{\infty}$ as a formal Wiener chaos, where

$$Z_n(s, x; t, y) = \bar{p}(s, x; t, y) \frac{1}{n!} h_n\left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}}\right) h_n\left(\frac{W_t - W_s}{\sqrt{t-s}}\right), \quad (3.30)$$

and $\bar{p}(s, x; t, y)$ is defined as in (3.9). $Z(s, x; t, y)$ is well defined as an element of L'_0 since each $h_n\left(\frac{W_t - W_s}{\sqrt{t-s}}\right)$ are orthogonal, thus each $Z_n(s, x; t, y)$ belongs to the Wiener chaos $\mathcal{H}_n^{s,t}$. Moreover,

$Z_n(s, x; t, y) \in L^2(\Omega, \mathcal{F}_{s,t}^W)$, in fact from Theorem 1.1.12 we have

$$\begin{aligned} h_n \left(\frac{W_t - W_s}{\sqrt{t-s}} \right) &= h_n \left(\frac{I(\mathbf{1}_{[s,t]})}{\|\mathbf{1}_{[s,t]}\|_{L^2}} \right) \\ &= \frac{1}{\|\mathbf{1}_{[s,t]}\|_{L^2}^n} I_n \left(\mathbf{1}_{[s,t]}^{\otimes n} \right). \end{aligned}$$

Hence, from the isometry (1.9),

$$\begin{aligned} \left\| h_n \left(\frac{W_t - W_s}{\sqrt{t-s}} \right) \right\|_{L^2} &= \frac{1}{(t-s)^{n/2}} \|I_n(\mathbf{1}_{[s,t]}^{\otimes n})\|_{L^2} \\ &= \frac{1}{(t-s)^{n/2}} \sqrt{n!} \|\mathbf{1}_{[s,t]}^{\otimes n}\|_{L^2([0,T]^n)} \\ &= \frac{1}{(t-s)^{n/2}} \sqrt{n!} (t-s)^{n/2} \\ &= \sqrt{n!}. \end{aligned}$$

Moreover, from standard estimates for Hermite polynomials, see [11], we have

$$\sup_{x \in \mathbb{R}} \left| \exp \left(-\frac{x^2}{4} \right) \frac{1}{\sqrt{n!}} h_n(x) \right| \leq C n^{-\frac{1}{12}}. \quad (3.31)$$

Then

$$\begin{aligned} \|Z_n(s, x; t, y)\|_{L^2} &= \bar{p}(s, x; t, y) \frac{1}{n!} \cdot \left| h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \right| \cdot \left\| h_n \left(\frac{W_t - W_s}{\sqrt{t-s}} \right) \right\|_{L^2(\Omega)} \\ &\leq \frac{C}{\sigma_t(y)(t-s)^{\frac{1}{2}} n!} \exp \left(-\frac{(y-x)^2}{4\sigma_t^2(y)(t-s)} \right) \cdot \sup_{x, y \in \mathbb{R}} \left| \exp \left(-\frac{(y-x)^2}{4\sigma_t^2(y)(t-s)} \right) h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \right| \sqrt{n!} \\ &\leq \frac{n^{-\frac{1}{12}}}{\sigma_t(y)} \frac{C}{(t-s)^{\frac{1}{2}}} \exp \left(-\frac{(y-x)^2}{4\sigma_t^2(y)(t-s)} \right) \\ &\leq \frac{n^{-\frac{1}{12}} \cdot C}{\sqrt{\inf_{(t,y)} \sigma_t^2(y)(t-s)}} \exp \left(-\frac{1}{4} \inf_{(t,y)} \sigma_t^{-2}(y) \frac{(y-x)^2}{t-s} \right). \end{aligned}$$

Since the coefficients $\sigma_t(y)$ are bounded by Assumption 1bis, $\|Z_n(s, x; t, y)\|_{L^2}$ is uniformly bounded in $(s, x; t, y) \in D$, for each $n \in \mathbb{N}$. $\square$

As a direct consequence of theorem 3.2.2, we have the smoothness of the parametrix in the (s, x) -variables when restricted on the Wiener chaos $\mathcal{H}_n$.

Corollary 3.2.3. *Let $Z(s, x; t, y) = \{Z_n(s, x; t, y)\}_{n=0}^\infty$ be the formal Wiener chaos of the parametrix. Then for any $t \in (0, T]$, $y \in \mathbb{R}^d$, we have $Z_n(\cdot, \cdot; t, y) \in \text{bC}_{0,t}^2(\mathbb{R}^d)$.*

Hereafter $C = C(\dots)$ denotes a constant depending only on quantities appearing in parenthesis. In a given context the same letter will be used to denote different constants depending on the same set of arguments.

Lemma 3.2.4 (Gaussian estimate). *Let Γ^λ the Gaussian kernel in (3.7). For all $p > 0$ and $\lambda_1 > \lambda_0$ there exists a constant $C = C(p, \lambda_1, \lambda_0, d)$ such that*

$$\left(\frac{|x|}{\sqrt{t}}\right)^p \Gamma^{\lambda_0}(t, x) \leq C \cdot \Gamma^{\lambda_1}(t, x), \quad t > 0, x \in \mathbb{R}^d \quad (3.32)$$

Proof. Let $z = \frac{|x|}{\sqrt{t}}$, we have

$$z^p \Gamma^{\lambda_0}(t, x) = \frac{z^p}{(2\pi\lambda_0 t)^{\frac{d}{2}}} \exp\left(-\frac{z^p}{2\lambda_0}\right) = \left(\frac{\lambda_1}{\lambda_0}\right)^d g(z) \Gamma^{\lambda_1}(t, x),$$

where

$$g(z) := z^p \exp\left(-\frac{1}{2}\left(\frac{1}{\lambda_0} - \frac{1}{\lambda_1}\right)z^2\right), \quad z > 0$$

g assumes its maximum at $z_0 = \sqrt{\frac{p}{\frac{1}{\lambda_0} - \frac{1}{\lambda_1}}}$, where it is equal to $g(z_0) = \left(\frac{p}{\frac{1}{\lambda_0} - \frac{1}{\lambda_1}}\right)^{\frac{p}{2}} \exp\left(-\frac{p}{2}\right)$. $\square$

Lemma 3.2.5. *Let $Z(s, x; t, y) = \{Z_n(s, x; t, y)\}_{n=0}^\infty$ be the formal Wiener chaos of the parametrix. Let λ_0 be the constant that appears in Assumption 2. Then there exists $\lambda > \lambda_0$ and a positive constant $C = C(T, \lambda_0, d, d_1, n)$ such that*

$$\|Z_n(s, x; t, y)\|_{L^2} \leq C \cdot \Gamma^\lambda(t - s, y - x), \quad (3.33)$$

$$\|\partial_i Z_n(s, x; t, y)\|_{L^2} \leq \frac{C}{\sqrt{t-s}} \Gamma^\lambda(t - s, y - x), \quad (3.34)$$

$$\|\partial_{ij} Z_n(s, x; t, y)\|_{L^2} \leq \frac{C}{t-s} \Gamma^\lambda(t - s, y - x), \quad (3.35)$$

for all $(s, x; t, y) \in D$, $k = 1, \dots, d_1$, where the derivatives act in the x -variable.

Proof. By the independence of the components of W and the isometry between norms (1.9), we have

$$\|Z_n(s, x; t, y)\|_{L^2} \leq \bar{p}(s, x; t, y) \sum_{|\alpha|=n} \frac{1}{\alpha!} \prod_{k=1}^{d_1} \left| h_{\alpha_k} \left(\left[\sigma_t(y)^{-1} \frac{y-x}{\sqrt{t-s}} \right]^k \right) \right| \cdot d_1 \sqrt{n!}$$

From standard estimates for Hermite polynomials, see [11], we have

$$\sup_{x \in \mathbb{R}} \left| \exp\left(-\frac{x^2}{4}\right) \frac{1}{\sqrt{n!}} h_n(x) \right| \leq C' n^{-\frac{1}{12}}. \quad (3.36)$$

so for any α_k we have

$$\sup_{x \in \mathbb{R}^d} \left| \exp\left(-\frac{1}{4} \left[\sigma_t(y)^{-1} \frac{y-x}{\sqrt{t-s}} \right]^k \cdot \left[\sigma_t(y)^{-1} \frac{y-x}{\sqrt{t-s}} \right]^k \right) h_{\alpha_k} \left(\left[\sigma_t(y)^{-1} \frac{y-x}{\sqrt{t-s}} \right]^k \right) \right| \leq C''$$

Then

$$\begin{aligned} \|Z_n(s, x; t, y)\|_{L^2} &\leq \Gamma^{\text{heat}}(2(t-s)(\sigma\sigma^*)_t(y), y-x) \sum_{|\alpha|=n} \frac{1}{\alpha!} (C'')^{d_1} \\ &\leq C \cdot \Gamma^{\text{heat}}(2(t-s)(\sigma\sigma^*)_t(y), y-x). \end{aligned}$$

By the uniformly parabolicity in assumption 2 we have

$$\frac{1}{\lambda_0} |y-x|^2 \leq \langle (\sigma\sigma^*)_t(y)(y-x), y-x \rangle \leq \lambda_0 |y-x|^2, \quad x \in \mathbb{R}^d \quad (3.37)$$

and as a consequence it holds

$$\frac{|y-x|^2}{\lambda_0} \leq \langle (\sigma\sigma^*)_t^{-1}(y)(y-x), y-x \rangle \leq \lambda_0 |y-x|^2, \quad x \in \mathbb{R}^d \quad (3.38)$$

and

$$\left(\frac{t-s}{\lambda_0}\right)^d \leq \det(t-s)(\sigma\sigma^*)_t(y) \leq (\lambda_0(t-s))^d.$$

For a matrix A , let $\|A\| := \max_{|\xi|=1} \langle A\xi, \xi \rangle$ be the spectral norm of A . Hence we can rewrite inequalities (3.37), (3.38) as

$$\frac{1}{\lambda_0} \leq \|(\sigma\sigma^*)_t(y)\| \leq \lambda_0, \quad \frac{1}{\lambda_0} \leq \|(\sigma\sigma^*)_t^{-1}(y)\| \leq \lambda_0.$$

We have

$$\begin{aligned} \|Z_n(s, x; t, y)\|_{L^2} &\leq C \cdot \Gamma^{\text{heat}}(2(t-s)\lambda_0 I_{d_1}, y-x) \\ &= C \cdot \Gamma^\lambda(t-s, y-x). \end{aligned}$$

About (3.34), since Hermite polynomials enjoy the property $h'_{\alpha_k}(\xi) = \alpha_k h_{\alpha_k-1}(\xi)$, we have

$$\begin{aligned} \|\partial_i Z_n(s, x; t, y)\|_{L^2} &\leq C \frac{\|(\sigma\sigma^*)_t^{-1}(y)(y-x)\|}{t-s} \bar{p}(s, x; t, y) \sum_{|\alpha|=n} \frac{1}{\alpha!} \prod_{k=1}^{d_1} \left| h_{\alpha_k}(\xi_k) \right| + \\ &\quad + C' \bar{p}(s, x; t, y) \sum_{|\alpha|=n} \frac{1}{\alpha!} \frac{1}{\sqrt{t-s}} \sum_{k=1}^{d_1} (\sigma_t(y)^{-1})_{ki} \alpha_k |h_{\alpha_k-1}(\xi_k)| \prod_{\substack{j=1 \\ j \neq k}}^{d_1} |h_{\alpha_j}(\xi_j)|, \end{aligned}$$

where $\xi_k = \left[\sigma_t(y)^{-1} \frac{y-x}{\sqrt{t-s}} \right]^k$ and we increased the first term with the norm of the gradient. By

using the same estimates in (3.33) we have

$$\|Z_n(s, x; t, y)\|_{L^2} \leq C \frac{\|(\sigma\sigma^*)_t^{-1}(y)\|}{\sqrt{t-s}} \left(\frac{|y-x|}{\sqrt{t-s}} \Gamma^\lambda(t-s, y-x) \right) + C' \frac{\lambda_0}{\sqrt{t-s}} \Gamma^\lambda(t-s, y-x) \leq$$

(by the Gaussian estimate (3.32))

$$\begin{aligned} &\leq C \frac{\lambda_0}{\sqrt{t-s}} \Gamma^\lambda(t-s, y-x) + \frac{C'}{\sqrt{t-s}} \Gamma^\lambda(t-s, y-x) \\ &\leq \frac{C}{\sqrt{t-s}} \Gamma^\lambda(t-s, y-x). \end{aligned}$$

Inequality (3.35) is obtained analogously. $\square$

Lemma 3.2.6. *Let $Z(s, x; t, y) = \{Z_n(s, x; t, y)\}_{n=0}^\infty$ be the formal Wiener chaos of the parametrix and let H, A^k be defined as in (3.18), (3.19). Let λ_0 be the constant that appears in Assumption 2. Then there exists $\lambda > \lambda_0$ and a positive constant $C = C(T, \lambda_0, d, d_1, n)$ such that*

$$\|A_n^k(s, x; t, y)\|_{L^2} \leq \frac{C}{(t-s)^{\frac{1}{2}-\frac{\beta}{2}}} \Gamma^\lambda(t-s, y-x), \quad (3.39)$$

$$\|H_n(s, x; t, y)\|_{L^2} \leq \frac{C}{(t-s)^{1-\frac{\beta}{2}}} \Gamma^\lambda(t-s, y-x), \quad (3.40)$$

for all $(s, x; t, y) \in D, k = 1, \dots, d_1$.

Proof. By Assumption 1bis we have

$$|\sigma_s^{ik}(x) - \sigma_t^{ik}(y)| \leq C \cdot |x-y|^\beta, \quad i = 1, \dots, d, \text{ and } k = 1, \dots, d_1,$$

for any $(s, x; t, y) \in D$. Then by lemma 3.2.5 we have

$$\begin{aligned} \|A_n^k(s, x; t, y)\|_{L^2} &= \left\| \left(G_s^k - \overline{G}^{k,t,y} \right) Z_n(s, x; t, y) \right\|_{L^2} \\ &\leq \sum_{i=1}^d \left\| \left(\sigma_s^{ik}(x) - \sigma_t^{ik}(y) \right) \partial_i Z_n(s, x; t, y) \right\|_{L^2} \\ &\leq \sum_{i=1}^d C \frac{|x-y|^\beta}{\sqrt{t-s}} \Gamma^\lambda(t-s, y-x) \\ &\leq C \frac{1}{(t-s)^{\frac{1}{2}-\frac{\beta}{2}}} \left(\frac{|x-y|}{\sqrt{t-s}} \right)^\beta \Gamma^\lambda(t-s, y-x) \leq \end{aligned}$$

(by the Gaussian estimate (3.32), up to rename λ)

$$\leq \frac{C}{(t-s)^{\frac{1}{2}-\frac{\beta}{2}}} \Gamma^\lambda(t-s, y-x)$$

By Assumption 1bis we have that also the coefficients of $(\sigma\sigma^*)_s(x)$ are β -uniformly Hölder continuous in x , i.e. there exists $C > 0$ such that

$$|(\sigma\sigma^*)_s^{ij}(x) - (\sigma\sigma^*)_t^{ij}(y)| \leq C \cdot |x - y|^\beta, \quad i, j = 1, \dots, d,$$

for any $(s, x; t, y) \in D$. Then by lemma 3.2.5 and the boundedness of b, σ , we have

$$\begin{aligned} \|H_n(s, x; t, y)\|_{L^2} &= \left\| \left(L_s - \bar{L}^{t, y} \right) Z_n(s, x; t, y) \right\|_{L^2} \\ &\leq \frac{1}{2} \sum_{i, j=1}^d \left\| \left((\sigma\sigma^*)_s^{ij}(x) - (\sigma\sigma^*)_t^{ij}(y) \right) \partial_{ij} Z_n(s, x; t, y) \right\|_{L^2} + \sum_{i=1}^d \|b_s^i(x) \partial_i Z_n(s, x; t, y)\|_{L^2} \\ &\leq C \frac{|x - y|^\beta}{t - s} \Gamma^\lambda(t - s, y - x) + \frac{C'}{\sqrt{t - s}} \Gamma^\lambda(t - s, y - x) \\ &\leq C \frac{1}{(t - s)^{1 - \frac{\beta}{2}}} \left(\frac{|x - y|}{\sqrt{t - s}} \right)^\beta \Gamma^\lambda(t - s, y - x) + \frac{C'}{\sqrt{t - s}} \Gamma^\lambda(t - s, y - x) \leq \end{aligned}$$

(by the Gaussian estimate (3.32), up to rename λ)

$$\begin{aligned} &\leq \frac{C}{(t - s)^{1 - \frac{\beta}{2}}} \Gamma^\lambda(t - s, y - x) + \frac{C'}{\sqrt{t - s}} \Gamma^\lambda(t - s, y - x) \\ &\leq \frac{C}{(t - s)^{1 - \frac{\beta}{2}}} \Gamma^\lambda(t - s, y - x). \end{aligned}$$

□

3.3 Convolution operators

In order to obtain the Duhamel representation for the fundamental solution $\Gamma = \Gamma(s, x; t, y)$ to (2.1), we need to define two convolution operators $\otimes$ and $\otimes_k$, $k = 1, \dots, d_1$. We postpone their rigorous definition in Definition 3.3.1. Intuitively, their meaning is what we have written in (3.21) and (3.22). In order to look for a suitable definition, we proceed formally, assuming for the moment that $Z, H \in L^2(\Omega)$. We shall define $(Z \otimes H)$ in (3.21) as a time-space convolution on the kernels z, h of Z, H . Let

$$\begin{aligned} Z(s, x; \tau, \xi) &= \sum_{l=0}^{\infty} I_l(z_l(s, x; \tau, \xi)), \\ H(\tau, \xi; t, y) &= \sum_{m=0}^{\infty} I_m(h_m(\tau, \xi; t, y)), \end{aligned}$$

be the Wiener-Itô chaos expansion of Z, H respectively, as given by Theorem 1.1.14. Unless otherwise specified, it is understood that $z_l(s, x; \tau, \xi) = z_l(s, x; \tau, \xi)^{(i_1, \dots, i_l)}[t_1, \dots, t_l]$ and

$h_m(\tau, \xi; t, y) = h_m(\tau, \xi; t, y)^{(i_1, \dots, i_m)}[t_1, \dots, t_m]$. Then

$$\begin{aligned} (Z \otimes H)(s, x; t, y) &= \int_s^t \int_{\mathbb{R}^d} Z(s, x; \tau, \xi) H(\tau, \xi; t, y) d\xi d\tau \\ &= \int_s^t \int_{\mathbb{R}^d} \sum_{l=0}^{\infty} I_l(z_l(s, x; \tau, \xi)) \cdot \sum_{m=0}^{\infty} I_m(h_m(\tau, \xi; t, y)) d\xi d\tau \\ &= \int_s^t \int_{\mathbb{R}^d} \sum_{n=0}^{\infty} \sum_{l+m=n} I_l(z_l(s, x; \tau, \xi)) I_m(h_m(\tau, \xi; t, y)) d\xi d\tau = \end{aligned}$$

(by proposition 1.1.11)

$$= \int_s^t \int_{\mathbb{R}^d} \sum_{n=0}^{\infty} \sum_{l+m=n} \sum_{r=0}^{\min\{l, m\}} r! \binom{l}{r} \binom{m}{r} I_{n-2r}(z_l(s, x; \tau, \xi) \boxtimes_r h_m(\tau, \xi; t, y)) d\xi d\tau.$$

Note that for $r = 0$ the expression simplifies as

$$\sum_{n=0}^{\infty} I_n \left(\sum_{l+m=n} \int_s^t \int_{\mathbb{R}^d} z_l(s, x; \tau, \xi) \boxtimes h_m(\tau, \xi; t, y) d\xi d\tau \right).$$

Notice that, since $\{Z(s, x; \tau, \xi)\}_{\tau \in [s, t]}$ is a $(\mathcal{F}_\tau^W)$ -adapted process, by lemma 1.4.5 it holds $z_l(s, x; \tau, \xi)^{(i_1, \dots, i_l)}[t_1, \dots, t_l] = 0$ if there exists $t_j > \tau$, for $j = 1, \dots, l$. While $\{H(\tau, \xi; t, y)\}_{\tau \in [s, t]}$ is a $(\mathcal{F}_t^{W, t-\tau})$ -adapted process, so by lemma 1.4.5 it holds $h_m(\tau, \xi; t, y)^{(i_1, \dots, i_m)}[t_1, \dots, t_m] = 0$ if there exists $t_j < \tau$, for $j = l, l+1, \dots, n$. Then we have

$$\sum_{n=0}^{\infty} I_n \left(\sum_{l+m=n} \int_{t_l}^{t_{l+1}} \int_{\mathbb{R}^d} z_l(s, x; \tau, \xi) \boxtimes h_m(\tau, \xi; t, y) d\xi d\tau \right). \quad (3.41)$$

In the operator $\otimes$ an anticipative stochastic integral is involved in the convolution $(Z \otimes_k A^k)$, so we argue as follows. We proceed formally, assuming for the moment that $Z, A^k \in L^2(\Omega)$ and generalize the Itô integral in (3.22) to non adapted processes by means of the Skorohod integral:

$$(Z \otimes_k A^k)(s, x; t, y) = \int_s^t \int_{\mathbb{R}^d} Z(s, x; \tau, \xi) A^k(\tau, \xi; t, y) d\xi \delta W_\tau^k. \quad (3.42)$$

Let $Z(s, x; \tau, \xi) = \sum_{l=0}^{\infty} I_l(z_l(s, x; \tau, \xi))$, $A^k(\tau, \xi; t, y) = \sum_{m=0}^{\infty} I_m(a_m^k(\tau, \xi; t, y))$, be the Wiener-Itô chaos expansion of Z, A^k respectively, as given by theorem 1.1.14. Then

$$\begin{aligned} Z(s, x; \tau, \xi) A^k(\tau, \xi; t, y) &= \sum_{l=0}^{\infty} I_l(z_l(s, x; \tau, \xi)) \cdot \sum_{m=0}^{\infty} I_m(a_m^k(\tau, \xi; t, y)) \\ &= \sum_{n=0}^{\infty} \sum_{l+m=n} I_l(z_l(s, x; \tau, \xi)) I_m(a_m^k(\tau, \xi; t, y)) = \end{aligned}$$

(by proposition 1.1.11)

$$= \sum_{n=0}^{\infty} \sum_{l+m=n} \sum_{r=0}^{\min\{l,m\}} r! \binom{l}{r} \binom{m}{r} I_{n-2r} (z_l(s, x; \tau, \xi) \boxtimes_r a_m^k(\tau, \xi; t, y)).$$

By linearity of the Skorohod integral and by formula (1.37) for the computation of the Wiener-Itô chaos expansion, we have that (3.42) becomes

$$\begin{aligned} (Z \otimes_k A^k)(s, x; t, y) &= \sum_{n=0}^{\infty} \sum_{l+m=n} \sum_{r=0}^{\min\{l,m\}} r! \binom{l}{r} \binom{m}{r} \delta \left(I_{n-2r} \left(\int_{\mathbb{R}^d} z_l(s, x; \tau, \xi) \boxtimes_r a_m^k(\tau, \xi; t, y) d\xi \right) \right) \\ &= \sum_{n=0}^{\infty} \sum_{l+m=n} \sum_{r=0}^{\min\{l,m\}} r! \binom{l}{r} \binom{m}{r} I_{n-2r+1} \left(\int_{\mathbb{R}^d} z_l(s, x; \cdot, \xi) \tilde{\boxtimes}_r a_m^k(\cdot, \xi; t, y) d\xi \right). \end{aligned}$$

Note that for $r = 0$, the expression simplifies as

$$\sum_{n=0}^{\infty} \sum_{l+m=n} I_{n+1} \left(\int_{\mathbb{R}^d} z_l(s, x; \cdot, \xi) \tilde{\boxtimes} a_m^k(\cdot, \xi; t, y) d\xi \right) = \sum_{n=0}^{\infty} I_{n+1} \left(\sum_{l+m=n} \int_{\mathbb{R}^d} z_l(s, x; \cdot, \xi) \tilde{\boxtimes} a_m^k(\cdot, \xi; t, y) d\xi \right) =$$

(by relabelling n index)

$$= \sum_{n=1}^{\infty} I_n \left(\sum_{l+m=n-1} \int_{\mathbb{R}^d} z_l(s, x; \cdot, \xi) \tilde{\boxtimes} a_m^k(\cdot, \xi; t, y) d\xi \right). \quad (3.43)$$

Then formulas (3.41), (3.43) suggest defining operators $\otimes$, $\otimes_k$ rigorously as follows.

Definition 3.3.1. Let $Z(s, x; t, y) = \{Z_n(s, x; t, y)\}_{n=0}^{\infty}$, $H(s, x; t, y) = \{H_n(s, x; t, y)\}_{n=0}^{\infty}$, $A^k(s, x; t, y) = \{A_n^k(s, x; t, y)\}_{n=0}^{\infty}$, be the formal Wiener chaos of Z , H and A^k , respectively, as defined as in (3.18), (3.19). Let

$$(Z \otimes H)(s, x; t, y) = \{(Z \otimes H)_n(s, x; t, y)\}_{n=0}^{\infty} \quad \text{in } L'_0,$$

according to corollary 1.2.1, be given by

$$\begin{aligned} (Z \otimes H)_n(s, x; t, y) &= I_n((z \otimes h)_n(s, x; t, y)) \\ &= \sum_{(i_1, \dots, i_n)} \int_{[s, t]^n} (z \otimes h)_n^{(i_1, \dots, i_n)}(s, x; t, y) [t_1, \dots, t_n] dW_{t_1}^{i_1} \dots dW_{t_n}^{i_n}. \end{aligned}$$

We define the kernels, when and only when the right-hand sides are integrable, as

$$(z \otimes h)_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] := \sum_{l+m=n} \int_{t_l}^{t_{l+1}} \int_{\mathbb{R}^d} z_l^{(i_1, \dots, i_l)}(s, x; \tau, \xi)[t_1, \dots, t_l] \times \\ \times h_m^{(i_{l+1}, \dots, i_n)}(\tau, \xi; t, y)[t_{l+1}, \dots, t_n] d\xi d\tau. \quad (3.44)$$

with the convention $t_0 = s, t_{n+1} = t$. Moreover, let

$$(Z \otimes_k A^k)(s, x; t, y) = \{(Z \otimes_k A^k)_n(s, x; t, y)\}_{n=0}^\infty \quad \text{in } L'_0, \quad \text{for all } k = 1, \dots, d_1,$$

where $(Z \otimes_k A^k)_n(s, x; t, y) \in L^2(\Omega)$ is given by

$$(Z \otimes_k A^k)_n(s, x; t, y) = I_n((z \otimes_k a^k)_n(s, x; t, y)) \\ = \sum_{(i_1, \dots, i_n)} \int_{[s, t]^n} (z \otimes_k a^k)_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] dW_{t_1}^{i_1} \dots dW_{t_n}^{i_n}.$$

We define the kernels, when and only when the right-hand sides are integrable, as

$$(z \otimes_k a^k)_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] := \begin{cases} 0 & \text{if } n = 0 \\ \sum_{l+m=n-1} \delta_{i_{l+1}, k} \int_{\mathbb{R}^d} z_l^{(i_1, \dots, i_l)}(s, x; t_{l+1}, \xi)[t_1, \dots, t_l] \times \\ \times a_m^{k, (i_{l+2}, \dots, i_n)}(t_{l+1}, \xi; t, y)[t_{l+2}, \dots, t_n] d\xi. & \text{if } n \in \mathbb{N}_{\geq 1} \end{cases} \quad (3.45)$$

Lemma 3.3.2. Given $0 \leq s < t \leq T$, $F = F(s, x; t, y), G = G(s, x; t, y) \in L^2(\Omega)^{s, t}$ r.v. which depend on $x, y \in \mathbb{R}^d$ and are $\mathcal{F}_s^{W, t}$ -measurable. Let $f_n = f_n(s, x; t, y), g_n = g_n(s, x; t, y)$ be the sequences of functions $f_n, g_n \in \tilde{L}^2([s, t]^n)$ defined in the corollary 1.2.1 such that

$$F = \sum_{n=0}^\infty I_n(f_n), \quad G = \sum_{n=0}^\infty I_n(g_n), \quad \text{in } L^2(\Omega), \quad (3.46)$$

and

$$I_n(f_n) = \sum_{(i_1, \dots, i_n)} \int_{[s, t]^n} f_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] dW_{t_1}^{i_1} \dots dW_{t_n}^{i_n}, \\ I_n(g_n) = \sum_{(i_1, \dots, i_n)} \int_{[s, t]^n} g_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] dW_{t_1}^{i_1} \dots dW_{t_n}^{i_n}.$$

Define, when and only when the right-hand sides are integrable,

$$h^{(i_1, \dots, i_{m+n})}(s, x; t, y)[t_1, \dots, t_{m+n}] := \quad (3.47)$$

$$\int_{t_m}^{t_{m+1}} \int_{\mathbb{R}^d} f_m^{(i_1, \dots, i_m)}(s, x; \tau, z)[t_1, \dots, t_m] g_n^{(i_{m+1}, \dots, i_{m+n})}(\tau, z; t, y)[t_{m+1}, \dots, t_{m+n}] dz d\tau, \quad (3.48)$$

and

$$k^{(i_1, \dots, i_{m+n+1})}(s, x; t, y)[t_1, \dots, t_{m+n+1}] := \quad (3.49)$$

$$\int_{\mathbb{R}^d} f_m^{(i_1, \dots, i_m)}(s, x; t_{m+1}, z)[t_1, \dots, t_m] g_n^{(i_{m+2}, \dots, i_{m+n+1})}(t_{m+1}, z; t, y)[t_{m+2}, \dots, t_{m+n+1}] dz, \quad (3.50)$$

Then

$$\|h^{(i_1, \dots, i_{m+n})}(s, x; t, y)\|_{L^2([s, t]^{m+n})} \leq \quad (3.51)$$

$$\int_s^t \int_{\mathbb{R}^d} \|f_m^{(i_1, \dots, i_m)}(s, x; \tau, z)\|_{L^2([s, \tau]^m)} \|g_n^{(i_{m+1}, \dots, i_{m+n})}(\tau, z; t, y)\|_{L^2([\tau, t]^n)} dz d\tau. \quad (3.52)$$

and

$$\|k^{(i_1, \dots, i_{m+n+1})}(s, x; t, y)\|_{L^2([s, t]^{m+n+1})} \leq \quad (3.53)$$

$$\left[\int_s^t \left(\int_{\mathbb{R}^d} \|f_m^{(i_1, \dots, i_m)}(s, x; \tau, z)\|_{L^2([s, \tau]^m)} \|g_n^{(i_{m+1}, \dots, i_{m+n})}(\tau, z; t, y)\|_{L^2([\tau, t]^n)} dz \right)^2 d\tau \right]^{\frac{1}{2}}. \quad (3.54)$$

Proof. In the following, for notational convenience, we omit the dependence on $(i_1, \dots, i_n)$. By the Fubini's theorem we have

$$\begin{aligned} & \|h(s, x; t, y)\|_{L^2([s, t]^{m+n})}^2 = \\ & \int_{[s, t]^{m+n}} h(s, x; t, y)[t_1, \dots, t_{m+n}]^2 dt_1 \dots dt_{m+n} (m+n)! \int_{S_{m+n}^{s, t}} h(s, x; t, y)[t_1, \dots, t_{m+n}]^2 dt_1 \dots dt_{m+n} \\ & = \int_{[s, t]^{m+n}} dt_1 \dots dt_{m+n} \int_{t_m}^{t_{m+1}} d\tau_1 \int_{t_m}^{t_{m+1}} d\tau_2 \int_{\mathbb{R}^d} dz_1 \int_{\mathbb{R}^d} dz_2 f_m(s, x; \tau_1, z_1)[t_1, \dots, t_m] \times \\ & \quad \times f_m(s, x; \tau_2, z_2)[t_1, \dots, t_m] g_n(\tau_1, z_1; t, y)[t_{m+1}, \dots, t_{m+n}] g_n(\tau_2, z_2; t, y)[t_{m+1}, \dots, t_{m+n}] \leq \end{aligned}$$

(by the Schwarz inequality with respect to variables (τ_1, z_1) and (τ_2, z_2))

$$\begin{aligned}
&\leq \int_{[s,t]^{m+n}} dt_1 \dots dt_{m+n} \left(\int_{\mathbb{R}^d} dz_1 \int_{t_m}^{t_{m+1}} d\tau_1 f_m(s, x; \tau_1, z_1)^2 g_n(\tau_1, z_1; t, y)^2 \right)^{\frac{1}{2}} \times \\
&\quad \times \left(\int_{\mathbb{R}^d} dz_2 \int_{t_m}^{t_{m+1}} d\tau_2 f_m(s, x; \tau_2, z_2)^2 g_n(\tau_2, z_2; t, y)^2 \right)^{\frac{1}{2}} \\
&\leq \int_{[s,t]^{m+n}} dt_1 \dots dt_{m+n} \int_{\mathbb{R}^d} dz \int_s^t d\tau f_m(s, x; \tau, z) [t_1, \dots, t_m]^2 g_n(\tau, z; t, y) [t_{m+1}, \dots, t_{m+n}]^2 \\
&= \int_s^t d\tau \int_{\mathbb{R}^d} dz \int_{[s,\tau]^m} dt_1 \dots dt_m \int_{[\tau,t]^n} dt_{m+1} \dots dt_{m+n} f_m(s, x; \tau, z) [t_1, \dots, t_m]^2 \times \\
&\quad \times g_n(\tau, z; t, y) [t_{m+1}, \dots, t_{m+n}]^2 \\
&= \int_s^t d\tau \int_{\mathbb{R}^d} dz \|f_m(s, x; \tau, z)\|_{L^2([s,\tau]^m)}^2 \cdot \|g_n(\tau, z; t, y)\|_{L^2([\tau,t]^n)}^2
\end{aligned}$$

(by the Schwarz inequality)

$$\leq \left(\int_s^t d\tau \int_{\mathbb{R}^d} dz \|f_m(s, x; \tau, z)\|_{L^2([s,\tau]^m)} \cdot \|g_n(\tau, z; t, y)\|_{L^2([\tau,t]^n)} \right)^2,$$

and inequality (3.51) is obtained.

$$\begin{aligned}
&\|k(s, x; t, y)\|_{L^2([s,t]^{m+n+1})}^2 = \\
&\int_{[s,t]^{m+n+1}} k(s, x; t, y) [t_1, \dots, t_{m+n+1}]^2 dt_1 \dots dt_{m+n+1} \\
&= \int_{[s,t]^{m+n+1}} dt_1 \dots dt_{m+n+1} \int_{\mathbb{R}^d} dz_1 \int_{\mathbb{R}^d} dz_2 f_m(s, x; t_{m+1}, z_1) [t_1, \dots, t_m] \times \\
&\quad \times f_m(s, x; t_{m+1}, z_2) [t_1, \dots, t_m] \cdot g_n(t_{m+1}, z_1; t, y) [t_{m+2}, \dots, t_{m+n+1}] g_n(t_{m+1}, z_2; t, y) [t_{m+2}, \dots, t_{m+n+1}] \\
&= \int_s^t dt_{m+1} \int_{\mathbb{R}^d} dz_1 \int_{\mathbb{R}^d} dz_2 \int_{[s,t_{m+1}]^m} dt_1 \dots dt_m \int_{[t_{m+1},t]^n} dt_{m+2} \dots dt_{m+n+1} f_m(s, x; t_{m+1}, z_1) [t_1, \dots, t_m] \times \\
&\quad \times f_m(s, x; t_{m+1}, z_2) [t_1, \dots, t_m] g_n(t_{m+1}, z_1; t, y) [t_{m+2}, \dots, t_{m+n+1}] g_n(t_{m+1}, z_2; t, y) [t_{m+2}, \dots, t_{m+n+1}]
\end{aligned}$$

(by the Schwarz inequality)

$$\begin{aligned}
&\leq \int_s^t dt_{m+1} \int_{\mathbb{R}^d} dz_1 \int_{\mathbb{R}^d} dz_2 \|f_m(s, x; t_{m+1}, z_1)\|_{L^2([s,t_{m+1}]^m)} \|f_m(s, x; t_{m+1}, z_2)\|_{L^2([s,t_{m+1}]^m)} \times \\
&\quad \times \|g_n(t_{m+1}, z_1; t, y)\|_{L^2([t_{m+1},t]^n)} \|g_n(t_{m+1}, z_2; t, y)\|_{L^2([t_{m+1},t]^n)} \\
&= \int_s^t dt_{m+1} \left(\int_{\mathbb{R}^d} dz \|f_m(s, x; t_{m+1}, z)\|_{L^2([s,t_{m+1}]^m)} \|g_n(t_{m+1}, z; t, y)\|_{L^2([t_{m+1},t]^n)} \right)^2,
\end{aligned}$$

and inequality (3.53) is obtained. $\square$

Proposition 3.3.3. *Let $H(s, x; t, y) = \{H_n(s, x; t, y)\}_{n=0}^\infty$, $A^k(s, x; t, y) = \{A_n^k(s, x; t, y)\}_{n=0}^\infty$,*

be the formal Wiener chaos of H and A^k , respectively. Let $F(s, x; t, y) = \{F_n(s, x; t, y)\}_{n=0}^\infty$ be the formal Wiener chaos of an element such that there exist a positive constant $C_F = C_F(F, T, \lambda_0, d, d_1, n)$ and $\lambda' > 0$, such that

$$\|F_n(s, x; t, y)\|_{L^2} \leq \frac{C_F}{(t-s)^{\frac{1}{2}-\alpha}} \Gamma^{\lambda'}(t-s, y-x). \quad (3.55)$$

Then, for each $n \in \mathbb{N}$ there exist a positive constant $C = C(T, \lambda_0, d, d_1, n)$ and $\lambda > 0$, such that

$$\|(F \otimes H)_n(s, x; t, y)\|_{L^2} \leq \frac{C_F \cdot C}{(t-s)^{-\alpha-\frac{\beta-1}{2}}} \frac{\Gamma_E(\alpha + \frac{1}{2}) \Gamma_E(\frac{\beta}{2})}{\Gamma_E(\alpha + \frac{\beta+1}{2})} \Gamma^\lambda(t-s, y-x), \quad (3.56)$$

$$\|(F \otimes_k A^k)_n(s, x; t, y)\|_{L^2} \leq \frac{C_F \cdot C}{(t-s)^{-\alpha-\frac{\beta-1}{2}}} \left(\frac{\Gamma_E(2\alpha) \Gamma_E(\beta)}{\Gamma_E(2\alpha + \beta)} \right)^{\frac{1}{2}} \Gamma^\lambda(t-s, y-x), \quad (3.57)$$

where Γ_E is the Euler Gamma function.

Proof. By the isometry in corollary 1.2.1 we have

$$\begin{aligned} \|(F \otimes H)_n(s, x; t, y)\|_{L^2}^2 &= n! \sum_{(i_1, \dots, i_n)} \|(f \otimes h)_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n]\|_{L^2([s, t]^n)}^2 \\ &= n! \sum_{(i_1, \dots, i_n)} \left\| \sum_{l+m=n} \int_{t_l}^{t_{l+1}} \int_{\mathbb{R}^d} f_l^{(i_1, \dots, i_l)}(s, x; \tau, \xi)[t_1, \dots, t_l] \times \right. \\ &\quad \times h_m^{(i_{l+1}, \dots, i_n)}(\tau, \xi; t, y)[t_{l+1}, \dots, t_n] d\xi d\tau \Big\|_{L^2([s, t]^n)}^2 \\ &\leq n! \sum_{(i_1, \dots, i_n)} \left(\sum_{l+m=n} \left\| \int_{t_l}^{t_{l+1}} \int_{\mathbb{R}^d} f_l^{(i_1, \dots, i_l)}(s, x; \tau, \xi)[t_1, \dots, t_l] \times \right. \right. \\ &\quad \times h_m^{(i_{l+1}, \dots, i_n)}(\tau, \xi; t, y)[t_{l+1}, \dots, t_n] d\xi d\tau \Big\|_{L^2([s, t]^n)} \Big)^2 \end{aligned}$$

(by inequality (3.51) in lemma 3.3.2)

$$\begin{aligned} &\leq n! \sum_{(i_1, \dots, i_n)} \left(\sum_{l+m=n} \int_s^t \int_{\mathbb{R}^d} \|f_l^{(i_1, \dots, i_l)}(s, x; \tau, \xi)\|_{L^2([s, \tau]^l)} \times \right. \\ &\quad \times \|h_m^{(i_{l+1}, \dots, i_n)}(\tau, \xi; t, y)\|_{L^2([\tau, t]^m)} \Big)^2. \end{aligned}$$

Since $l \leq n$, w.l.o.g. we can use (3.55) to estimate

$$\|f_l^{(i_1, \dots, i_l)}(s, x; \tau, \xi)\|_{L^2([s, \tau]^l)} \leq \|F_l(s, x; \tau, \xi)\|_{L^2} \leq \frac{C_F}{(\tau-s)^{\frac{1}{2}-\alpha}} \Gamma^{\lambda'}(\tau-s, \xi-x),$$

and similarly, we use (3.40)

$$\|h_m^{(i_{l+1}, \dots, i_n)}(\tau, \xi; t, y)\|_{L^2([\tau, t]^m)} \leq \|H_m(\tau, \xi; t, y)\|_{L^2} \leq \frac{C}{(t - \tau)^{\frac{1}{2} - \frac{\beta-1}{2}}} \Gamma^\lambda(t - \tau, y - \xi).$$

We have

$$\begin{aligned} \|(F \otimes H)_n(s, x; t, y)\|_{L^2}^2 &\leq n! d_1^n \left((n+1) \int_s^t \int_{\mathbb{R}^d} \frac{C_F}{(\tau - s)^{\frac{1}{2} - \alpha}} \frac{C}{(t - \tau)^{\frac{1}{2} - \frac{\beta-1}{2}}} \times \right. \\ &\quad \left. \times \Gamma^{\lambda'}(\tau - s, \xi - x) \Gamma^\lambda(t - \tau, y - \xi) d\xi d\tau \right)^2 \leq \end{aligned}$$

(by the Chapman-Kolmogorov property for Γ^λ)

$$\leq C_F \cdot C \left(\Gamma^{\max\{\lambda, \lambda'\}}(t - s, y - x) \int_s^t \frac{1}{(\tau - s)^{\frac{1}{2} - \alpha}} \frac{1}{(t - \tau)^{\frac{1}{2} - \frac{\beta-1}{2}}} d\tau \right)^2 \leq$$

(by the well-known properties of the Gamma function and up to rename λ)

$$\leq \left(\frac{C_F \cdot C}{(t - s)^{-\alpha - \frac{\beta-1}{2}}} \frac{\Gamma_E(\alpha + \frac{1}{2}) \Gamma_E(\frac{\beta}{2})}{\Gamma_E(\alpha + \frac{\beta+1}{2})} \Gamma^\lambda(t - s, y - x) \right)^2.$$

Hence (3.56) holds. By the isometry in corollary 1.2.1 we have

$$\begin{aligned} \|(F \otimes_k A^k)_n(s, x; t, y)\|_{L^2}^2 &= n! \sum_{(i_1, \dots, i_n)} \|(f \otimes_k h)_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n]\|_{L^2([s, t]^n)}^2 \\ &= n! \sum_{(i_1, \dots, i_n)} \left\| \sum_{l+m=n-1} \delta_{i_{l+1}, k} \int_{\mathbb{R}^d} f_l^{(i_1, \dots, i_l)}(s, x; t_{l+1}, \xi) [t_1, \dots, t_l] \times \right. \\ &\quad \left. \times a_m^{k, (i_{l+2}, \dots, i_n)}(t_{l+1}, \xi; t, y) [t_{l+2}, \dots, t_n] d\xi \right\|_{L^2([s, t]^n)}^2 \\ &\leq n! \sum_{(i_1, \dots, i_n)} \left(\sum_{l+m=n-1} \delta_{i_{l+1}, k} \left\| \int_{\mathbb{R}^d} f_l^{(i_1, \dots, i_l)}(s, x; t_{l+1}, \xi) [t_1, \dots, t_l] \times \right. \right. \\ &\quad \left. \left. \times a_m^{k, (i_{l+2}, \dots, i_n)}(t_{l+1}, \xi; t, y) [t_{l+2}, \dots, t_n] d\xi \right\|_{L^2([s, t]^n)} \right)^2 \leq \end{aligned}$$

(by inequality (3.53) in lemma 3.3.2)

$$\begin{aligned} &\leq n! \sum_{(i_1, \dots, i_n)} \left(\sum_{l+m=n-1} \delta_{i_{l+1}, k} \left(\int_s^t \left(\int_{\mathbb{R}^d} \|f_l^{(i_1, \dots, i_l)}(s, x; \tau, \xi)\|_{L^2([s, \tau]^l)} \times \right. \right. \right. \\ &\quad \left. \left. \times \|a_m^{k, (i_{l+2}, \dots, i_{n-1})}(\tau, \xi; t, y)\|_{L^2([\tau, t]^m)} d\xi \right)^2 d\tau \right)^{\frac{1}{2}} \right)^2 \end{aligned}$$

(by Cauchy-Schwarz inequality)

$$\begin{aligned} &\leq n! \sum_{(i_1, \dots, i_n)} \left(\sum_{l+m=n-1} \delta_{i_{l+1}, k} \left(\int_s^t \int_{\mathbb{R}^d} \|f_l^{(i_1, \dots, i_l)}(s, x; \tau, \xi)\|_{L^2([s, \tau]^l)}^2 \right. \right. \\ &\quad \left. \left. \times \|a_m^{k, (i_l, \dots, i_{n-1})}(\tau, \xi; t, y)\|_{L^2([\tau, t]^m)}^2 d\xi d\tau \right)^{\frac{1}{2}} \right)^2. \end{aligned}$$

Since $l \leq n$, w.l.o.g. we can use (3.55) to estimate

$$\|f_l^{(i_1, \dots, i_l)}(s, x; \tau, \xi)\|_{L^2([s, \tau]^l)}^2 \leq \|F_l(s, x; \tau, \xi)\|_{L^2}^2 \leq \frac{C_F}{(\tau - s)^{1-2\alpha}} \Gamma^{\lambda'}(\tau - s, \xi - x),$$

and similarly, we use (3.39)

$$\|a_m^{k, (i_{l+1}, \dots, i_{n-1})}(\tau, \xi; t, y)\|_{L^2([\tau, t]^m)}^2 \leq \|A_m^k(\tau, \xi; t, y)\|_{L^2}^2 \leq \frac{C}{(t - \tau)^{1-\beta}} \Gamma^\lambda(t - \tau, y - \xi).$$

We have

$$\|(F \otimes_k A^k)_n(s, x; t, y)\|_{L^2}^2 \leq C' d_1^m \cdot n^2 \int_s^t \int_{\mathbb{R}^d} \frac{C_F}{(\tau - s)^{1-2\alpha}} \frac{C}{(t - \tau)^{1-\beta}} \Gamma^{\lambda'}(\tau - s, \xi - x) \Gamma^\lambda(t - \tau, y - \xi) d\xi d\tau \leq$$

(by the Chapman-Kolmogorov property for Γ^λ)

$$\leq C_F C \cdot \Gamma^{\max\{\lambda, \lambda'\}}(t - s, y - x) \int_s^t \frac{1}{(\tau - s)^{1-2\alpha}} \frac{1}{(t - \tau)^{1-\beta}} d\tau \leq$$

(by the well-known properties of the Gamma function and up to rename λ)

$$\leq \frac{C_F \cdot C}{(t - s)^{1-2\alpha-\beta}} \frac{\Gamma_E(2\alpha) \Gamma_E(\beta)}{\Gamma_E(2\alpha + \beta)} \Gamma^\lambda(t - s, y - x),$$

hence (3.57) follows by taking the square root. $\square$

Definition 3.3.4. Let $H(s, x; t, y) = \{H_n(s, x; t, y)\}_{n=0}^\infty$, $A^k(s, x; t, y) = \{A_n^k(s, x; t, y)\}_{n=0}^\infty$, be the formal Wiener chaos of H and A^k , respectively. Let $F(s, x; t, y) = \{F_n(s, x; t, y)\}_{n=0}^\infty$ the formal Wiener chaos of an element in L'_0 . We define

$$\Lambda(F)(s, x; t, y) = \{\Lambda(F)_n(s, x; t, y)\}_{n=0}^\infty, \quad \text{in } L'_0$$

for $(s, x; t, y) \in D$ by

$$\Lambda(F)_n(s, x; t, y) := (F \otimes H)_n(s, x; t, y) + \sum_{k=1}^{d_1} (F \otimes_k A^k)_n(s, x; t, y), \quad (3.58)$$

for every $n \in \mathbb{N}$. Moreover, we define

$$\Lambda^r(F)_n(s, x; t, y) := \begin{cases} F(s, x; t, y), & \text{if } r = 0, \\ \Lambda(\Lambda^{r-1}(F))_n(s, x; t, y), & \text{if } r \in \mathbb{N}_{\geq 1}, \end{cases}$$

Theorem 3.3.5. *Let $(s, x; t, y) \in D$. There exist positive constant $C = C(T, \lambda_0, d, d_1, n)$, $M_q = M(q, \beta)$ such that*

$$\|\Lambda^r(Z)_n(s, x; t, y)\|_{L^2} \leq \frac{C^r}{(t-s)^{-\frac{r\beta}{2}}} \prod_{q=1}^r M_q \cdot \Gamma^\lambda(t-s, y-x), \quad (3.59)$$

and the series

$$\sum_{r=0}^{\infty} \Lambda^r(Z)(s, x; t, y)$$

converges in L'_0 compact uniformly in $(s, x; t, y) \in D$. Let $\Gamma(s, x; t, y)$ be such limit, then there exist constants $C' = C'(T, \lambda_0, d, d_1, n)$, $\lambda > 0$ such that

$$\|\Gamma_n(s, x; t, y)\|_{L^2} \leq C' \cdot \Gamma^\lambda(t-s, y-x), \quad (3.60)$$

for every $n \in \mathbb{N}$.

Proof. The series $\sum_{r=0}^{\infty} \Lambda(Z)^r$ converges in L'_0 if $\sum_{r=0}^q \Lambda(Z)^r_n$ converges in $L^2(\Omega)$ to some Γ_n as $q \rightarrow \infty$, for every $n \in \mathbb{N}$. We prove the preliminary estimate

$$\|\Lambda(Z)^r_n(s, x; t, y)\|_{L^2} \leq \frac{C^r}{(t-s)^{-\frac{r\beta}{2}}} \prod_{q=1}^r M_q \cdot \Gamma^\lambda(t-s, y-x), \quad (3.61)$$

where $C = C(T, \lambda_0, d, d_1, n)$, $\lambda > 0$ and

$$M_r := \max \left\{ \frac{\Gamma_E(\frac{r-1}{2}\beta + \frac{1}{2})\Gamma_E(\frac{\beta}{2})}{\Gamma_E(\frac{1}{2} + \frac{r}{2}\beta)}, \left(\frac{\Gamma_E(1 + (r-1)\beta)\Gamma_E(\beta)}{\Gamma_E(1 + r\beta)} \right)^{\frac{1}{2}} \right\},$$

where Γ_E is the Euler Gamma function. Fix $n \in \mathbb{N}$, for $r=1$ we have

$$\|\Lambda(Z)_n(s, x; t, y)\|_{L^2} \leq \|(Z \otimes H)_n(s, x; t, y)\|_{L^2} + d_1 \|(Z \otimes_k A^k)_n(s, x; t, y)\|_{L^2} \leq$$

(by estimates (3.33),(3.56) and (3.57))

$$\begin{aligned} &\leq \frac{CM_1}{(t-s)^{-\frac{1}{2}-\frac{\beta-1}{2}}} \Gamma^\lambda(t-s, y-x) \\ &\leq \frac{CM_1}{(t-s)^{-\frac{\beta}{2}}} \Gamma^\lambda(t-s, y-x), \end{aligned}$$

and this proves (3.61) for $k = 1$. Now we assume that (3.61) holds for k and prove it for $k + 1$: we have

$$\|\Lambda(Z)_n^{r+1}(s, x; t, y)\|_{L^2} \leq \|(\Lambda(Z)^r \otimes H)_n(s, x; t, y)\|_{L^2} + d_1 \|(\Lambda(Z)^r \otimes_k A^k)_n(s, x; t, y)\|_{L^2} \leq$$

(by inductive hypothesis and estimates (3.56) and (3.57))

$$\begin{aligned} &\leq \frac{C \cdot C^r M_{r+1}}{(t-s)^{-\frac{r\beta+1}{2}-\frac{\beta-1}{2}}} \prod_{q=1}^r M_q \cdot \Gamma^\lambda(t-s, y-x) \\ &\leq \frac{C^{r+1}}{(t-s)^{-\frac{(r+1)\beta}{2}}} \prod_{q=1}^{r+1} M_q \cdot \Gamma^\lambda(t-s, y-x), \end{aligned}$$

and this proves (3.61). Notice that, from asymptotic estimates for Γ_E , we have

$$\left(\frac{\Gamma_E(1 + (r-1)\beta) \Gamma_E(\beta)}{\Gamma_E(1 + r\beta)} \right)^{\frac{1}{2}} \sim \frac{(r-1)^{\frac{(r-1)\beta}{2}}}{r^{\frac{r\beta}{2}}} \cdot \beta^{-\frac{1}{4}} \sim r^{-\frac{\beta}{2}} \quad \text{as } r \rightarrow \infty,$$

and it coincides with M_r as $r \rightarrow \infty$. Then

$$\prod_{q=1}^r M_q \sim (2\pi)^{-\frac{\beta}{4}} \left(\frac{r}{e} \right)^{-\frac{r\beta}{2}} \quad \text{as } r \rightarrow \infty.$$

As a consequence, there exists $r_0 \in \mathbb{N}$ such that

$$\begin{aligned} \sum_{r=r_0}^{\infty} \|\Lambda^r(Z)_n(s, x; t, y)\|_{L^2} &\leq C \cdot \Gamma^\lambda(t-s, y-x) \sum_{r=r_0}^{\infty} \frac{C^{r-r_0}}{(t-s)^{-\frac{(r-r_0)\beta}{2}}} \left(\frac{r-r_0}{e} \right)^{-\frac{(r-r_0)\beta}{2}} \\ &\leq C \cdot \Gamma^\lambda(t-s, y-x) \sum_{\bar{r}=1}^{\infty} \frac{C^{\bar{r}}}{(t-s)^{-\frac{\bar{r}\beta}{2}}} \left(\frac{\bar{r}}{e} \right)^{-\frac{\bar{r}\beta}{2}} \\ &\leq C \cdot \Gamma^\lambda(t-s, y-x) \sum_{\bar{r}=1}^{\infty} \left(\frac{C^{-\frac{2}{\beta}} \bar{r}}{(t-s)e} \right)^{-\frac{\bar{r}\beta}{2}}, \end{aligned} \tag{3.62}$$

which is finite compact uniformly in $(s, x; t, y) \in D$. Then $\sum_{r=0}^q \Lambda^r(Z)_n$ converges in $L^2(\Omega)$ to

some Γ_n as $q \rightarrow \infty$, for every $n \in \mathbb{N}$ and it is also obvious by the same estimate (3.62) that (3.60) holds. $\square$

Remark 3.3.6. *It should be taken into account the case $n = 0$. From theorem 3.2.2 we have $Z_0(s, x; t, y) = \bar{p}(s, x; t, y)$, where $\bar{p}(s, x; t, y)$, defined as in (3.9), is the parametrix for the deterministic heat equation. By definition of $\otimes$ and $\otimes_k$ it holds*

$$\begin{aligned} (Z \otimes H)_0(s, x; t, y) &= \int_s^t \int_{\mathbb{R}^d} \bar{p}(s, x; \tau, \xi) h(\tau, \xi; t, y) d\xi d\tau, \quad H_0 = h, \\ (Z \otimes_k A^k)_0(s, x; t, y) &= 0. \end{aligned}$$

hence, $(Z \otimes H)_0$ coincides with the convolution operator in the Duhamel representation for the fundamental solution of the classical heat equation. When we take the generalized expectation of Γ as an element in $\mathbb{D}^{-\infty}$, we have

$$\begin{aligned} \mathbb{E}[\Gamma(s, x; t, y)] &= \langle \Gamma(s, x; t, y), 1 \rangle \\ &= \Gamma_0(s, x; t, y) \\ &= \sum_{r=0}^{\infty} \Lambda(\bar{p})_0^r(s, x; t, y) \\ &= \bar{p}(s, x; t, y) + \sum_{r=1}^{\infty} \left((\bar{p} \otimes H)_0^{\otimes(r-1)} \otimes H \right)_0(s, x; t, y) \\ &= \bar{p}(s, x; t, y) + \sum_{r=1}^{\infty} (\bar{p} \otimes H^{\otimes r})_0 \\ &= p(s, x; t, y). \end{aligned}$$

Hence, as we expect, the generalized expectation of Γ , i.e. the projection of Γ on the 0-th order Wiener chaos, coincides with the classical fundamental solution of the heat equation.

Chapter 4

The fundamental solution

In this chapter we show that the candidate Γ obtained as a limit in Theorem 3.3.5 is indeed a fundamental solution for the backward SPDE (2.1), according to definition 2.0.2.

Lemma 4.0.1. *Let $(s, x; t, y) \in D$ and $Z(s, x; t, y) = \{Z_n(s, x; t, y)\}_{n=0}^\infty$ be the parametrix, given by*

$$Z_n(s, x; t, y) = I_n(z_n(s, x; t, y)) = \sum_{(i_1, \dots, i_n)} \int_{[s, t]^n} z_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] dW_{t_1}^{i_1} \dots dW_{t_n}^{i_n}.$$

For each $n \in \mathbb{N}$, $(i_1, \dots, i_n) \in \{1, \dots, d_1\}^n$, kernel $z_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n]$ satisfies the Cauchy problem

$$\left(\frac{\partial}{\partial s} + \bar{\mathbb{L}}^{t, y} \right) z_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = 0, \quad x \in \mathbb{R}^d, \quad s \in [0, t), \quad (4.1)$$

with boundary conditions

$$\lim_{s \rightarrow t_1^-} z_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = \begin{cases} \delta_y(x) & \text{if } n = 0, \\ \bar{\mathbb{G}}^{i_1, t, y} z_{n-1}^{(i_2, \dots, i_n)}(t_1, x; t, y)[t_2, \dots, t_n] & \text{if } n \in \mathbb{N}_{\geq 1}. \end{cases} \quad (4.2)$$

Proof. For notational convenience, we give the proof for the case $d_1 = d = 1$. By formula (1.15) we have

$$h_n \left(\frac{W_t - W_s}{\sqrt{t-s}} \right) = \frac{1}{\|\mathbf{1}_{[s, t]}\|_{L^2}^n} I_n \left(\mathbf{1}_{[s, t]}^{\otimes n} \right),$$

then, from Theorem 3.2.2 we deduce

$$z_n(s, x; t, y)[t_1, \dots, t_n] = \frac{1}{n!(t-s)^{\frac{n}{2}}} \bar{p}(s, x; t, y) h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \mathbf{1}_{[s, t]^n}[t_1, \dots, t_n].$$

Since Hermite polynomials enjoy the relationships

$$h'_n(x) = nh_{n-1}(x), \quad h_{n+1}(x) = xh_n(x) - nh_{n-1}(x), \quad \text{for any } n \in \mathbb{N}_{\geq 1}, \quad (4.3)$$

it holds

$$\begin{aligned} \partial_x z_n(s, x; t, y)[t_1, \dots, t_n] &= \\ &= \frac{1}{n!(t-s)^{\frac{n}{2}}} \bar{p}(s, x; t, y) \left[\frac{y-x}{\sigma_t^2(y)(t-s)} h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) - h'_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \frac{1}{\sigma_t(y)\sqrt{t-s}} \right] \mathbf{1}_{[s,t]^n}[t_1, \dots, t_n] \\ &= \frac{1}{n!\sigma_t(y)(t-s)^{\frac{n+1}{2}}} \bar{p}(s, x; t, y) \left[\frac{y-x}{\sigma_t(y)\sqrt{t-s}} h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) - nh_{n-1} \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \right] \mathbf{1}_{[s,t]^n}[t_1, \dots, t_n] \\ &= \frac{1}{n!\sigma_t(y)(t-s)^{\frac{n+1}{2}}} \bar{p}(s, x; t, y) h_{n+1} \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \mathbf{1}_{[s,t]^n}[t_1, \dots, t_n]. \end{aligned} \quad (4.4)$$

As a consequence, we deduce

$$\partial_{xx} z_n(s, x; t, y)[t_1, \dots, t_n] = \frac{1}{n!\sigma_t^2(y)(t-s)^{\frac{n}{2}+1}} \bar{p}(s, x; t, y) h_{n+2} \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \mathbf{1}_{[s,t]^n}[t_1, \dots, t_n] \quad (4.5)$$

Moreover,

$$\begin{aligned} \partial_s z_n(s, x; t, y)[t_1, \dots, t_n] &= \\ &= \frac{1}{n!(t-s)^{\frac{n}{2}}} \bar{p}(s, x; t, y) \left[\frac{n}{2(t-s)} h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) + \left(\frac{1}{2(t-s)} - \frac{(y-x)^2}{2\sigma_t^2(y)(t-s)^2} \right) h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) + \right. \\ &\quad \left. - h'_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \frac{y-x}{2\sigma_t(y)(t-s)\sqrt{t-s}} \right] \mathbf{1}_{[s,t]^n}[t_1, \dots, t_n] \\ &= -\frac{1}{2n!(t-s)^{\frac{n}{2}+1}} \bar{p}(s, x; t, y) \left[-(n+1)h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) + \frac{(y-x)^2}{\sigma_t^2(y)(t-s)} h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) + \right. \\ &\quad \left. - nh_{n-1} \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right] \mathbf{1}_{[s,t]^n}[t_1, \dots, t_n] \\ &= -\frac{1}{2n!(t-s)^{\frac{n}{2}+1}} \bar{p}(s, x; t, y) \left[-(n+1)h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) + \frac{y-x}{\sigma_t(y)\sqrt{t-s}} h_{n+1} \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \right] \times \\ &\quad \times \mathbf{1}_{[s,t]^n}[t_1, \dots, t_n] \\ &= -\frac{1}{2n!(t-s)^{\frac{n}{2}+1}} \bar{p}(s, x; t, y) h_{n+2} \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \mathbf{1}_{[s,t]^n}[t_1, \dots, t_n]. \end{aligned}$$

Hence $z_n(s, x; t, y)[t_1, \dots, t_n]$ solves

$$\left(\frac{\partial}{\partial s} + \frac{1}{2} \sigma_t^2(y) \partial_{xx} \right) z_n(s, x; t, y)[t_1, \dots, t_n] = 0, \quad x \in \mathbb{R}, \quad s \in [0, t],$$

As for the boundary conditions (4.2), we see from (4.4) that it obviously holds for $n = 0$ and, for $n \geq 1$,

$$\begin{aligned}
\overline{G}^{t,y} z_{n-1}(t_1, x; t, y)[t_2, \dots, t_n] &= \\
&= \sigma_t(y) \frac{\partial}{\partial x} \frac{1}{(n-1)!(t-t_1)^{\frac{n-1}{2}}} \overline{p}(t_1, x; t, y) h_{n-1} \left(\frac{y-x}{\sigma_t(y)\sqrt{t-t_1}} \right) \mathbf{1}_{[t_1, t]^{n-1}}[t_2, \dots, t_n] \\
&= \frac{1}{(n-1)!(t-t_1)^{\frac{n}{2}}} \overline{p}(t_1, x; t, y) h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-t_1}} \right) \mathbf{1}_{[t_1, t]^{n-1}}[t_2, \dots, t_n] \\
&= \lim_{s \rightarrow t_1^-} \frac{1}{(n-1)!(t-s)^{\frac{n}{2}}} \overline{p}(s, x; t, y) h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \mathbf{1}_{[s, t]^{n-1}}[t_2, \dots, t_n] \\
&= \lim_{s \rightarrow t_1^-} \frac{1}{(n-1)!(t-s)^{\frac{n}{2}}} \overline{p}(s, x; t, y) h_n \left(\frac{y-x}{\sigma_t(y)\sqrt{t-s}} \right) \frac{1}{n} \tilde{\mathbf{1}}_{[s, t]^n}[t_1, t_2, \dots, t_n] \\
&= \lim_{s \rightarrow t_1^-} z_n(s, x; t, y)[t_1, \dots, t_n].
\end{aligned}$$

This completes the proof of equation (4.2). $\square$

Theorem 4.0.2. *Let $(s, x; t, y) \in D$ and $\Gamma(s, x; t, y) = \{\Gamma_n(s, x; t, y)\}_{n=0}^\infty$ be the series defined in Theorem 3.3.5. Let*

$$\Gamma_n(s, x; t, y) = I_n(\gamma_n(s, x; t, y)) = \sum_{(i_1, \dots, i_n)} \int_{[s, t]^n} \gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] dW_{t_1}^{i_1} \dots dW_{t_n}^{i_n}.$$

Then, for $n \geq 1$, we have

$$\gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = \int_{\mathbb{R}^d} \dots \int_{\mathbb{R}^d} p(s, x; t_1, y_1) \prod_{j=1}^n G_{t_j}^{i_j} p(t_j, y_j; t_{j+1}, y_{j+1}) dy_1 \dots dy_n, \quad (4.6)$$

with the convention $t_{n+1} = t, y_{n+1} = y$; where $p(s, x; t, y) = \Gamma_0(s, x; t, y)$ is the fundamental solution of the backward Cauchy problem (3.3), operator G_{t_j} is defined as in (2.3) and acts on the backward variables (t_j, y_j) .

Proof. It is well-known that if $\varphi \in \mathcal{S}(\mathbb{R}^d)$ then there exists a unique solution $u_s(x)$ to the Cauchy problem for the backward heat equation

$$\begin{cases} \left(\frac{\partial}{\partial s} + L_s \right) u_s(x) = 0, & x \in \mathbb{R}^d, \text{ a.e. } s \in [0, t], \\ \lim_{s \rightarrow t^-} u_s(x) = \varphi(x), & x \in \mathbb{R}^d, \end{cases} \quad (4.7)$$

and is given by

$$u_s(x) = \int_{\mathbb{R}^d} p(s, x; t, y) \varphi(y) dy,$$

where $p(s, x; t, y)$ is the fundamental solution of $\left(\frac{\partial}{\partial s} + L_s \right)$. Hence, to show equation (4.6), it is suf-

ficient to prove that, for each $n \in \mathbb{N}$ and $(i_1, \dots, i_n) \in \{1, \dots, d_1\}^n$, kernel $\gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n]$ satisfies the Cauchy problem

$$\left(\frac{\partial}{\partial s} + \mathbb{L}_s\right) \gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = 0, \quad x \in \mathbb{R}^d, \quad s \in [0, t), \quad (4.8)$$

with boundary conditions

$$\lim_{s \rightarrow t_1^-} \gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = \begin{cases} \delta_y(x) & \text{if } n = 0, \\ G_{t_1}^{i_1} \gamma_{n-1}^{(i_2, \dots, i_n)}(t_1, x; t, y)[t_2, \dots, t_n] & \text{if } n \in \mathbb{N}_{\geq 1}. \end{cases} \quad (4.9)$$

From Lemma 4.0.1, we have that, for every $n \in \mathbb{N}$, $(i_1, \dots, i_n) \in \{1, \dots, d_1\}^n$, the kernels of the parametrix $Z(s, x; t, y) = \{Z_n(s, x; t, y)\}_{n=0}^\infty$ satisfies the Cauchy problem

$$\left(\frac{\partial}{\partial s} + \bar{\mathbb{L}}^{t, y}\right) z_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = 0, \quad x \in \mathbb{R}^d, \quad s \in [0, t), \quad (4.10)$$

with boundary conditions

$$\lim_{s \rightarrow t_1^-} z_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = \begin{cases} \delta_y(x) & \text{if } n = 0, \\ G_{t_1}^{i_1} z_{n-1}^{(i_2, \dots, i_n)}(t_1, x; t, y)[t_2, \dots, t_n] & \text{if } n \in \mathbb{N}_{\geq 1}. \end{cases} \quad (4.11)$$

Given operators H , A^k , Λ , defined as in (3.18), (3.19), (3.58), we define

$$\Phi^k(s, x; t, y) = \{\Phi_n^k(s, x; t, y)\}_{n=0}^\infty, \quad \Psi(s, x; t, y) = \{\Psi_n^k(s, x; t, y)\}_{n=0}^\infty,$$

as elements in L'_0 by

$$\Phi^k = \sum_{r=0}^\infty \Lambda^r(A^k), \quad (4.12)$$

$$\Psi = \sum_{r=0}^\infty \Lambda^r(H), \quad (4.13)$$

in the same way we defined Γ in Theorem 3.3.5. Note that, on each Wiener chaos $\mathcal{H}_n[s, t]$, it holds

$$\Phi_n^k(s, x; t, y) = \left(G_s^k - \bar{G}^{k, t, y}\right) \Gamma_n(s, x; t, y), \quad k = 1, \dots, d_1.$$

and

$$\Psi_n(s, x; t, y) = \left(\mathbb{L}_s - \bar{\mathbb{L}}^{t, y}\right) \Gamma_n(s, x; t, y).$$

so, at the kernels, we have the identity

$$\psi_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = \left(L_s - \bar{L}^{t, y} \right) \gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n]. \quad (4.14)$$

Then, it follows that

$$\Gamma_n(s, x; t, y) = Z_n(s, x; t, y) + (Z \otimes \Psi)_n(s, x; t, y) + \sum_{k=1}^{d_1} (Z \otimes_k \Phi^k)_n(s, x; t, y). \quad (4.15)$$

Therefore,

$$\begin{aligned} \gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = & \\ & z_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] + \mathbf{1}_{\{n \geq 1\}} \sum_{l+m=n-1} \int_{\mathbb{R}^d} z_l^{(i_1, \dots, i_l)}(s, x; t_l, \xi)[t_1, \dots, t_l] \times \\ & \times \phi_m^{i_{l+1}, (i_{l+2}, \dots, i_n)}(t_{l+1}, \xi; t, y)[t_{l+2}, \dots, t_n] d\xi + \sum_{l+m=n} \int_{\mathbb{R}^d} \int_{t_l}^{t_{l+1}} z_l^{(i_1, \dots, i_l)}(s, x; \tau, \xi)[t_1, \dots, t_l] \times \\ & \times \psi_m^{(i_{l+1}, \dots, i_n)}(\tau, \xi; t, y)[t_{l+1}, \dots, t_n] d\tau d\xi. \end{aligned} \quad (4.16)$$

Hence, noting that

$$\frac{\partial}{\partial s} + L_s = \frac{\partial}{\partial s} + \bar{L}^{t, y} + \left(L_s - \bar{L}^{t, y} \right),$$

from equations (4.10), (4.14) and by the Dominated convergence theorem we deduce that

$$\begin{aligned} \left(\frac{\partial}{\partial s} + L_s \right) \gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = & \\ = - \int_{\mathbb{R}^d} z_0(s, x; s, \xi) \psi_n^{(i_1, \dots, i_n)}(s, \xi; t, y)[t_1, \dots, t_n] d\xi + \left(L_s - \bar{L}^{t, y} \right) \gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] & \\ = - \int_{\mathbb{R}^d} \delta_x(\xi) \psi_n^{(i_1, \dots, i_n)}(s, \xi; t, y)[t_1, \dots, t_n] d\xi + \left(L_s - \bar{L}^{t, y} \right) \gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] & \\ = - \psi_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] + \left(L_s - \bar{L}^{t, y} \right) \gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = 0. & \end{aligned}$$

and problem (4.8) is proved.

As for the boundary conditions (4.9), we see from equations (4.11), (4.14) and (4.16) that it

obviously holds for $n = 0$ and, if $n \geq 1$ by the Dominated convergence theorem,

$$\begin{aligned}
& \lim_{s \rightarrow t_1} \gamma_n^{(i_1, \dots, i_n)}(s, x; t, y)[t_1, \dots, t_n] = \\
& = \bar{G}^{i_1, t, y}_{z_{n-1}^{(i_2, \dots, i_n)}}(s, x; t, y)[t_2, \dots, t_n] + \Phi_{n-1}^{i_1, (i_2, \dots, i_n)}(t_1, x; t, y)[t_2, \dots, t_n] + \\
& + \sum_{\substack{l+m=n-1 \\ l \geq 1}} \int_{\mathbb{R}^d} \bar{G}^{i_1, t, y}_{z_{l-1}^{(i_2, \dots, i_l)}}(t_1, x; t_{l+1}, \xi)[t_2, \dots, t_l] \phi_m^{i_{l+1}, (i_{l+2}, \dots, i_n)}(t_{l+1}, \xi; t, y)[t_{l+2}, \dots, t_n] d\xi + \\
& + \sum_{\substack{l+m=n \\ l \geq 1}} \int_{\mathbb{R}^d} \int_{t_l}^{t_{l+1}} \bar{G}^{i_1, t, y}_{z_{l-1}^{(i_2, \dots, i_l)}}(t_1, x; \tau, \xi)[t_2, \dots, t_l] \psi_m^{(i_{l+1}, \dots, i_n)}(\tau, \xi; t, y)[t_{l+1}, \dots, t_n] d\tau d\xi \\
& = \bar{G}^{i_1, t, y}_{z_{n-1}^{(i_2, \dots, i_n)}}(t_1, x; t, y)[t_2, \dots, t_n] + \Phi_{n-1}^{i_1, (i_2, \dots, i_n)}(t_1, x; t, y)[t_2, \dots, t_n] \\
& = G_{t_1}^{i_1} \gamma_{n-1}^{(i_2, \dots, i_n)}(t_1, x; t, y)[t_2, \dots, t_n] + \left(\bar{G}^{i_1, t, y} - G_{t_1}^{i_1} \right) \gamma_{n-1}^{(i_2, \dots, i_n)}(t_1, x; t, y)[t_2, \dots, t_n] + \\
& + \Phi_{n-1}^{i_1, (i_2, \dots, i_n)}(t_1, x; t, y)[t_2, \dots, t_n] \\
& = G_{t_1}^{i_1} \gamma_{n-1}^{(i_2, \dots, i_n)}(t_1, x; t, y)[t_2, \dots, t_n].
\end{aligned}$$

This completes the proof of equation (4.9). $\square$

Remark 4.0.3. The system of equations (4.8), together with the boundary conditions (4.9), may be regarded as a Wiener chaos version of the Backward diffusion SPDE

$$\begin{cases} -du_s(x) = L_s u_s(x) ds + G_s^k u_s(x) \star dW_s^k, & x \in \mathbb{R}^d, \quad s \in [0, t), \\ \lim_{s \rightarrow t^-} u_s(x) = \delta_y(x), & x \in \mathbb{R}^d, \end{cases}$$

since it is obtained when we project it on the Wiener chaos subspace $\mathcal{H}_n[s, t]$.

Corollary 4.0.4. For the fundamental solution $p(s, x; t, y)$ of the heat equation (3.3), the right-hand side of equation (4.6) belongs to $\tilde{L}^2([s, t]^n)$.

4.1 The Zvonkin-Krylov representation for strong solutions of SDEs

From Chapter 1, we know by Theorem 1.1.14 that any random variable in $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ is representable as a series of multiple stochastic Itô integrals of nonrandom functions. Under Assumptions (1), (2) on b, σ , we shall write a specific representation for the random variable $X_t^{s, x}$, where $X_t^{s, x}$ is the strong solution of SDE

$$\begin{cases} dX_t^{s, x} = b_t(X_t^{s, x}) dt + \sigma_t(X_t^{s, x}) dW_t, & t \in [s, T], \\ X_s^{s, x} = x, & x \in \mathbb{R}^d, \end{cases} \quad (4.17)$$

Definition 4.1.1. Let $(W_t)_{t \in [0, T]}$ be a d_1 -dimensional Brownian motion on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$

and $\lambda \in L^\infty([0, T]; \mathbb{R}^{d_1})$, we define the exponential martingale $M_t^{s, \lambda}$ as the process of the form

$$M_t^{s, \lambda} := \exp \left(\int_s^t \lambda_\tau dW_\tau - \frac{1}{2} \int_s^t |\lambda_\tau|^2 d\tau \right), \quad t \in [s, T]. \quad (4.18)$$

Remark 4.1.2. Note that if $\lambda \in L^\infty([0, T]; \mathbb{R}^{d_1})$ then M_t^λ is a true martingale since the Novikov's condition is obviously satisfied.

Lemma 4.1.3. The space of linear combination of exponential martingales $M_t^{s, \lambda}$, for $\lambda \in L^\infty([0, T]; \mathbb{R}^{d_1})$, is dense in $L^2([s, T], \mathcal{F}_{s, T}^W, \mathbb{P})$, i.e. if $F \in L^2([s, T], \mathcal{F}_{s, T}^W)$, and for every $\lambda \in L^\infty([0, T]; \mathbb{R}^{d_1})$ it holds

$$\mathbb{E} \left[F M_t^{s, \lambda} \right] = 0,$$

then $F = 0$ $\mathbb{P}$ -a.s.

Proof. See [32], Theorem 17.4.1. □

Theorem 4.1.4 (Girsanov). Let $(W_t)_{t \in [0, T]}$ be a d_1 -dimensional Brownian motion and M_t^λ be an exponential martingale on $(\Omega, \mathcal{F}, (\mathcal{F}_t^W), \mathbb{P})$. Define the measure $\mathbb{Q}$ by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = M_T^{s, \lambda}.$$

Then

$$M_t^{s, \lambda} = \mathbb{E}^\mathbb{P} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \middle| \mathcal{F}_t^{s, W} \right], \quad \mathbb{P} - \text{a.s.}, \quad t \in [s, T], \quad (4.19)$$

and the process

$$W_t^{s, \lambda} := W_t - W_s - \int_s^t \lambda_\tau d\tau, \quad t \in [s, T], \quad (4.20)$$

is a Brownian motion on $(\Omega, \mathcal{F}, (\mathcal{F}_t^W), \mathbb{Q})$.

Proof. See [32], Theorem 17.3.3. □

We consider the backward Cauchy problem

$$\begin{cases} \left(\frac{\partial}{\partial s} + L_s \right) v_s(x) = -\lambda_s G_s v_s(x), & s \in [0, t), \quad x \in \mathbb{R}^d, \\ \lim_{s \rightarrow t^-} v_s(x) = \varphi(x), & x \in \mathbb{R}^d, \end{cases} \quad (4.21)$$

where $\varphi \in \mathcal{S}(\mathbb{R}^d)$ and $\lambda \in L^\infty([0, T]; \mathbb{R}^{d_1})$. Although we can apply several methods to find the solution, we shall implement an adaption of the classical parametrix method. The key idea is to transform the non-homogeneous problem into a series of homogeneous problems. We define a sequence of functions $\{v_s^{(n)}(x)\}_{n=0}^\infty$ as follows. Let $v_s^{(0)}(x)$ be the unique solution to the

homogeneous backward Cauchy problem

$$\begin{cases} \left(\frac{\partial}{\partial s} + L_s\right) v_s^{(0)}(x) = 0, & s \in [0, t), \quad x \in \mathbb{R}^d, \\ \lim_{s \rightarrow t^-} v_s^{(0)}(x) = \varphi(x), & x \in \mathbb{R}^d. \end{cases}$$

The solution is given by

$$v_s^{(0)}(x) = \int_{\mathbb{R}^d} p(s, x; t, y) \varphi(y) dy,$$

where $p(s, x; t, y)$ is the fundamental solution of $\left(\frac{\partial}{\partial s} + L_s\right)$. Let $n \in \mathbb{N}_{\geq 1}$, we define $v_s^{(0)}(x)$ as the unique solution to the non-homogeneous backward Cauchy problem

$$\begin{cases} \left(\frac{\partial}{\partial s} + L_s\right) v_s^{(n)}(x) = -\lambda_s G_s v_s^{(n-1)}(x), & s \in [0, t), \quad x \in \mathbb{R}^d, \\ \lim_{s \rightarrow t^-} v_s^{(n)}(x) = 0, & x \in \mathbb{R}^d. \end{cases}$$

Then the solution is given by

$$v_s^{(n)}(x) = \int_s^t \int_{\mathbb{R}^d} p(s, x; \tau, y) \lambda_\tau G_\tau v_\tau^{(n-1)}(y) dy d\tau.$$

By iterating this procedure, we get the following result.

Theorem 4.1.5 (Duhamel formula). *Let $\varphi \in \mathcal{S}(\mathbb{R}^d)$, $\lambda \in L^\infty([0, T]; \mathbb{R}^{d_1})$. Under Assumptions 1, 2, the series $\sum_{n=0}^{\infty} v_s^{(n)}(x)$ converges in $(s, x) \in [0, t] \times \mathbb{R}^d$, for $t \in (0, T]$, and defines the function*

$$v_s(x) = \sum_{n=0}^{\infty} v_s^{(n)}(x), \quad (4.22)$$

which belongs to $C_{0,t}^2(\mathbb{R}^d)$ and satisfies the backward Cauchy problem

$$\begin{cases} \left(\frac{\partial}{\partial s} + L_s\right) v_s(x) = -\lambda_s G_s v_s(x), & s \in [0, t), \quad x \in \mathbb{R}^d, \\ \lim_{s \rightarrow t^-} v_s(x) = \varphi(x), & x \in \mathbb{R}^d, \end{cases} \quad (4.23)$$

Proof. See [5], Theorem 12. □

Theorem 4.1.6 (Zvonkin-Krylov). *Under Assumptions 1, 2, given $\varphi \in \mathcal{S}(\mathbb{R}^d)$ and $X_t^{s,x}$ solution of SDE (4.17), we have*

$$\mathcal{J}_n \varphi(X_t^{s,x}) = I_n \left(\int_{\mathbb{R}^d} \gamma_n(s, x; t, y) \varphi(y) dy \right), \quad \mathbb{P} - a.s. \quad (4.24)$$

for every $n \in \mathbb{N}$.

Proof. For notational convenience, let us consider the case $d_1 = 1$. Let $v_s(x)$ be the function

defined in (4.22), which is solution of problem (4.23). Then

$$\begin{aligned} v_s(x) &= \int_{\mathbb{R}^d} p(s, x; t, y) \varphi(y) dy + \\ &\quad + \sum_{n=1}^{\infty} \int_{S_n[s, t]} \prod_{j=1}^n \lambda_{t_j} dt_1 \dots dt_n \int_{\mathbb{R}^d} \dots \int_{\mathbb{R}^d} p(s, x; t_1, y_1) \prod_{j=1}^n G_{t_j} p(t_j, y_j; t_{j+1}, y_{j+1}) \varphi(y) dy_1 \dots dy_n dy \\ &= \int_{\mathbb{R}^d} p(s, x; t, y) \varphi(y) dy + \sum_{n=1}^{\infty} \int_{S_n[s, t]} \prod_{j=1}^n \lambda_{t_j} dt_1 \dots dt_n \int_{\mathbb{R}^d} \gamma_n(s, x; t, y) [t_1, \dots, t_n] \varphi(y) dy, \end{aligned}$$

since, by Theorem 4.6 we have

$$\gamma_n(s, x; t, y) [t_1, \dots, t_n] = \int_{\mathbb{R}^d} \dots \int_{\mathbb{R}^d} p(s, x; t_1, y_1) \prod_{j=1}^n G_{t_j} p(t_j, y_j; t_{j+1}, y_{j+1}) dy_1 \dots dy_n.$$

By the Girsanov Theorem 4.1.4, given $n \in \mathbb{N}$ and $h \in L^2(S_n[s, t])$, we have

$$\begin{aligned} \mathbb{E}^{\mathbb{P}} \left[M_t^{s, \lambda} \int_{S_n[s, t]} h(t_1, \dots, t_n) dW_{t_1} \dots dW_{t_n} \right] &= \\ &= \mathbb{E}^{\mathbb{Q}} \left[\int_{S_n[s, t]} h(t_1, \dots, t_n) \left(dW_{t_1}^{s, \lambda} + \lambda_{t_1} dt_1 \right) \dots \left(dW_{t_n}^{s, \lambda} + \lambda_{t_n} dt_n \right) \right] \\ &= \int_{S_n[s, t]} h(t_1, \dots, t_n) \lambda_{t_1} dt_1 \dots \lambda_{t_n} dt_n. \end{aligned}$$

Therefore,

$$\begin{aligned} v_s(x) &= \mathbb{E}^{\mathbb{P}} \left[M_t^{s, \lambda} \left(\int_{\mathbb{R}^d} p(s, x; t, y) \varphi(y) dy + \sum_{n=1}^{\infty} \int_{S_n[s, t]} \int_{\mathbb{R}^d} \gamma_n(s, x; t, y) [t_1, \dots, t_n] \varphi(y) dy dW_{t_1} \dots dW_{t_n} \right) \right] \\ &= \mathbb{E}^{\mathbb{P}} \left[M_t^{s, \lambda} \cdot \sum_{n=0}^{\infty} \int_{S_n[s, t]} \int_{\mathbb{R}^d} \gamma_n(s, x; t, y) [t_1, \dots, t_n] \varphi(y) dy dW_{t_1} \dots dW_{t_n} \right] \\ &= \mathbb{E}^{\mathbb{P}} \left[M_t^{s, \lambda} \cdot \sum_{n=0}^{\infty} J_n \left(\int_{\mathbb{R}^d} \gamma_n(s, x; t, y) \varphi(y) dy \right) \right] \end{aligned}$$

Let $X_t^{s, x}$ be the strong solution of SDE (4.17). We prove that the process $(Y_\tau)_{\tau \in [s, t]}$, $Y_\tau = M_\tau^{s, \lambda} v_\tau(X_\tau^{s, x})$ is a martingale. To this end we apply the Itô formula to Y_τ , since

$$dM_\tau^{s, \lambda} = M_\tau^{s, \lambda} \lambda_\tau dW_\tau,$$

we have

$$\begin{aligned}
dY_\tau &= dM_\tau^{s,\lambda} v_\tau(X_\tau^{s,x}) + M_\tau^{s,\lambda} dv_\tau(X_\tau^{s,x}) + \langle M_\tau^{s,\lambda}, v(X_\tau^{s,x}) \rangle_\tau \\
&= M_\tau^{s,\lambda} \lambda_\tau v_\tau(X_\tau^{s,x}) dW_\tau + M_\tau^{s,\lambda} \left(\frac{\partial}{\partial \tau} + L_\tau \right) v_\tau(X_\tau^{s,x}) d\tau + M_\tau^{s,\lambda} G_\tau v_\tau(X_\tau^{s,x}) dW_\tau + M_\tau^{s,\lambda} \lambda_\tau G_\tau v_\tau(X_\tau^{s,x}) d\tau \\
&= M_\tau^{s,\lambda} [\lambda_\tau v_\tau(X_\tau^{s,x}) + G_\tau v_\tau(X_\tau^{s,x})] dW_\tau + M_\tau^{s,\lambda} \left[\left(\frac{\partial}{\partial \tau} + L_\tau \right) v_\tau(X_\tau^{s,x}) + \lambda_\tau G_\tau v_\tau(X_\tau^{s,x}) \right] d\tau \\
&= M_\tau^{s,\lambda} [\lambda_\tau v_\tau(X_\tau^{s,x}) + G_\tau v_\tau(X_\tau^{s,x})] dW_\tau.
\end{aligned}$$

Therefore, $Y_\tau = M_\tau^{s,\lambda} v_\tau(X_\tau^{s,x})$ is a martingale and

$$\mathbb{E}^\mathbb{P} \left[M_t^{s,\lambda} \varphi(X_t^{s,x}) \right] = \mathbb{E}^\mathbb{P} \left[M_t^{s,\lambda} v_t(X_t^{s,x}) \right] = \mathbb{E}^\mathbb{P} \left[M_s^{s,\lambda} v_s(X_s^{s,x}) \right] = v_s(x).$$

As a consequence,

$$\mathbb{E}^\mathbb{P} \left[M_t^{s,\lambda} \varphi(X_t^{s,x}) \right] = \mathbb{E}^\mathbb{P} \left[M_t^{s,\lambda} \cdot \sum_{n=0}^{\infty} J_n \left(\int_{\mathbb{R}^d} \gamma_n(s, x; t, y) \varphi(y) dy \right) \right],$$

and by Lemma 4.1.3 we have

$$\varphi(X_t^{s,x}) = \sum_{n=0}^{\infty} J_n \left(\int_{\mathbb{R}^d} \gamma_n(s, x; t, y) \varphi(y) dy \right), \quad \mathbb{P} - \text{a.s.}$$

Moreover, up to scaling, we have

$$\varphi(X_t^{s,x}) = \sum_{n=0}^{\infty} I_n \left(\int_{\mathbb{R}^d} \gamma_n(s, x; t, y) \varphi(y) dy \right), \quad \mathbb{P} - \text{a.s.} \quad (4.25)$$

Note that the terms in the right-hand side of formula (4.25) are orthogonal thanks to (1.10). $\square$

Corollary 4.1.7. *Under Assumptions 1, 2, it holds*

$$\varphi(X_t^{s,x}) = \int_{\mathbb{R}^d} \Gamma(s, x; t, y) \varphi(y) dy, \quad \mathbb{P} - \text{a.s.} \quad (4.26)$$

and moreover,

$$\Gamma(s, x; t, y) = \delta_y(X_t^{s,x}) \quad \text{in } L'_0, \quad (4.27)$$

Hence, Γ is the fundamental solution of the equation (2.1). In particular, if $\frac{d(p-1)}{p} < \min \{\alpha, 1\}$ then $\Gamma(s, x; t, y) \in \mathbb{D}^{-\alpha, p}$

Proof. By Theorem 4.1.6 we have

$$\mathcal{J}_n \varphi(X_t^{s,x}) = I_n \left(\int_{\mathbb{R}^d} \gamma_n(s, x; t, y) \varphi(y) dy \right), \quad \mathbb{P} - \text{a.s.} \quad (4.28)$$

for every $n \in \mathbb{N}$. On the other hand, since the integrand processes are deterministic functions, we can apply the Fubini's theorem with respect to the Itô-Lebesgue integrals. Then

$$\mathcal{J}_n \int_{\mathbb{R}^d} \Gamma(s, x; t, y) \varphi(y) dy = \int_{\mathbb{R}^d} I_n(\gamma_n(s, x; t, y)) \varphi(y) dy = I_n \left(\int_{\mathbb{R}^d} \gamma_n(s, x; t, y) \varphi(y) dy \right),$$

for every $n \in \mathbb{N}$. By the uniqueness of the Wiener decomposition, we have the thesis and in L'_0 we have the equality

$$\Gamma(s, x; t, y) = \delta_y(X_t^{s,x}),$$

and by Theorem 2.0.6, they belong to $\mathbb{D}^{-\alpha,p}$ if $\frac{d(p-1)}{p} < \min\{\alpha, 1\}$. □

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